

Passivization and composite A/ $\bar{A}$ -movement in the Mandarin BEI-construction

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The BEI-construction in Mandarin is a well studied construction known for exhibiting both passive-like properties and *tough*-movement-like properties (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.). In this paper, I argue for a novel analysis of the BEI-construction in Mandarin as a passive construction where the passive head/BEI hosts a composite probe $[\phi + \bar{A}]$, which triggers composite A/ $\bar{A}$ -movement, in the sense of Van Urk (2015). The subject in the BEI-construction is derived via (successive-cyclic) composite A/ $\bar{A}$ -movement, followed by a terminating step of A-movement, similar to Longenbaugh's (2017) analysis of English *tough*-movement. Under the proposed analysis, the mixed A/ $\bar{A}$ -properties associated with the BEI-construction are direct consequences of composite A/ $\bar{A}$ -movement (following Van Urk 2015; Longenbaugh 2017). The proposed analysis of the BEI-construction accounts for two restrictions on long-distance dependencies in the BEI-construction – the requirement that no overt, case-less NPs should intervene between the subject of BEI and the gap in agent-less BEI-constructions, and the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI.

Keywords: BEI-construction; composite A/ $\bar{A}$ -movement; passive; Mandarin; syntax

1 Introduction

Passivization in a canonical passive construction, such as the English *be*-passive, derives three characteristics of a canonical passive construction that are apparently different from its corresponding simple transitive construction (in the active voice): object promotion, agent/external argument demotion, and the presence of a passive marker (while constructions that lack at least one of these characteristics are referred to as non-canonical passive constructions; see e.g., Legate 2021). Traditionally, passivization is motivated as an instance of *case-driven movement*. According to *Burzio's generalization* (Burzio 1986), all and only the verbs that can assign a theta-role to the (logical) subject can assign accusative case to an object. In the English *be*-passive, the passivized verb does not assign a theta-role to its logical subject, nor

does it assign accusative case; as a result, the object of the passivized verb, which cannot get case from the passivized verb, must move to the grammatical subject position where it can get case (see e.g., Baker, Johnson & Roberts 1989).¹ More recently, Bruening (2013) proposes that the passive construction involves a passive head, which selects a projection of the agent/external-argument-introducing Voice head (Kratzer 1996). In the English *be*-passive, when the agent/external argument of the passivized verb is overtly expressed, it is introduced in a *by*-phrase, and the passive head is semantically vacuous; when the agent/external argument is non-overt and is interpreted as existentially bound, the passive head is responsible for existentially binding the agent/external argument.

It is commonly assumed that A-movement, such as subject-to-subject raising and passivization (in a canonical passive construction), and $\bar{A}$ -movement, such as *wh*-movement, are associated with distinct properties (see e.g., Postal 1971; Chomsky 1977, 1981; a.o. and Richards 2014 for a comprehensive overview of these properties). From a *featural view of the A/ $\bar{A}$ -distinction*, the distinct properties associated with A-movement and $\bar{A}$ -movement are derived from the distinct ϕ - and $\bar{A}$ -features which trigger A-movement and $\bar{A}$ -movement, respectively (Van Urk 2015). Furthermore, the possibility of *composite probing* allows for ϕ - and $\bar{A}$ -features present on the same head to probe together, attracting the closest NP with both a matching ϕ -feature and a matching $\bar{A}$ -feature (Van Urk 2015). The featural view of the A/ $\bar{A}$ -distinction and the possibility of composite probing together predict the existence of *composite A/ $\bar{A}$ -movement*, triggered by the composite probe $[\phi + \bar{A}]$, and that such movement should be associated with mixed A/ $\bar{A}$ -properties. Positive evidence has been found in languages such as Dinka Bor, a Nilotic language, where movement to Spec, CP (e.g., topicalization and relativization) exhibits properties of both A-movement and $\bar{A}$ -movement under the standard diagnostics (Van Urk 2015), and English, where *tough*-movement exhibits the same mix of properties as Dinka movement to Spec, CP (Longenbaugh 2017; see also Chomsky 1977, 1981; Brody 1993; Rezac 2006; Hicks 2009; Takahashi 2011; Hartman 2011; Keine & Poole 2017; a.o.).

In this paper, I argue that the featural view of the A/ $\bar{A}$ -distinction and the possibility of composite probing by the composite probe $[\phi + \bar{A}]$ also allow for a *passive* construction to involve composite A/ $\bar{A}$ -movement, if the passive head hosts a composite probe $[\phi + \bar{A}]$. Specifically, I argue for a novel analysis of the BEI-construction in Mandarin, which exhibits both passive-like properties and *tough*-movement-like properties, as a passive construction where the passive head/BEI hosts a composite probe $[\phi + \bar{A}]$, which triggers composite A/ $\bar{A}$ -movement, in the sense of Van Urk (2015). The derivation of the subject in the BEI-construction involves composite A/ $\bar{A}$ -movement, which proceeds successive-cyclically, followed by a terminating step of A-movement, akin to the analysis proposed by Longenbaugh (2017) for English *tough*-movement. Consequently, the mixed A/ $\bar{A}$ -properties associated with the BEI-construction emerge as direct consequences of composite A/ $\bar{A}$ -movement (following Van Urk 2015; Longenbaugh 2017).

¹ According to Baker, Johnson & Roberts (1989), it is actually the passive suffix that ‘absorbs’ both the agent theta-role and the accusative case of the passivized verb.

The proposed analysis of the BEI-construction as a passive construction where the subject in the BEI-construction is derived via A-movement after (successive-cyclic) composite A/ $\bar{A}$ -movement diverges from a widely accepted alternative approach that derives the dependency involved in the BEI-construction via base-generation of the subject of BEI as an argument of BEI and $\bar{A}$ -movement of a null operator (NOP) in BEI's complement, on a par with Chomsky's (1977, 1981) analysis of English *tough*-movement (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.).²

I will argue that two restrictions on long-distance dependencies in the BEI-construction provide evidence for the proposed analysis of the BEI-construction as a passive construction where the subject in the BEI-construction is derived via A-movement after (successive-cyclic) composite A/ $\bar{A}$ -movement and, crucially, are not accounted for by the various proposals under the alternative approach to the BEI-construction involving base-generation of the subject of BEI and NOP movement in BEI's complement. The first restriction involves a ban on any overt, case-less NPs intervening between the subject of BEI and the gap in agent-less BEI-constructions. This restriction can be accounted for under the proposed analysis of the BEI-construction as a passive construction and Burzio's generalization (Burzio 1986), which states that all and only the verbs that can assign a theta-role to the (logical) subject can assign accusative case to an object. Specifically, in agent-less BEI-constructions, when there is an overt NP that cannot be assigned case by the matrix Voice head, that NP must become the subject of BEI, where it can receive case from Infl; in such cases, it is predicted that long-distance dependencies between the subject of BEI and a deeply embedded gap in BEI's complement is impossible. The second restriction lies in a contrast when the BEI-construction involves a cross-finite-clause dependency between the subject of BEI and a subject vs. object gap. This contrast can be derived from the possibility of raising to subject via A-movement to Spec, CP, or *hyper-raising to subject* (see e.g., Fong 2019; Wurmbrand 2019; Lohninger, Kovač & Wurmbrand 2022; a.o.), and the ban on improper $\bar{A}$ -movement to Spec, CP followed by composite A/ $\bar{A}$ -movement (see Longenbaugh 2017).

The rest of this paper is organized as follows: In section 2, I will provide a primer on the BEI-construction, with the aim of establishing the syntactic properties of the overt or non-overt agent/external argument of the matrix verb, BEI, and its complement, as well as demonstrating both the possibility of and two restrictions on long-distance dependencies between the subject of BEI and the gap in BEI's complement. In section 3, I will show that the dependency involved in the BEI-construction is established via movement and that the BEI-construction exhibits the same mix of properties as Dinka movement to Spec, CP and English *tough*-movement. In section 4, I will provide the details of the proposed analysis of the BEI-construction as a passive construction where the passive head/BEI hosts a composite probe $[\phi + \bar{A}]$ and the subject in the BEI-construction is derived via A-movement after (successive-cyclic) composite A/ $\bar{A}$ -movement. In section 5, I will account for the two restrictions on long-distance dependencies in the BEI-construction – namely, the requirement that no

²These various proposals differ in their specific analyses of BEI-constructions with and without an overt agent/external argument of the matrix verb. See section 4.3 for an overview and section 6 for a more detailed review of each specific proposal.

overt, case-less NPs should intervene between the subject of BEI and the gap in agent-less BEI-constructions, and the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI – under the proposed analysis, and argue against the various proposals under the alternative approach to the BEI-construction. In section 6, I will review the alternative analyses in more detail and recapitulate their problems. In section 7, I will reconcile two conflicting arguments regarding the base-generated vs. derived status of the subject of BEI presented in the literature. In section 8, I will extend the proposed analysis of the BEI-construction to BEI-constructions where the subject of BEI is identified with an indirect object in BEI’s complement (i.e., the so-called *indirect passives*; see e.g., Huang, Li & Li 2009). Finally, section 9 will conclude.

Throughout the paper, the sources of linguistic examples and judgements are cited when they are based on external references; any minor adaptations, such as changes to lexical items among other simplifications and/or modifications, are made without altering the intended purpose of the data. Uncited examples and judgements are my own and have been checked with 15+ additional native speakers.^{3,4}

2 A primer on the bei-construction

In this section, I will provide a primer on the BEI-construction in Mandarin, which is a well studied construction known for exhibiting both passive-like properties and *tough*-movement-like properties (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.). As schematized in (1), a BEI-construction has a subject, followed by BEI, followed by one or multiple (extended) verbal projections. The agent/external argument of the matrix verb may be overtly expressed, in which case it immediately follows BEI, or it may be non-overt, in which case it is interpreted as existentially bound. The BEI-construction involves a dependency between the subject and a gap embedded in the verbal projection(s).

- (1) BEI-construction
NP_i BEI (NP) V (... V ...) ___i (...)

Concretely, like a canonical passive construction, such as the English *be*-passive, the BEI-construction appears to involve object promotion, agent/external argument demotion, and the presence of BEI. Compared with the simple transitive construction (in the active voice) in (2-a), both the *overt-agent* BEI-construction in (2-b) and the *agent-less* BEI-construction in (2-c) appear to involve the promotion of the direct object of the simple transitive verb from the post-verbal position to the grammatical

³During the development of the paper, the examples were constructed, and judgements determined, in consultation with 3 additional native speakers: one (non-linguist) speaking the Beijing/northern dialect and two (linguists) speaking a southern dialect. Additionally, in view of some existing controversies regarding the grammaticality of certain examples in the literature, two judgement-check sessions were conducted on December 27 and 30, 2024, during which judgements on the Mandarin examples presented throughout the paper were checked with 6 additional native speakers (non-linguists) of the Beijing/northern dialect (many of whom offered to consult additional speakers, resulting in 10+ judgement reports on a few controversial examples, which are noted throughout the paper) and 7 additional native speakers (linguists) of various dialects, respectively. I myself am a speaker of the Beijing/northern dialect.

⁴Abbreviations in glosses follow the Leipzig glossing conventions, with the following additions: C=complementizer; CL=classifier; CS=construct state; DIST=distributivity marker; EXP=experiential aspect; HAB=habitual; MOD=modification marker; NF=nonfinite; OV=object voice; P=preposition; SV=subject voice.

subject position. In the overt-agent BEI-construction in (2-b), the agent/external argument of the simple transitive verb is overtly expressed and immediately follows BEI, instead of surfacing in the grammatical subject position. In the agent-less BEI-construction in (2-c), the agent/external argument of the simple transitive verb is non-overt and is interpreted as existentially bound.

(2) a. *Simple transitive (active voice)*

Wo ma-le Lisi.
1SG scold-PRF Lisi
'I scolded Lisi.'

b. *Overt-agent BEI-construction*

Lisi_i bei wo ma-le ___i.
Lisi BEI 1SG scold-PRF
'Lisi was scolded by me.'

c. *Agent-less BEI-construction*

Lisi_i bei ma-le ___i.
Lisi BEI scold-PRF
'Lisi was scolded.'

But unlike the English *be*-passive and like English *tough*-movement, the BEI-construction can also involve multiple verbal projections and a long-distance dependency between the subject of BEI and a deeply embedded gap, as seen in (3).

(3) a. *Long-distance dependency in BEI-construction (object control)*

Na-feng xin_i bei *(wo) jiao Li_j [PRO_j qing Wang_k [PRO_k tuo
that-CL letter BEI 1SG order Li ask Wang entrust
Zhang_l [PRO_l ji-chu-le ___i]].
Zhang send-out-PRF
Lit. 'That letter was ordered Li to ask Wang to entrust Zhang to send out
*(by me).' (Based on Huang, Li & Li 2009: 132: ex. 47b, 48b)

b. *Long-distance dependency in BEI-construction (subject control)*

Gongsi-de wangluo_i ceng bei (heike_j) changshi/qitu [PRO_j ruqin ___i].
company's network once BEI hacker try/attempt hack
Lit. 'The company's network was once tried/attempted to hack (by the
hacker).' (Based on Her 2009: ex. 21a)

The aim of this section is to establish the syntactic properties of the overt or non-overt agent/external argument of the matrix verb, BEI, and its complement, as well as to demonstrate both the possibility of and two restrictions on long-distance dependencies between the subject of BEI and the gap in BEI's complement. In section 2.1, I will present evidence that in overt-agent BEI-constructions, the agent/external argument of the matrix verb, which immediately follows BEI, remains in the thematic subject position as it originates in a simple transitive construction (in the active

voice), that is, Spec, VoiceP. In section 2.2, I will present evidence that in agent-less BEI-constructions, the non-overt agent/external argument of the matrix verb is semantically present. In section 2.3, I will present additional evidence that BEI selects a VoiceP complement (with an overtly expressed agent/external argument of the matrix verb introduced in Spec, VoiceP), and is best analyzed as the spell-out of a passive head, in the sense of Bruening (2013). In section 2.4, I will demonstrate both the possibility of and two restrictions on long-distance dependencies in the BEI-construction. Finally, in section 2.5, I will remark on the adversity reading of the BEI-construction, which, as I will show, is no longer a requirement in modern Mandarin.

2.1 Overt-agent bei-construction

A distinctive property of overt-agent BEI-constructions is that the agent/external argument of the matrix verb, which immediately follows BEI, remains in the thematic subject position as it originates in a simple transitive construction (in the active voice), that is, Spec, VoiceP. Effectively, this means that BEI is best analyzed as the spell-out of a predicate taking an extended verbal projection as its complement (which must at least be as large as a VoiceP, with an overtly expressed agent/external argument of the matrix verb introduced in Spec, VoiceP), as schematized in (4-a) (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.). Crucially, BEI should not be analyzed as a preposition taking the agent/external argument of the matrix verb as its complement and projecting a PP adjunct, as schematized in (4-b) (contra Chao 1968; Cheng 1987; Li 1990; a.o.).

- (4) a. *VoiceP-complementation analysis of BEI*

$$\text{NP}_i \text{ [BEI [VoiceP (NP) V (... V ...) } __i \text{ (...)]}]$$
- b. *NP-complementation analysis of BEI (BEI as preposition)*

$$\text{NP}_i \text{ [VoiceP [PP BEI (NP)] V (... V ...) } __i \text{ (...)]}]$$

The most compelling evidence has come from reflexive binding (see e.g., Ting 1995, 1998; Huang 1999; Huang, Li & Li 2009); additionally, the status of BEI as a predicate taking an extended verbal projection as its complement and crucially not as a preposition taking the agent/external argument of the matrix verb as its complement is supported by its ability to incorporate into Infl to form a matrix A-not-A question in the form BEI-not-BEI (see Huang 1991) and its inability to move with the agent/external argument of the matrix verb as a constituent (see e.g., Huang 1999; Huang, Li & Li 2009).

Specifically, as seen in (5-a), in the BEI-construction, either the subject of BEI (*Lisi*) or the agent/external argument of the matrix verb (*Zhangsan*) can bind the bare reflexive *ziji* ‘self’, which is exclusively bound by subjects when it is bound anaphorically (see e.g., Tang 1989; Cole, Hermon & Sung 1990; Huang & Tang 1991; Huang, Li & Li 2009; a.o.) although it may also be bound logophorically under certain conditions (see e.g., Huang et al. 1984; Huang & Liu 2001; Huang, Li & Li 2009; a.o.), as well as the compound reflexive *ta-ziji* ‘3SG-self’, which is a local anaphor constrained by Principle A (see e.g., Huang & Tang 1991; Huang & Liu

2001): the ability for the agent/external argument of the matrix verb (*Zhangsan*) to bind the subject-oriented, bare reflexive *ziji* ‘self’ (anaphorically) indicates that the agent/external argument of the matrix verb (*Zhangsan*) must not only c-command the subject-oriented, bare reflexive *ziji* ‘self’ but also occupy the thematic subject position, that is, Spec, VoiceP; and the ability for the agent/external argument of the matrix verb (*Zhangsan*) to bind the compound reflexive *ta-ziji* ‘3SG-self’ further suggests that this binding is not (necessarily) achieved logophorically. Additionally, as seen in (5-b), BEI can incorporate into an interrogative Infl to form a matrix A-not-A question in the form BEI-not-BEI, which follows if BEI is a predicate taking an extended verbal projection as its complement (which must at least be as large as a VoiceP, with an overtly expressed agent/external argument of the matrix verb introduced in Spec, VoiceP); in contrast, as seen in (5-c), BEI cannot move with the agent/external argument of the matrix verb as a constituent, indicating that BEI is not a preposition taking the agent/external argument of the matrix verb as its complement.

- (5) *Context: Zhangsan_i introduced Lisi_j to his_{i/j} friends.*
- a. Lisi_i [bei [Zhangsan_j jieshao-gei-le (ta-)ziji_{i/j}-de pengyou]].
 Lisi BEI Zhangsan introduce-to-PRF 3SG-self’s friend
 ‘Lisi_i was introduced by Zhangsan_j to his_{i/j} friend.’ (Based on Ting 1998: ex. 35a)
 - b. Lisi_i bei-meí-bei (jingcha) zhuazou ___i?
 Lisi BEI-not.have-BEI police arrest
 ‘Was Lisi arrested or not (by the police)?’
 - c. *Bei jingcha, Lisi_i zhuazou-le ___i.
 BEI police, Lisi arrest-PRF
 Int. ‘By the police, Lisi was arrested.’

To further support this line of argumentation, consider the contrast between (5) and (6). In (6-a), the pre-verbal applicative argument (*Zhangsan*) is introduced by an applicative head/preposition (*dui* ‘to’); while the applicative argument (*Zhangsan*) fails to bind the subject-oriented, bare reflexive *ziji* ‘self’ (either anaphorically or logophorically), there is speaker variation with respect to whether the applicative argument (*Zhangsan*) can bind the compound reflexive *ta-ziji* ‘3SG-self’. This, I suggest, stems from two possible structures for (6-a): the applicative argument (*Zhangsan*) could be introduced in the specifier of an applicative projection, in which case it c-commands, and hence can bind, the compound reflexive *ta-ziji* ‘3SG-self’, or the applicative argument (*Zhangsan*) could be introduced by a preposition in a PP adjunct, in which case it does not c-command, and hence cannot bind, the compound reflexive *ta-ziji* ‘3SG-self’. These two structures are independently attested in (6-b) and (6-c): In (6-b), the applicative argument introducer (*dui* ‘to’) is an applicative head, hence it can incorporate into an interrogative Infl to form an A-not-A question in the form *dui*-not-*dui*; in this case, the applicative argument (*Zhangsan*) can bind the compound reflexive *ta-ziji* ‘3SG-self’. In (6-c), the applicative argument introducer (*dui* ‘to’) must be a preposition, hence it can move with the applicative argument (*Zhangsan*) as a constituent; in this case, the applicative argument (*Zhangsan*) cannot bind the compound reflexive *ta-ziji* ‘3SG-self’.

- (6) *Context: Lisi said something about Lisi/Zhangsan to Zhangsan.*
- a. Lisi_i dui Zhangsan_j shuo-le {ziji_i/*_j, ta-ziji_i/%_j}-de shi.
 Lisi to Zhangsan say-PRF self 3SG-self's matter
 ‘Lisi_i, to Zhangsan_j, said his_i/*_j matter.’ (Based on Ting 1998: ex. 35b)
 - b. Lisi_i dui-meì-duì Zhangsan_j shuo ta-ziji_i/?_j-de shi?
 Lisi to-not.have-to Zhangsan say 3SG-self's matter
 ‘Did Lisi_i, to Zhangsan_j, said his_i/*_j matter or not?’
 - c. [PP Dui Zhangsan_j], Lisi_i shuo-le ta-ziji_i/*_j-de shi.
 to Zhangsan Lisi say-PRF 3SG-self's matter
 ‘To Zhangsan_j, Lisi_i said his_i/*_j matter.’

As a preview, in section 4, I will analyze overt-agent BEI-constructions as (non-canonical) passive constructions where the passive head/BEI selects a projection of a Voice_[+agent] head (with the [+agent] feature borrowed from Erlewine & Smith 2024) and is responsible for case-licensing the overtly expressed agent/external argument of the matrix verb introduced in Spec, VoiceP, albeit being semantically vacuous.

2.2 Agent-less bei-construction

Like other passive constructions, agent-less BEI-constructions are distinguished by the semantic presence of a non-overt agent/external argument of the matrix verb. To provide context, in the English *be*-passive, a non-overt agent/external argument of the passivized verb must nevertheless be semantically present, because it can be modified by a ‘deliberately’-type adverb, as seen in (7-a), and can control the PRO subject of an infinitival purpose clause, as seen in (7-b) (see e.g., Bhatt & Pancheva 2006, 2017).

- (7) *Implicit agent in English be-passive*
- a. The boat was sunk deliberately. (Bhatt & Pancheva 2017: ex. 25a)
 - b. The boat was sunk to collect the insurance. (Bhatt & Pancheva 2017: ex. 24b)

Similarly, in agent-less BEI-constructions, the non-overt agent/external argument of the matrix verb must nevertheless be semantically present, as supported by its ability to be modified by a ‘deliberately’-type adverb, as seen in (8-a), and its ability to control the PRO subject of an infinitival purpose clause (headed by *lai* ‘in order to’), as seen in (8-b).

- (8) *Implicit agent in BEI-construction*
- a. Chuan_i bei (Lisi) guyi chen-le ___i.
 boat BEI Lisi deliberately sink-PRF
 ‘The boat was sunk (by Lisi) deliberately.’
 - b. Chuan_i bei (Lisi_j) chen-le ___i [lai PRO_j huode peichang].
 boat BEI Lisi sink-PRF in order to receive compensation
 ‘The boat was sunk (by Lisi) in order to receive compensation.’

- c. Lisi_i guyi chen-le zhe-sou chuan [lai PRO_i huode
 Lisi deliberately sink-PRF this-CL boat in order to receive
 peichang].
 compensation
 ‘Lisi deliberately sank the boat in order to receive compensation.’

By contrast, neither modification by ‘deliberately’-type adverbs nor control into purpose clauses is possible with unaccusative constructions, which lack an implicit agent/external argument, both in English, as seen in (9) (see e.g., Bhatt & Pancheva 2006, 2017), and in Mandarin, as seen in (10).

- (9) *No implicit agent in English unaccusative construction*
 a. *The boat sank deliberately. (Bhatt & Pancheva 2017: ex. 25b)
 b. *The boat sank to collect the insurance. (Bhatt & Pancheva 2017: ex. 24a)
- (10) *No implicit agent in Mandarin unaccusative construction*
 a. Chuan_i (*guyi) chen-le ___i.
 boat deliberately sink-PRF
 ‘The boat sank (*deliberately).’
 b. *Chuan_i chen-le ___i [lai PRO huode peichang].
 boat sink-PRF in order to receive compensation
 Int. ‘The boat sank in order to receive compensation.’

In addition, it is important to note that in agent-less BEI-constructions, while the non-overt agent/external argument of the matrix verb is generally interpreted existentially, a null pronoun reading may be enforced if given a strong context. Hence, without any context, (11) would be judged true if and only if Lisi was not scolded by *anyone* and, crucially, cannot mean Lisi was not scolded by *someone*; however, if a strong context is provided, a null pronoun reading becomes possible, as seen in (11) (see Bruening & Tran 2015: 169 for a similar report on the Vietnamese *bi*-construction and related discussion).⁵

- (11) *Context: Zhangsan_j scolded many people today. Did he_j scold Lisi?*
 Lisi_i mei-you bei ma ___i.
 Lisi not-have BEI scold
 ‘Lisi was not scolded (by *pro_j*).’

As a preview, in section 4, I will analyze agent-less BEI-constructions as (canonical) passive constructions where the passive head/BEI selects a projection of a Voice_[-agent] head (with the [-agent] feature borrowed from Erlewine & Smith 2024) and is responsible for existentially binding the agent/external argument (following Bruening 2013; see also Bach 1980; Keenan 1980, 1985; Williams 1987; a.o.).

⁵Ruling out a null pronoun reading of an apparently agent-less BEI-construction will be crucial when investigating the possibility of and restrictions on long-distance dependencies in agent-less BEI-constructions (see section 5.1.1).

2.3 Bei and its complement

In section 2.1, I have presented evidence that BEI is best analyzed as the spell-out of a predicate taking an extended verbal projection – which must at least be as large as a VoiceP, with an overtly expressed agent/external argument of the matrix verb introduced in Spec, VoiceP – as its complement. Here, I will present two pieces of additional evidence that BEI selects a VoiceP complement, and is best analyzed as the spell-out of a passive head, in the sense of Bruening (2013).

First, BEI’s complement must be structurally smaller than an IP/AspP (but nevertheless be as large as a VoiceP, with an overtly expressed agent/external argument of the matrix verb introduced in Spec, VoiceP); more importantly, BEI must project within the domain of extended verbal projections/event predicates, making the passive head a compelling candidate for analysis. The evidence comes from the distribution of tense-aspect-modality elements and event modifiers within a BEI-construction. Specifically, all of temporal adverbs (*zuotian* ‘yesterday’), aspectual adverbs (*yijing*, ‘already’), modal verbs (*hui* ‘will’, *yinggai* ‘should’, *neng* ‘can (be able)’, *keyi* ‘can (be permissible)’), the progressive aspect (*(zheng)zai* and the perfective negation *mei-you* ‘not-have’ must precede/be structurally higher than BEI and cannot follow/be structurally lower than the agent/external argument of the matrix verb (in Spec, VoiceP), as seen in (12-a-c); in contrast, event-internal adverbs (e.g., manner adverbs and subject-oriented adverbs) can either precede/be structurally higher than BEI or follow/be structurally lower than the agent/external argument of the matrix verb (in Spec, VoiceP), as seen in (12-d) (see also Ernst 2010).

- (12) a. Chuan_i {zuotian/yijing} bei (Lisi) {*zuotian/*yijing}
boat yesterday/already BEI Lisi yesterday/already
ji-chen-le __i.
hit-be.sunk-PRF
‘The boat yesterday/already was hit sunk (by Lisi).’
- b. Chuan {hui/yinggai/neng/keyi} bei (Lisi)
boat will/should/can/can BEI Lisi
{*hui/*yinggai/*neng/*keyi} ji-chen __i.
will/should/can/can hit-be.sunk
‘The boat will/should/can/can be hit sunk (by Lisi).’
- c. Chuan {(zheng)zai/mei-you} bei (Lisi) {*zhengzai/*zai/*mei-you}
boat PROG/not-have BEI Lisi PROG/PROG/not-have
ji-chen __i.
hit-be.sunk
‘The boat is being/has not been hit sunk (by Lisi).’
- d. Chuan {manman-de/toutou-de} bei (Lisi) {manman-de/toutou-de}
boat slowly/secretly BEI Lisi slowly/secretly
ji-chen-le __i.
hit-be.sunk-PRF
‘The boat was hit sunk slowly/secretly (by Lisi).’

Second, BEI imposes a selectional restriction on the matrix verb, requiring it to be transitive, which in turn entails the presence of a VoiceP; effectively, this means that BEI selects a VoiceP complement. Specifically, the so-called impersonal passive of unergative, which is possible in languages like German and Dutch but impossible in languages like English, is also impossible in Mandarin, as seen in (14-b). If one adopts the proposal that the agent/external argument of an unergative verb is introduced by a *v* (which categorizes the unergative verbal root as a VP, see e.g., Massam 2009; Tollan 2018; a.o.) and crucially not by the Voice head, then the incompatibility of BEI and an unergative verb would follow if BEI (spells out a passive head which) selects a VoiceP complement.

- (13) *Dutch impersonal passive of unergative*
 Er wordt hier door de jonge lui veel gedanst.
 there be here by the young people a lot danced
 Lit. ‘It is danced here a lot by the young people.’ (Perlmutter & Postal 1984: 107)
- (14) a. *Unergative construction*
 (Zheli you) henduo ren_i [_{VP} ___i ku/xiao-le].
 here have many person cry/laugh-PRF
 Lit. ‘Here cried/laughed many people.’ / ‘Many people cried/laughed.’
- b. *Zheli bei (henduo ren_i) [_{VP} ___i ku/xiao-le].
 here BEI many person cry/laugh-PRF
 Int. ‘Here was laughed/cried (by many people).’

Similarly, BEI is also incompatible with an unaccusative verb that lacks a transitive variant, as seen in (15-b) and (16-b). Assuming that unaccusative verbs, which lack a thematic subject, also lack a VoiceP, the incompatibility of BEI and an unaccusative verb also follows if BEI (spells out a passive head which) selects a VoiceP complement (see also Bruening 2013).

- (15) a. *Unaccusative construction*
 (Zheli you) henduo ren_i [_{VP} si/lai-le ___i].
 here have many person die/come-PRF
 Lit. ‘Here died/came many people.’ / ‘Many people died/came.’
- b. *Zheli bei (henduo ren_i) [_{VP} si/lai-le ___i].
 here BEI many person die/come-PRF
 Int. ‘Here was died/come (by many people).’
- (16) a. *Unaccusative construction*
 (Zheli) [_{VP} si/lai-le henduo ren].
 here die/come-PRF many person
 Lit. ‘(Here) died/came many people.’
- b. *Henduo ren_i bei (zheli) [_{VP} si/lai-le ___i].
 many person BEI here die/come-PRF

Int. ‘Many people were died/come (by here).’

Another related observation of BEI’s selectional restriction is its incompatibility with hyper-raising predicates, which are CP-taking verbs that lack a thematic subject and effectively a VoiceP. This observation and its analysis will be presented in section 5.2.1.

2.4 Restricted long-distance dependencies

The possibility of long-distance dependencies in the BEI-construction has long been recognized in the literature (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Her 2009; Bruening & Tran 2015; a.o.). Specifically, the BEI-construction allows for a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI’s complement *across non-finite clause boundaries*, as seen in (17-b) and (18-b), which involve object control verbs.⁶

- (17) a. *Object control construction*
 Lisi bipo na-ge jingcha_i [PRO_i zhuzou-le Zhangsan].
 Lisi force that-CL police arrest-PRF Zhangsan
 ‘Lisi forced that police to arrest Zhangsan.’
- b. *Long-distance dependency in BEI-construction (object control)*
 Zhangsan_i bei *(Lisi) bipo na-ge jingcha_j [PRO_j zhuzou-le ___i].
 Zhangsan BEI Lisi force that-CL police arrest-PRF
 Lit. ‘Zhangsan was forced that police to arrest *(by Lisi).’
- (18) a. *Object control construction*
 Wo jiao Li_i [PRO_i qing Wang_j [PRO_j tuo Zhang_k [PRO_k
 1SG order Li ask Wang entrust Zhang
 ji-chu-le na-feng xin]]].
 send-out-PRF that-CL letter
 ‘I ordered Li to ask Wang to entrust Zhang to send out that letter.’
- b. *Long-distance dependency in BEI-construction (object control)*
 Na-feng xin_i bei *(wo) jiao Li_j [PRO_j qing Wang_k [PRO_k tuo
 that-CL letter BEI 1SG order Li ask Wang entrust
 Zhang_i [PRO_i ji-chu-le ___i]].
 Zhang send-out-PRF
 Lit. ‘That letter was ordered Li to ask Wang to entrust Zhang to send out *(by me).’ (Based on Huang, Li & Li 2009: 132: ex. 47b, 48b)

Similarly, the BEI-constructions in (19-b) and (20-b), which involve a subject control matrix verb, also involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI’s complement across non-finite clause boundaries

⁶Following Huang (1989) and many others, I assume that object control verbs, including *bi(po)* ‘force’, *pai* ‘send’, *jiao* ‘order’, *qing(qiu)* ‘ask’, *(bai)tuo* ‘entrust’, take a non-finite clausal complement, because their complement cannot contain an aspect marker or a modal. See section 5 for more details.

(note that the original examples from Her 2009, where the agent/external argument of the matrix verb is non-overt, are found on the internet).⁷

- (19) a. *Subject control construction*
Heike_i ceng changshi/qitu [PRO_i ruqin gongsi-de wangluo].
hacker once try/attempt hack company's network
'The hacker once tried/attempted to hack the company's network.'
- b. *Long-distance dependency in BEI-construction (subject control)*
Gongsi-de wangluo_i ceng bei (heike_j) changshi/qitu [PRO_j ruqin ___i].
company's network once BEI hacker try/attempt hack
Lit. 'The company's network was once tried/attempted to hack (by the hacker).' (Based on Her 2009: ex. 21a)
- (20) a. *Subject control construction*
Xiaotou_i shefa [PRO_i kaobei-le ziliao].
thief manage copy-PRF document
'The thief managed to copy the documents.'
- b. *Long-distance dependency in BEI-construction (subject control)*
Ziliao_i bei (xiaotou_j) shefa [PRO_j kaobei-le ___i].
document BEI thief manage copy-PRF
Lit. 'The documents were managed to copy (by the thief).' (Based on Her 2009: ex. 21b)

However, the possibility of long-distance dependencies in *agent-less* BEI-constructions has been a controversy in the literature: On the one hand, in (17-b) and (18-b), the agent/external argument of the matrix verb must be overtly expressed and cannot be non-overt – Huang, Li & Li (2009) (see also Ting 1995, 1998; Huang 1999; a.o.) take this to indicate that only overt-agent BEI-constructions but not agent-less BEI-constructions can involve a long-distance dependency between the subject of BEI and a deeply embedded gap in BEI's complement. On the other hand, in (19-b) and (20-b), the agent/external argument of the matrix verb can be overtly expressed or non-overt – Her (2009) and subsequently Bruening & Tran (2015) and Ngui (2024) take this to indicate that both overt-agent and agent-less BEI-constructions can *generally* involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI's complement across non-finite clause boundaries (see e.g., Her 2009; Bruening & Tran 2015; a.o.).

Contra both sets of proposals, I argue that the contrast between (17-b), (18-b) and (19-b), (20-b) instantiate a restriction on long-distance dependencies in agent-less BEI-constructions, which lies in a ban on any overt NP that would be assigned case by the matrix verb (which would be case-less in agent-less BEI-constructions, under

⁷Following Huang (1989) and many others, I assume that subject control verbs, including *changshi* 'try', *qitu* 'attempt', *shefa* 'manage (Lit. find a way)', take a non-finite clausal complement, because their complement cannot contain an aspect marker or a modal. See section 5 for more details.

the proposed analysis) intervening between the subject of BEI and the gap in agent-less BEI-constructions.⁸ Specifically, (17-b) and (18-b), which are ill-formed when the agent/external argument of the matrix verb is non-overt, involve an overt NP, the matrix object, intervening between the subject of BEI and the deeply embedded object gap; in contrast, (19-b) and (20-b), which are well-formed when the agent/external argument of the matrix verb is non-overt, involve no overt NPs intervening between the subject of BEI and the deeply embedded object gap. As is pointed out by a reviewer, to fully demonstrate the stated empirical generalization on the restriction, a further contrast must be considered: (21-b), like (17-b) and (18-b), is ill-formed when the agent/external argument of the matrix verb is non-overt; in contrast, (22-b) is like (19-b) and (20-b) in that it is well-formed when the agent/external argument of the matrix verb is non-overt, but, crucially, is unlike (19-b) and (20-b) in that there is an overt NP, *Lisi*, which is the object of the embedded verb (*pai/jiao/qing(qiu)/(bai)tu* ‘send/order/ask/entrust’), intervening between the subject of BEI and the deeply embedded object gap.⁹ Importantly, based on examples like (22-b), it is incorrect to simply state the restriction as that no overt NPs should intervene between the subject of BEI and the gap in agent-less BEI-constructions; more examples like (22-b) will be presented in section 5.1.

- (21) a. Zhangsan *pai/jiao/qing(qiu)/(bai)tu* Lisi_i [PRO_i shefa [PRO_i Zhangsan send/order/ask/entrust Lisi manage ji-chu-le na-feng xin]].
 send-out-PRF that-CL letter
 ‘Zhangsan sent/ordered/asked/entrusted Lisi to manage to send out that letter.’
 b. Na-feng xin_i bei *(Zhangsan) *pai/jiao/qing(qiu)/(bai)tu* Lisi_j [PRO_j that-CL letter BEI Zhangsan send/order/ask/entrust Lisi shefa [PRO_j ji-chu-le ___i]].
 manage send-out-PRF
 Lit. ‘That letter was asked Lisi to manage to send out *(by Zhangsan).’
- (22) a. Zhangsan_i shefa [PRO_i *pai/jiao/qing(qiu)/(bai)tu* Lisi_j [PRO_j Zhangsan manage send/order/ask/entrust Lisi ji-chu-le na-feng xin]].
 send-out-PRF that-CL letter

⁸Despite Mandarin lacking overt morphological marking for case, as I will argue extensively in section 5.1, it is crucial to appeal to case in this empirical generalization.

⁹A few speakers I have consulted reported that while (21-b) is completely unacceptable when agent-less, (22-b) is also not (entirely) acceptable when agent-less (with varying degree of degradedness reported); interestingly, the degradedness of (22-b) can be mitigated by embedding a future-oriented adverb such as *ming-tian* ‘tomorrow’ within the infinitival complement of the subject control matrix verb, as in (i):

- (i) Na-feng xin_i bei (Zhangsan_j) shefa [PRO_j *ming-tian pai/jiao/qing(qiu)/(bai)tu* Lisi_k [PRO_k that-CL letter BEI Zhangsan manage next-day send/order/ask/entrust Lisi ji-chu ___i]].
 send-out-PRF
 Lit. ‘That letter was managed to send/order/ask/entrust Lisi to send out tomorrow (by Zhangsan).’

This piece of data will be accounted for in section 5.1.4.

‘Zhangsan managed to send/order/ask/entrust Lisi to send out that letter.’

- b. Na-feng xin_i bei %(Zhangsan_j) shefa [PRO_j
that-CL letter BEI Zhangsan manage
pai/jiao/qing(qiu)/(bai)tuo Lisi_k [PRO_k ji-chu-le ___i].
send/order/ask/entrust Lisi send-out-PRF
Lit. ‘That letter was managed to send/order/ask/entrust Lisi to send out
%(by Zhangsan).’

Another restriction on long-distance dependencies in the BEI-construction lies in a contrast when the BEI-construction involves a *cross-finite-clause* dependency between the subject of BEI and a subject vs. object gap. Specifically, the BEI-construction does not allow for a cross-finite-clause dependency between the subject of BEI and an object gap, as seen in (23-b) (see e.g., Ting 1995, 1998; a.o.), but allows for a cross-finite-clause dependency between the subject of BEI and a subject gap, as seen in (23-c) (see e.g., Her 2009).¹⁰

- (23) a. Jingcha renwei/huaiyi/xiangxin [CP Zhangsan hui mousha Lisi].
police think/suspect/believe Zhangsan will murder Lisi
‘The police thought/suspected/believed that Zhangsan will murder Lisi.’
- b. *Cross-finite-clause dependency in BEI-construction (finite clause object gap)*
*Lisi_i bei (jingcha) renwei/huaiyi/xiangxin [CP Zhangsan hui mousha
Lisi BEI police think/suspect/believe Zhangsan will murder
___i].
- Int. ‘Lisi was thought/suspected/believed that Zhangsan will murder (him) (by the police).’ (Based on Ting 1998: ex. 28c)
- c. *Cross-finite-clause dependency in BEI-construction (finite clause subject gap)*
Zhangsan_i bei (jingcha) renwei/huaiyi/xiangxin [CP ___i hui mousha
Zhangsan BEI police think/suspect/believe will murder
Lisi].
Lisi
Lit. ‘Zhangsan was thought/suspected/believed that (he) will murder Lisi (by the police).’ (Based on Her 2009: ex. 25a)

In section 5, I will argue that the two restrictions on long-distance dependencies in the BEI-construction – namely, the requirement that no overt, case-less NPs should intervene between the subject of BEI and the gap in agent-less BEI-constructions, and the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI – provide evidence for the proposed analysis of the BEI-construction as a passive construction where the subject in the

¹⁰Following Huang (1989) and many others, I assume that verbs like *renwei* ‘think’, *huaiyi* ‘suspect’, *xiangxin* ‘believe’, take a finite CP complement, because their complement can contain an aspect marker as well as a modal. See section 5 for more details.

BEI-construction is derived via A-movement after (successive-cyclic) composite A/ $\bar{A}$ -movement and, crucially, are not accounted for by the various proposals under the alternative approach to the BEI-construction involving base-generation of the subject of BEI and NOP movement in BEI’s complement.

2.5 A note on the adversity reading

Before proceeding, a note is in order on the adversity reading of the BEI-construction. Traditionally, BEI is described as being confined primarily to signal *adversity* (e.g., *bei ma* ‘be scolded’) (see e.g., Chao 1968; Li & Thompson 1981; Cheng 1987; and especially Li 2007, for the historical development of BEI from a lexical verb meaning ‘suffer, receive’ in the pre-Qin period). However, in modern Mandarin, the semantics of BEI has become entirely obscure, which is characteristic of functional categories, and the use of BEI-constructions in non-adversative contexts is fully productive (e.g., *bei biaoYang* ‘be praised’, *bei jiu* ‘be rescued’, *bei anzhuang-hao* ‘be assembled’, *bei xiu-hao* ‘be repaired’, etc.; such instances are abundant and result in a large number of hits in a Google search) (see also Chao 1968; Li & Thompson 1981).

- (24) a. Lisi_i bei (wo) biaoYang/jiu-le ___i.
 Lisi BEI 1SG praise/rescue-PRF
 ‘Lisi was praised/rescued (by me).’
 b. Diannaoli_i bei (wo) anzhuang-hao/xiu-hao-le ___i.
 computer BEI 1SG assemble-be.good/repair-be.good-PRF
 ‘The computer was assembled/repaired (by me).’

In addition, BEI is also compatible with stative predicates, in both adverse and non-adverse contexts:

- (25) Lisi_i bei (henduo ren) hen/(xi)ai/renke/chongbai ___i.
 Lisi BEI many person hate/love/accept/admire
 ‘Lisi is hated/loved/accepted/admire (by many people).’

For present purposes, I take the fully productive use of BEI-constructions in (both adverse and) non-adverse contexts to indicate that BEI has been fully grammaticalized as a functional category in modern Mandarin (albeit possibly originating from a lexical category associated with adversity semantics).

3 Mixed A/ $\bar{A}$ -properties as direct consequences of composite A/ $\bar{A}$ -movement

As mentioned previously, it has long been recognized that A-movement and $\bar{A}$ -movement are associated with distinct properties (see e.g., Postal 1971; Chomsky 1977, 1981; a.o. and Richards 2014 for a comprehensive overview of these properties). More specifically, A-movement, such as subject-to-subject raising and passivization (in a canonical passive construction), (i) is restricted to noun phrases; (ii) is local/cannot cross c-commanding noun phrases; (iii) creates new antecedents for anaphor binding;

(iv) is not subject to weak crossover; (v) does not reconstruct for Principle C; (vi) does not license parasitic gaps; and (vii) feeds $\bar{A}$ -movement. By contrast, $\bar{A}$ -movement, such as *wh*-movement, (i) is not restricted to noun phrases; (ii) can cross c-commanding noun phrases and finite clause boundaries to establish long-distance dependencies; (iii) does not create new antecedents for anaphor binding; (iv) is subject to weak crossover; (v) obligatorily reconstructs for Principle C; (vi) licenses parasitic gaps; and (vii) does not feed A-movement/only feeds $\bar{A}$ -movement (the so-called *Ban on Improper Movement*; see e.g., May 1979; Chomsky 1981; Abels 2007; Neeleman & Van De Koot 2010; Williams 2011).

The *positional* view of the A/ $\bar{A}$ -distinction holds that the distinct properties associated with A-movement and $\bar{A}$ -movement are derived from the distinct A-positions and $\bar{A}$ -positions that A-movement and $\bar{A}$ -movement target, respectively (see e.g., Chomsky 1981, 1995; Mahajan 1990; Déprez 1990; Miyagawa 2009). By contrast, the *featural* view of the A/ $\bar{A}$ -distinction, namely, that the distinct properties associated with A-movement and $\bar{A}$ -movement are derived from the distinct ϕ - and $\bar{A}$ -features which trigger A-movement and $\bar{A}$ -movement, respectively, and the possibility of composite probing by the composite probe $[\phi + \bar{A}]$, which attracts the closest NP with both a matching ϕ -feature and a matching $\bar{A}$ -feature, together predict that mixed properties of both A-movement and $\bar{A}$ -movement emerge as direct consequences of composite A/ $\bar{A}$ -movement (Van Urk 2015). Positive evidence has come from the Nilotic language Dinka Bor, where movement targeting Spec, CP, e.g., topicalization and relativization, exhibits properties of both A-movement and $\bar{A}$ -movement under the standard diagnostics (Van Urk 2015), as well as English *tough*-movement, which exhibits the same mix of properties as Dinka movement to Spec, CP (Longenbaugh 2017; see also Chomsky 1977, 1981; Brody 1993; Rezac 2006; Hicks 2009; Takahashi 2011; Hartman 2011; Keine & Poole 2017; a.o.).

This section (i) reviews the mixed A/ $\bar{A}$ -properties and composite A/ $\bar{A}$ -movement analyses of Dinka movement to Spec, CP and English *tough*-movement, which provide the foundation for (ii) examining the mixed A/ $\bar{A}$ -properties of the BEI-construction and developing a composite A/ $\bar{A}$ -movement analysis for the BEI-construction. The remainder of the section is organized as follows: In sections 3.1 and 3.2, I will review the mixed A/ $\bar{A}$ -properties associated with Dinka movement to Spec, CP and Van Urk's (2015) analysis of Dinka movement to Spec, CP as involving (successive-cyclic) composite A/ $\bar{A}$ -movement, triggered by a composite probe $[\phi + \bar{A}]$ on the C head (and the Voice head(s) involved in the successive-cyclic chain), as well as the same mix of properties associated with English *tough*-movement and Longenbaugh's (2017) analysis of *tough*-movement as involving (successive-cyclic) composite A/ $\bar{A}$ -movement, triggered by a composite probe $[\phi + \bar{A}]$ on the Voice head(s) (involved in the path of *tough*-movement). In section 3.3, I will show that the dependency involved in the BEI-construction is established via movement and that the BEI-construction exhibits the same mix of properties as Dinka movement to Spec, CP and English *tough*-movement. Under the proposed analysis of the BEI-construction as a passive construction involving composite A/ $\bar{A}$ -movement, the mixed A/ $\bar{A}$ -properties associated with the BEI-construction are direct consequences of (successive-cyclic)

composite A/ $\bar{A}$ -movement, triggered by a composite probe $[\phi + \bar{A}]$ on the passive head/BEI (and the Voice head(s) involved in the successive-cyclic chain).

As a preview, the mixed A/ $\bar{A}$ -properties associated with Dinka movement to Spec, CP, English *tough*-movement, and the Mandarin BEI-construction are summarized in (26).¹¹

- (26) *Mixed A/ $\bar{A}$ -properties associated with Dinka movement to Spec, CP, English tough-movement, and the Mandarin BEI-construction*

Properties	A	$\bar{A}$	Dinka mvmt to Spec, CP	English <i>tough</i> -mvmt	Mandarin $\bar{A}$ -mvmt	Mandarin BEI-constrn
New antecedents for anaphors	✓	*	✓	✓	✓	✓
No weak crossover	✓	*	✓	✓	✓	✓
No reconstruction for Principle C	✓	*	✓	✓	✓	✓
Long-distance, ... crossing c-commanding NPs	*	✓	✓	✓	*	✓
Long-distance, ... crossing finite CPs	*	✓	✓	?non-subj *subj	*	✓subj *non-subj
Islands for extraction	*	✓	✓	✓	*	✓
Parasitic gap licensing	*	✓	NA	✓	NA	NA

3.1 Dinka

In Dinka, movement targeting Spec, CP, e.g., topicalization and relativization, behaves like A-movement in that it (i) creates new antecedents for anaphor binding, as seen in (27); (ii) is not subject to weak crossover, as seen in (28); and (iii) does not show reconstruction effects for Principle C, as seen in (29).

- (27) *Dinka topicalization: new antecedents for anaphor binding*
- a. Bol_i a-citii [DP akekool-ti e rɔt-de_i] __ piɔɔlic.
Bol 3S-PRF.OV story-that P self-SG.3SG criticize.NF
‘Bol, that story about himself has criticized (him).’ (Van Urk 2015: 111, ex. 37a)
- b. Bol_i a-cii [DP thura e rɔt-de_i] nyɔɔth [CP ke cuuku __
Bol 3S-PRF.OV picture P self-SG.3SG show.NF C PRF.1PL
tiɪŋ].
see.NF
‘Bol, a picture of himself has shown that we have seen (him).’ (Van Urk 2015: 111, ex. 37b)

- (28) *Dinka topicalization: no weak crossover*

¹¹I leave it to future research to determine whether the BEI-construction can license parasitic gaps. Despite some existing work arguing for the existence of parasitic gaps in Mandarin (see e.g., Lin 2005; Ting & Huang 2008), given the prevalence of null object constructions in Mandarin (see e.g., Aoun & Li 2008; Li 2014; Li & Wei 2014), I contend that it is still necessary to examine the relevant constructions with scrutiny, so as to establish that parasitic gaps indeed exist in Mandarin, before drawing further conclusions.

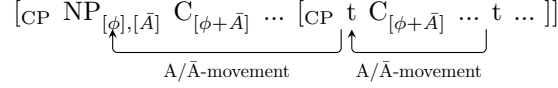
- a. Dhuk eβen_i a-cii thək-de_i __ kaac.
 boy every 3S-PRF.OV goat-CS-SG.SG bite.NF
 ‘Every boy_i, his_i goat bit (him).’ (Van Urk 2015: 110, ex. 35a)
- b. Moc eβen_i a-yii tieeŋ-de_i lueeel [CP e __ thet].
 man every 3S-HAB.OV wife-SG.3SG say.NF C cook.SV
 ‘Every man_i, his_i wife says (he) is cooking.’ (Van Urk 2015: 110, ex. 36a)
- (29) *Dinka topicalization: no reconstruction for Principle C*
- a. [DP Manh e [Mayen ku Ayen]_i] ciike_i __ tiij.
 brother P Mayen.GEN and Ayen PRF.3PL see.NF
 ‘The brother of [Mayen and Ayen]_i, they_i have seen (him).’ (Van Urk 2015: 114, ex. 43a)
- b. [DP Manh e [Mayen ku Ayen]_i] a-yuuku taak [CP
 brother P Mayen.GEN and Ayen 3S-HAB.PL think.NF
 ciike_i __ tiij].
 PRF.3PL see.NF
 ‘The brother of [Mayen and Ayen]_i, we think they_i have seen (him).’ (Van Urk 2015: 114, ex. 44a)

Dinka movement to Spec, CP behaves like $\bar{A}$ -movement in that (i) topicalization can be long-distance, crossing both other c-commanding NPs, as seen in (27), (28), and (29), and finite clause boundaries, as seen in (27-b), (28-b) and (29-b); and (ii) relativization induces islands for extraction, as seen in (30).

- (30) *Dinka relativization: islands for extraction*
- a. Ayen a-ce [DP raan [CP mer toony]] tiij.
 Ayen 3S-PRF.SV person-CS decorate.SV pot see.NF
 ‘Ayen has seen someone who is decorating a pot.’ (Van Urk 2015: 99, ex. 13a)
- b. *Ye ŋo [CP Op cii Ayen [DP raan [CP mer __]]
 be what PRF.OV Ayen.GEN person-CS decorate.SV
 tiij]?
 see.NF
 Lit. ‘What has Ayen seen someone [who is decorating __]?’ (Van Urk 2015: 99, ex. 13b)
- c. *Toony a-cii Ayen [DP raan [CP mer __]] tiij.
 pot 3S-PRF.OV Ayen.GEN person-CS decorate.SV see.NF
 Lit. ‘A pot, Ayen has seen someone who is decorating __.’ (Van Urk 2015: 99, ex. 13c)

Van Urk (2015) proposes that the mixed A/ $\bar{A}$ -properties associated with Dinka movement to Spec, CP emerge as direct consequences of (successive-cyclic) composite A/ $\bar{A}$ -movement, triggered by a composite probe $[\phi + \bar{A}]$ on the C head (and the Voice head(s), for purposes of successive-cyclic composite A/ $\bar{A}$ -movement), as illustrated in (31).

(31) *Van Urk's (2015) analysis of Dinka movement to Spec, CP*



3.2 English

As noted by Longenbaugh (2017), English *tough*-movement exhibits the same mix of properties as Dinka movement to Spec, CP. Specifically, English *tough*-movement behaves like A-movement in that it (i) creates new antecedents for anaphor binding, as seen in (32-a); (ii) is immune to weak crossover, as seen in (32-b); and (iii) does not show reconstruction effects for Principle C, as seen in (32-c).

(32) *English tough-movement: A-properties*

- a. *New antecedents for anaphor binding*
[Jon and Mary]_i were hard for each other_i's friends to get along with ___i. (Longenbaugh 2017: ex. 14b; see also Ruys 2000; Pesetsky 2013)
- b. *No weak crossover*
No employee_i will be easy for us to get his_i boss to fire ___i. (Longenbaugh 2017: ex. 14a; see also Lasnik & Stowell 1991)
- c. *No reconstruction for Principle C*
[Mary_i's father]_j is tough for her_i to get along with ___j. (Longenbaugh 2017: ex. 14c; see also Mulder & den Dikken 1992; Takahashi 2011)

English *tough*-movement behaves like $\bar{A}$ -movement in that it (i) can be long-distance, crossing other c-commanding NPs, as seen in (33-a); (ii) induces weak islands for *wh*-adjunct extraction, as seen in (33-b);¹² and (iii) licenses parasitic gaps, as seen in (33-c).

(33) *English tough-movement: $\bar{A}$ -properties*

- a. *Long-distance*
Aspects_i was annoying to be asked by Joan to convince Matt to read ___i. (Longenbaugh 2017: ex. 15a)
- b. *Islands for extraction*
*Where_i was *Syntactic Structures*_j enjoyable to read ___j ___i?
(Longenbaugh 2017: ex. 15c; see also Chomsky 1977; Rezac 2006)
- c. *Parasitic gap licensing*
?On Raising_i is easy to admire ___i without having read ___i.
(Longenbaugh 2017: ex. 15b; see also Chomsky 1982)

¹²Note that it is possible to extract a *wh*-argument from the *tough*-predicate's complement, as seen in (i).

(i) What balalaika_i are these partitas_j easy to play ___j on ___i. (Pesetsky 1982: ex. 38a)

However, unlike Dinka movement to Spec, CP, which can cross finite clause boundaries, English *tough*-movement, which is possible across non-finite clause boundaries (which arguably lack a CP projection, see e.g., Wurmbrand 2014), as seen in (33-a), is degraded (but possible) for non-subjects and impossible for subjects across a phasal CP-projection, as seen in (34) and (35) (Longenbaugh 2017; see also Postal 1971; Bresnan 1972; Chomsky 1973; Lasnik & Fiengo 1974; Browning 1987; Rezac 2006).¹³

- (34) *Long-distance dependency in English tough-movement (finite object gap)*
- a. ??[*]John_i was easy to show [_{CP} that Bill killed ___i]. (Longenbaugh 2017: ex. 38a; see also Lasnik & Fiengo 1974)
 - b. ?[*]Kim_i is tough for me to believe [_{CP} that Sandy would ever marry ___i]. (Longenbaugh 2017: ex. 38b; see also Hukari & Levine 1991)
 - c. ?[]Mary_i is tough for me to believe [_{CP} that John would ever marry ___i]. (Longenbaugh 2017: ex. 38c; see also Kaplan & Bresnan 1982)
 - d. ?[%]Mary_i is hard for me to believe [_{CP} Leslie kissed ___i]. (Longenbaugh 2017: ex. 38d; see also Dalrymple & King 2000)
 - e. ?[?]This boulder_i would be easy for me to claim [_{CP} that I had lifted ___i]. (Longenbaugh 2017: ex. 38e; see also Heycock 1991)
 - f. ?[]This book_i is difficult to convince people [_{CP} that they ought to read ___i]. (Longenbaugh 2017: ex. 38f; see also Chomsky 1981)
- (35) *Cross-finite-clause dependency in English tough-movement (finite clause subject gap)*
- a. *Jon_i is hard to believe [_{CP} ___i liked Sue]. (Longenbaugh 2017: ex. 37a)
 - b. *That book_i was easy to show [_{CP} ___i sold well when it was first released]. (Longenbaugh 2017: ex. 37b)

Following Van Urk (2015), Longenbaugh (2017) proposes that the mixed A/ $\bar{A}$ -properties associated with English *tough*-movement are direct consequences of (successive-cyclic) composite A/ $\bar{A}$ -movement, triggered by a composite probe [$\phi + \bar{A}$] on the Voice head(s) (involved in the path of *tough*-movement); the *tough*-subject undergoes (successive-cyclic) composite A/ $\bar{A}$ -movement to the matrix Spec, VoiceP, triggered by the composite probe [$\phi + \bar{A}$] on the matrix Voice head (as well as the Voice head(s) involved in the successive-cyclic chain), as illustrated in (36-a), followed by a terminating step of A-movement to Spec, IP, which, Longenbaugh (2017) assumes,

¹³Judgements immediately preceding the square brackets and in the square brackets and are provided by Longenbaugh (2017) and the other cited author(s), respectively. In addition, there seems to also be speakers who generally accept cross-finite-clause *tough*-movement (e.g., David Pesetsky, p.c.)

does not violate the ban on improper (A- after $\bar{A}$ -) movement.¹⁴ The derivation is illustrated in (36).

- (36) *Longenbaugh's (2017) analysis of English tough-movement*
- $$\begin{array}{c}
 [\text{IP NP}_{[\phi],[\bar{A}]} \text{ Infl}_{[\phi]} [\text{VoiceP t Voice}_{[\phi+\bar{A}]} \text{ tough ... V (... V ...) t (...)}]] \\
 \underbrace{\hspace{10em}}_{\text{A-movement}} \quad \underbrace{\hspace{10em}}_{\text{A}/\bar{\text{A}}\text{-movement}}
 \end{array}$$

In addition, Longenbaugh (2017) assumes a ban on composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement, which extends from the ban on improper A-after- $\bar{A}$ -movement.¹⁵ Specifically, to account for the restrictions on long-distance dependencies with English *tough*-movement, Longenbaugh (2017) proposes that the distribution of composite probes can be different in different languages: In Dinka, both the C head and the Voice head host a composite probe $[\phi + \bar{A}]$; hence, composite A/ $\bar{A}$ -movement can cross finite clause boundaries (Van Urk 2015). In English, only the Voice head (involved in the path of *tough*-movement) hosts a composite probe $[\phi + \bar{A}]$ while the C head only hosts a pure $\bar{A}$ -probe; hence, composite A/ $\bar{A}$ -movement can proceed successive-cyclically through the specifiers of successive VoicePs, but crucially cannot proceed from Spec, CP, i.e., following a step of $\bar{A}$ -movement to Spec, CP triggered by the pure $\bar{A}$ -probe on the C head, due to the ban on improper composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement, as illustrated in (37).¹⁶

- (37) *No improper composite-after- $\bar{A}$ -movement in English tough-movement*
- $$\begin{array}{c}
 *[\text{VoiceP NP}_{[\phi],[\bar{A}]} \text{ Voice}_{[\phi+\bar{A}]} \text{ tough ... } [\text{CP t C}_{[\bar{A}]} [\text{IP NP ... V (...)} \text{ t (...)}]] \\
 \underbrace{\hspace{10em}}_{* \text{A}/\bar{\text{A}}\text{-movement}} \quad \underbrace{\hspace{10em}}_{\bar{\text{A}}\text{-movement}}
 \end{array}$$

¹⁴For Longenbaugh (2017), this assumption builds on Neeleman & van De Koot's (2010) insight that A-movement can feed $\bar{A}$ -movement, because A-movement does not reconstruct, but $\bar{A}$ -movement cannot feed A-movement, because $\bar{A}$ -movement must reconstruct, which renders the highest copy of an $\bar{A}$ -movement chain unavailable for (carrying the relevant selectional feature for) further A-movement. Because composite A/ $\bar{A}$ -movement does not show reconstruction effects – it is equivalent to A-movement from the perspective of Neeleman & van De Koot (2010) – it should be able to feed A-movement.

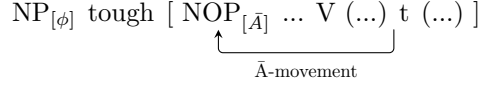
In addition, the assumption that (successive-cyclic) composite A/ $\bar{A}$ -movement can feed A-movement can be made on the basis of Obata & Epstein's (2011) insight that the ban on improper (A- after $\bar{A}$ -) movement follows if a pure $\bar{A}$ -probe only triggers $\bar{A}$ -movement (implemented as copying) of the matching $\bar{A}$ -feature on the goal and crucially not the A-features ($[\phi]$ and $[\text{CASE}]$) on the goal: if it is only the (goal with its) $\bar{A}$ -feature that undergoes $\bar{A}$ -movement, it is expected that the (goal with its) $\bar{A}$ -feature alone cannot undergo further A-movement. Because a composite probe $[\phi + \bar{A}]$ must at least trigger movement/copying of both the matching ϕ -feature and the matching $\bar{A}$ -feature on the goal, it is expected that this goal (with both of its ϕ -feature and $\bar{A}$ -feature) can undergo further A-movement.

¹⁵One can derive the ban on composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement from the perspective of either Neeleman & van De Koot (2010) or Obata & Epstein (2011). From the perspective of Neeleman & van De Koot (2010), if $\bar{A}$ -movement must reconstruct, which renders the highest copy of an $\bar{A}$ -movement chain unavailable for (carrying the relevant selectional feature for) further A-movement, then it should also render the highest copy of an $\bar{A}$ -movement chain unavailable for (carrying the relevant selectional feature for) further composite A/ $\bar{A}$ -movement. From the perspective of Obata & Epstein (2011), if it is only the $\bar{A}$ -feature that is present on the higher copy of the goal in an $\bar{A}$ -movement chain, then it is predicted that this copy of the goal only with an $\bar{A}$ -feature cannot undergo further composite A/ $\bar{A}$ -movement (just like how it cannot undergo further A-movement).

¹⁶Longenbaugh (2017: 10) suggests that degraded (but possible) instances of *tough*-movement, as seen in (34), are derived without an intermediate step of $\bar{A}$ -movement to Spec, CP. Alternatively, for speakers who generally accept cross-finite-clause *tough*-movement, one might assume that the C head (involved in the path of *tough*-movement) also hosts a composite probe $[\phi + \bar{A}]$.

Longenbaugh’s (2017) analysis of English *tough*-movement diverges from Chomsky’s (1977, 1981) analysis of English *tough*-movement, in which the *tough*-predicate is analyzed as a two-place predicate, both introducing the *tough*-subject and selecting a secondary predicate of the *tough*-subject, which contains an $\bar{A}$ -moved NOP, which is co-indexed with the *tough*-subject, as illustrated in (38); at the level of Logical Form (LF), the NOP serves as a lambda operator, which turns a proposition into a predicate via lambda abstraction.

(38) *Chomsky’s (1977, 1981) analysis of English tough-movement*



Chomsky’s (1977, 1981) analysis of English *tough*-movement is bipartite, in the sense that it attempts to derive the A-properties associated with *tough*-movement by base-generating the *tough*-subject as an argument of the *tough*-predicate, and derives the $\bar{A}$ -properties associated with *tough*-movement via $\bar{A}$ -movement of a NOP in the *tough*-predicate’s complement. It is worth noting that weak crossover and Principle C reconstruction might still be expected solely from NOP movement, given that $\bar{A}$ -movement of quantifier phrases and *wh*-phrases is subject to weak crossover and that $\bar{A}$ -movement in general, including topicalization, is subject to Principle C reconstruction. To address these issues, Lasnik & Stowell (1991) suggest that first, weak crossover is restricted to $\bar{A}$ -movement of quantifier phrases and *wh*-phrases, while NOP movement and topicalization involve $\bar{A}$ -movement of a *non-quantificational* R-expression; second, principle C reconstruction is exempted in the case of predication – because predication requires that the *tough*-subject must A-bind the NOP in the *tough*-predicate’s complement, principle C does not apply to A-binding relations between the *tough*-subject and the trace of the NOP (Lasnik & Stowell 1991: 714). One problem with Chomsky’s (1977, 1981) analysis of English *tough*-movement involving base-generation of the *tough*-subject and NOP movement in the *tough*-predicate’s complement is that it does not straightforwardly account for the contrast between (34) and (35). Specifically, if *tough*-movement involves $\bar{A}$ -movement of a NOP, then cross-finite-clause dependencies between the *tough*-subject and either a finite subject gap or a finite object gap should be possible (or impossible, depending on independent assumptions/evidence regarding whether or not NOP movement can cross a finite CP boundary).

3.3 Mandarin

Turning now to Mandarin. In section 3.3.1, I will establish the baseline that IP-external topicalization/focalization and relativization in Mandarin are instances of $\bar{A}$ -movement, which will provide a basis for comparing the $\bar{A}$ -properties exhibited by these constructions with the mixed A/ $\bar{A}$ -properties exhibited by the BEI-construction. In section 3.3.2, I will show that the dependency involved in the BEI-construction is established via movement and that the BEI-construction exhibits the same mix of properties as Dinka movement to Spec, CP and English *tough*-movement. Under

the proposed analysis of the BEI-construction as a passive construction involving composite $A/\bar{A}$ -movement, the mixed $A/\bar{A}$ -properties associated with the BEI-construction are direct consequences of (successive-cyclic) composite $A/\bar{A}$ -movement, triggered by a composite probe $[\phi + \bar{A}]$ on the passive head/BEI (and the Voice head(s) involved in the successive-cyclic chain). In section 3.3.3, I will present additional evidence that the $\bar{A}$ -feature triggering the movement in the BEI-construction is flat, which can be satisfied by any $\bar{A}$ -feature on the goal (see Rizzi 1997, 2004; Abels 2012; a.o.).

3.3.1 $\bar{A}$ -movement

In Mandarin, IP-external topicalization/focalization, which involves a (cross-finite-clause) dependency between a topicalized/focalized NP, which surfaces IP-externally, and a gap, as exemplified by (39-a),¹⁷ and relativization, which involves a (cross-finite-clause) dependency between a relativized NP and a gap, as exemplified by (40-a), are established via movement; this is a well established fact in the literature (see e.g., Qu 1994; Shyu 1995; Ting 1995; Paul 2002, 2005; Kuo 2009). One piece of evidence is their sensitivity to syntactic islands, as seen in (39-b), (39-c) for IP-external topicalization/focalization and (40-b), (40-c) for relativization.

- (39) a. *IP-external topicalization/focalization*
 (Lian) Lisi_i, Zhangsan (dou) renwei/huaiyi/xiangxin [_{CP} wo ma-guo
 even Lisi Zhangsan DIST think/suspect/believe 1SG scold-EXP
 __i].
 ‘(Even) Lisi, Zhangsan thought/suspected/believed that I once scolded (him).’
- b. *IP-external topicalization/focalization: sensitive to complex NP island*
 *Lisi_i/*Lian Lisi_i, jingcha (dou) zhuazou-le [_{NP} na-ge [___j da-le
 Lisi/even Lisi police DIST arrest-PRF that-CL hit-PRF
 __i] de ren_j].
 REL person
 Int. ‘(Even) Lisi, the police arrested that person who hit (him).’
- c. *IP-external topicalization/focalization: sensitive to adjunct island*
 *Lisi_i/*Lian Lisi_i, jingcha (dou) zhuazou-le Zhangsan_j, [yinwei ta_j
 Lisi/even Lisi police DIST arrest-PRF Zhangsan because 3SG
 da-le __i].
 hit-PRF
 Int. ‘(Even) Lisi, the police arrested Zhangsan, because he hit (him).’
- (40) a. *Relativization*

¹⁷In the *lian ... dou* focalization constructions, *lian* ‘even’ is a focus marker and *dou* is a predicate quantifier, according to Shyu (1995). The syntax and semantics of *dou* have been a topic of much discussion but they are not the focus of this paper; for an overview of the distribution and interpretations of *dou*, see e.g., Lin (1998); Xiang (2008); a.o.

na-ge [Zhangsan renwei/huaiyi/xiangxin [CP jingcha zhuazou-le
that-CL Zhangsan think/suspect/believe police arrest-PRF
__i]] de ren_i
REL person

‘that person who Zhangsan thought/suspected/believed that the police arrested’

b. *Relativization: sensitive to complex NP island*

*na-ge [wo zhidao na-ge [__i zhuazou-le __j] de ren_j] de
that-CL 1SG know that-CL arrest-PRF REL person REL
jingcha_i
police

Int. ‘that police who I know that person who (he) arrested’

c. *Relativization: sensitive to adjunct island*

*na-ge [jingcha zhuazou-le Zhangsan_i, [yinwei ta_i da-le __j]] de
that-CL police arrest-PRF Zhangsan because 3SG hit-PRF REL
ren_j
person

Int. ‘that person who the police arrested Zhangsan, because he hit (him)’

IP-external topicalization/focalization and relativization exhibit properties of $\bar{A}$ -movement, which is also a well established fact in the literature (see e.g., Qu 1994; Shyu 1995; Ting 1995; Paul 2002, 2005; Kuo 2009).¹⁸ Specifically, IP-external topicalization/focalization (i) does not create new antecedents for anaphor binding, as seen in (41); (ii) is subject to weak crossover, as seen in (42); and (iii) shows reconstruction effects for Principle C, as seen in (43) (see e.g., Huang 1993; Qu 1994; Shyu 1995; Kuo 2009; a.o.).^{19,20}

(41) *IP-external topicalization/focalization: no new antecedents for anaphor binding*

- a. *Lisi_i/*Lian Lisi_i, (ta-)ziji_i-de pengyou dou ma-guo __i.
Lisi/even Lisi 3SG-self’s friend DIST scold-EXP

¹⁸It is worth mentioning here that topicalization and focalization in Mandarin may also target an IP-internal topic and IP-internal focus position, respectively. In contrast to IP-external topicalization/focalization, IP-internal topicalization/focalization exhibit mixed A/ $\bar{A}$ -properties and are clause-bound – that is, they allow for dependencies across non-finite clause boundaries but not finite ones (see e.g., Qu 1994; Shyu 1995; Ting 1995; Paul 2002, 2005; Kuo 2009) (see section 5.2.2 for further discussion).

¹⁹Principle C reconstruction effects with topicalized phrases exhibit speaker variation (see Huang 1993, footnote 17), but they are relatively robust with focalized phrases, which are reported here.

²⁰Note that IP-external topicalization/focalization also shows reconstruction effects for Principle A, as seen in (i).

(i) *IP-external topicalization/focalization: reconstruction for Principle A*

- a. (Lian) (ta-)ziji_i-de pengyou_j, Lisi_i (dou) ma-guo __j.
even 3SG-self’s friend Lisi DIST scold-EXP
Int. ‘(Even) his_i’s friend_j, Lisi_i DIST once scolded (him_j).’
- b. (Lian) (ta-)ziji_i-de tonghuo_j, Lisi_i (dou) bipo Zhangsan pai jingcha zhuazou-le __j.
even 3SG-self’s complice Lisi DIST force Zhangsan send police arrest-PRF
Int. ‘(Even) his_i complice_j, Lisi_i forced Zhangsan to send the police to arrest (him_j).’

- Int. ‘(Even) Lisi_i, his_i friend once scolded (him_i).’
- b. *Lisi_i/*Lian Lisi_i, (ta-)ziji_i-de tonghuo dou bipo Zhangsan pai
 Lisi/even Lisi 3SG-self’s complice DIST force Zhangsan send
 jingcha zhuazou-le ___i.
 police arrest-PRF
 Int. ‘(Even) Lisi_i, his_i complice forced Zhangsan to send the police to
 arrest (him_i).’
- (42) *IP-external topicalization (of quantifier phrase): weak crossover*
- a. *Mei-ge ren_i, ta_i-de pengyou dou ma-guo ___i.
 every-CL person 3SG’s friend DIST scold-EXP
 Int. ‘Every person_i, his_i friend once scolded (him_i).’
- b. *Mei-ge xiaotou_i, ta_i-de tonghuo dou bipo Zhangsan pai jingcha
 every-CL thief 3SG’s complice DIST force Zhangsan send police
 zhuazou-le ___i.
 arrest-PRF
 Int. ‘Every thief_i, his_i complice forced Zhangsan to send the police to
 arrest (him_i).’
- (43) *IP-external focalization: reconstruction for Principle C*
- a. *Lian zhe-ge ren_i-de pengyou_j, ta_i dou ma-guo ___j.
 even this-CL person’s friend 3SG DIST scold-EXP
 Int. ‘Even this person_i’s friend_j, he_i once scolded (him_j).’ (Based on
 Huang 1993: ex. 54a)
- b. *Lian zhe-ge xiaotou_i-de tonghuo_j, ta_i dou bipo Zhangsan pai
 even this-CL thief’s complice 3SG DIST force Zhangsan send
 jingcha zhuazou-le ___j.
 police arrest-PRF
 Int. ‘Even this thief_i’s complice_j, he_i forced Zhangsan to send the police
 to arrest (him_j).’

In addition, relativization induces (strong) islands for both argument and non-argument extraction. As seen previously in (39-b) and (40-b), extraction of an argument out of a relative clause (via topicalization/focalization or relativization) is impossible. In addition, a relative clause cannot contain the *wh*-adjunct *weishenme* ‘why’, which undergoes covert movement to its scope position in the matrix Spec, CP, hence is subject to island constraints, as seen in (44-a) (cf. English *wh*-phrases, which undergo overt movement to the matrix Spec, CP, hence are subject to island constraints) (Huang 1982; Tsai 1994/2014); in contrast, a relative clause can contain the *wh*-argument *shei* ‘who’, which is subject to unselective binding without needing to move outside the island, hence is not subject to island constraints, as seen in (44-b) (Tsai 1994/2014).

- (44) *Relativization: islands for non-argument extraction*
- a. *Wh-adjunct: island-sensitive*

- *Ni xiang-zhidao [Zhangsan weishenme mousha __i] de ren_i?
 you want-know Zhangsan why murder REL person
 Int. ‘What is the reason x such that you want to know about the person whom Zhangsan murdered for x?’
- b. *Wh-argument: island-insensitive*
 Ni xiang-zhidao [shei mousha __i] de ren_i?
 you want-know who murder REL person
 ‘Who is x such that you want to know about the person whom x murdered?’

Similarly, a relative clause cannot contain a so-called *A-not-A* question, which involves an interrogative Infl of the form *A-not-A*, which undergoes covert head-movement to its scope position in the matrix *C*, hence is subject to island constraints, as seen in (45-a); in contrast, a relative clause can contain a disjunctive question, which involves conjunct reduction and no movement, hence is not subject to island constraints, as seen in (45-b) (Huang 1991).

- (45) *Relative clause: island for non-argument extraction*
- a. *A-not-A question: island-sensitive*
 ??Xiaotou tou-le [shi-bu-shi jia-de __i] de na-fu hua_i?
 thief steal-PRF be-not-be fake-MOD REL that-CL painting
 Int. ‘Did the thief steal that painting which is fake or (that painting which is) not fake?’
- b. *Disjunctive question: island-insensitive*
 Xiaotou tou-le [shi jia-de haishi bu-shi jia-de __i] de
 thief steal-PRF be fake-MOD or not-be fake-MOD REL
 na-fu hua_i?
 that-CL painting
 ‘Did the thief steal that painting which is fake or (that painting which is) not fake?’

3.3.2 The bei-construction

In the BEI-construction, the dependency between the subject of BEI and the gap in BEI’s complement is also derived (entirely or partially) via movement, as supported by its sensitivity to syntactic islands, as seen in (46).²¹

- (46) a. *BEI-construction: sensitive to complex NP island*

²¹Note that the ungrammaticality of (46-a) can be repaired with resumption: (i) is acceptable, with the gap in (46-a) filled with an overt, coreferent pronoun (see also Huang, Li & Li 2009: 125, ex. 27). Also note that (i) is acceptable as a neutral statement of fact – in particular, the subject of BEI (*Zhangsan*) needs not be negatively affected by the event specified by the matrix verb (*zhuazou* ‘arrest’).

- (i) ?Lisi_i bei (jingcha) zhuazou-le [NP na-ge [__j da-le tai_i] de ren_j].
 Lisi BEI police arrest-PRF that-CL hit-PRF 3SG REL person
 Lit. ‘Lisi was arrested that person who hit (him) (by the police).’

By contrast, the ungrammaticality in (46-b) cannot be repaired, even with resumption. This, I suggest, is because the head probing the movement in the BEI-construction (which, under the proposed analysis, is the passive head/BEI) cannot target an NP introduced in an adjunct (see section 8 for related discussion).

*Lisi_i bei (jingcha) zhuazou-le [NP na-ge [____j da-le ____i] de ren_j].
 Lisi BEI police arrest-PRF that-CL hit-PRF 3CL REL person
 Int. ‘Lisi was arrested that person who hit him (by the police).’

b. BEI-construction: sensitive to adjunct island

*Lisi_i bei (jingcha) zhuazou-le Zhangsan_j, [yinwei ta_j da-le ____i].
 Lisi BEI police arrest-PRF Zhangsan because 3SG hit-PRF
 Int. ‘Lisi was arrested Zhangsan, because he hit (him) (by the police).’

Unlike IP-external topicalization/focalization and relativization, the BEI-construction exhibits properties of both A-movement and $\bar{A}$ -movement under the standard diagnostics. Like A-movement and unlike $\bar{A}$ -movement, the BEI-construction (i) creates new antecedents for anaphor binding, as seen in (47-a); (ii) is immune to weak crossover, as seen in (47-b); and (iii) does not show reconstruction effects for Principle C, as seen in (47-c) (see also Kuo 2009).

(47) BEI-construction: A-properties

a. New antecedents for anaphor binding

(Lian) Lisi_i (dou) bei (ta-)ziji_i-de pengyou ma-guo ____i.
 even Lisi DIST BEI 3SG-self’s friend scold-EXP
 ‘(Even) Lisi_i was once scolded by his_i friend.’

b. No weak crossover

Mei-ge ren_i dou bei ta_i-de pengyou ma-guo ____i.
 every-CL person DIST BEI 3SG’s friend scold-EXP
 ‘Every person_i was once scolded by his_i friend.’

c. No reconstruction for Principle C

(Lian) zhe-ge ren_i-de pengyou_j (dou) bei ta_i ma-guo ____j.
 even this-CL person’s friend DIST BEI 3SG scold-EXP
 ‘(Even) this person_i’s friend was once scolded by him_i.’

As mentioned previously in section 2, like $\bar{A}$ -movement and unlike A-movement, the BEI-construction can also involve multiple verbal projections and a long-distance dependency between the subject of BEI and a deeply embedded gap (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Her 2009; Bruening & Tran 2015; a.o.). Note that in such cases, the BEI-construction also exhibits the same A-movement properties, as seen in (48).²²

²²Also note that, unlike $\bar{A}$ -movement, the BEI-construction does not show reconstruction effects for Principle A, whether it involves just a simple transitive verbal projection or multiple verbal projections, as seen in (i).

(i) BEI-construction: no reconstruction for Principle A

- a. *Ta-ziji_i/*Ziji_i-de pengyou_j bei Lisi_i ma-guo ____j.
 3SG-self/self’s friend BEI Lisi scold-EXP
 Int. ‘His_i friend was once scolded by Lisi_i.’
- b. *Ta-ziji_i/*Ziji_i-de tonghuo_j bei Lisi_i bipo Zhangsan pai jingcha zhuazou-le ____j.
 3SG-self/self’s complice BEI Lisi force Zhangsan send police arrest-PRF
 Int. ‘His_i complice_j was forced Zhangsan to send the police to arrest (him_j) by Lisi_i.’

- (48) BEI-construction: *A-properties*
- a. *New antecedents for anaphor binding*
 (Lian) Lisi_i (dou) bei (ta-)ziji_i-de tonghuo bipo Zhangsan pai
 even Lisi DIST BEI 3SG-self's complice force Zhangsan send
 jingcha zhuazou-le ___i.
 police arrest-PRF
 Lit. '(Even) Lisi was forced Zhangsan to send the police to arrest by his
 complice.'
 - b. *No weak crossover*
 Mei-ge xiaotou_i dou bei ta_i-de tonghuo bipo Zhangsan pai jingcha
 every-CL thief DIST BEI 3SG's complice force Zhangsan send police
 zhuazou-le ___i.
 arrest-PRF
 Lit. 'Every thief_i was forced Zhangsan to send the police to arrest (him_i)
 by his_i complice.'
 - c. *No reconstruction for Principle C*
 (Lian) zhe-ge xiaotou_i-de tonghuo_j (dou) bei ta_i bipo Zhangsan pai
 even this-CL thief's complice DIST BEI 3SG force Zhangsan send
 jingcha zhuazou-le ___j.
 police arrest-PRF
 Lit. '(Even) this thief_i's complice was forced Zhangsan to send the police
 to arrest by him_i.'

Cross-linguistically, $\bar{A}$ -movement shows reconstruction effects for Principle A, while languages differ with respect to whether A-movement also shows reconstruction effects for Principle A. In languages like English, both $\bar{A}$ -movement and A-movement show reconstruction effects for Principle A (and for scope), as seen in (ii) (see also Fox 1999, 2000, 2003; Wurmbrand & Bobaljik 1999).

- (ii) a. [Which photos of herself_i] did Bill hear that Mary_i liked __ best? (Pesetsky 2013: 12a)
 b. [This aspect of herself_i] seemed to Mary_i [__ to be a virtue]. (Pesetsky 2013: ex. 27a)

In languages like Dutch, $\bar{A}$ -movement shows reconstruction effects for Principle A, as seen in (iii).

- (iii) Piet zei dat [aan zichzelf_i]_j Jan_i nooit ___j gedacht heeft.
 Peter said that on self John never thought has
 'Peter said that John never thought about himself' (Neeleman & Van De Koot 2010: ex. 13)

However, A-movement does not show reconstruction effects for Principle A, as seen in (iv).

- (iv) Jan_i ziet [[een foto van zichzelf_i/*]_j] Marie_j __ getoond worden].
 John sees a photo of self Mary shown be
 'John sees a photo of himself being shown to Mary.' (Neeleman & Van De Koot 2010: ex. 17b)

In addition, scrambling in languages like German and Japanese is famously known for exhibiting reconstruction effects for binding (and scope) in cases of A-scrambling (also referred to as long/medium or IP scrambling), but a lack of reconstruction effects for binding in cases of A-scrambling (short or VP scrambling); notably, reconstruction effects for scope remain present with A-scrambling (see Saito 1989, 2003; Mahajan 1990; Tada 1993; Nemoto 1993; Lasnik 1999; a.o. see also Wurmbrand 2010 for references therein and further discussion).

I suggest that Mandarin is like Dutch, German and Japanese and unlike English in that only $\bar{A}$ -movement shows reconstruction effects for Principle A. I leave it to future research to explore reconstruction effects for scope in Mandarin.

Also, like $\bar{A}$ -movement and unlike A-movement, the BEI-construction induces weak islands for non-argument extraction.²³ As seen previously in section 3.3.1, the *wh*-adjunct *weishenme* ‘why’ contrasts with the *wh*-argument *shei* ‘who’ in that the former undergoes covert movement to its scope position in the matrix Spec, CP and hence is island-sensitive (Huang 1982; Tsai 1994/2014), while the latter is subject to unselective binding without movement and hence is island-insensitive (Tsai 1994/2014); hence, the ill-formedness of (49-a) indicates that extraction of a non-argument out of a BEI-construction is impossible.

- (49) BEI-construction: islands for non-argument extraction
- a. *Wh-adjunct: island-sensitive*
 ??Wo xiang-zhidao Zhangsan_i bei (jingcha) renwei/huaiyi/xiangxin ___i
 1SG want-know Zhangsan BEI police think/suspect/believe
 weishenme mousha Lisi.
 why murder Lisi
 Int. ‘I want to know the reason x such that Zhangsan was thought/suspected/believed to murder Lisi for x (by the police).’
- b. *Wh-argument: island-insensitive*
 Wo xiang-zhidao Zhangsan_i bei (jingcha) renwei/huaiyi/xiangxin ___i
 1SG want-know Zhangsan BEI police think/suspect/believe
 mousha-le shei.
 murder-PRF who
 ‘I want to know who is x such that Zhangsan was thought/suspected/believed to murder x (by the police).’

Also recall that an A-not-A question contrasts with a disjunctive question in that the former involves covert head-movement of an interrogative Infl of the form A-not-A to its scope position in the matrix C, and hence is island-sensitive, while the latter involves conjunct reduction and no movement, hence is island-insensitive (Huang 1991); hence, the ill-formedness of (50-a) also indicates that extraction of a non-argument out of a BEI-construction is impossible.

- (50) BEI-construction: islands for non-argument extraction
- a. *A-not-A question: island-sensitive*
 *Zhe-fu-hua_i bei (xiaotou) renwei/huaiyi/xiangxin ___i shi-bu-shi
 this-CL-painting BEI thief think/suspect/believe be-not-be
 jia-de?
 fake
 Int. ‘Is this painting thought/suspected/believed to be fake or not fake (by the thief)?’
- b. *Disjunctive question: island-insensitive*

²³The islands induced by the BEI-construction are deemed weak, because it is possible to extract an NP-argument from BEI’s complement, e.g., with a double-object construction, it is possible that the direct object becomes the subject of BEI while the indirect object undergoes $\bar{A}$ -movement; similarly, with a construction involving multiple object control verbs, it is possible that the matrix/more local object becomes the subject of BEI while the embedded/more remote object undergoes $\bar{A}$ -movement. Such cases will be discussed in detail in section 3.3.3.

Zhe-fu-hua_i bei (xiaotou) renwei/huaiyi/xiangxin _____i shi jia-de
 this-CL-painting BEI thief think/suspect/believe be fake
 haishi bu-shi jia-de?
 or not-be fake
 ‘Is this painting thought/suspected/believed to be fake or not fake (by
 the thief)?’

At this point, it can be concluded that the BEI-construction exhibits the same mix of properties as Dinka movement to Spec, CP and English *tough*-movement. Under the proposed analysis of the BEI-construction as a passive construction involving composite A/ $\bar{A}$ -movement, the mixed A/ $\bar{A}$ -properties associated with the BEI-construction are direct consequences of (successive-cyclic) composite A/ $\bar{A}$ -movement, triggered by a composite probe $[\phi + \bar{A}]$ on the passive head/BEI (and the Voice head(s) involved in the successive-cyclic chain).

3.3.3 A flat $\bar{A}$ -feature on bei

Before proceeding to the proposed analysis of the BEI-construction, I will present additional evidence that the $\bar{A}$ -feature triggering the movement in the BEI-construction is flat, which can be satisfied by any $\bar{A}$ -feature on the goal (see Rizzi 1997, 2004; Abels 2012; a.o.).

To provide context, cross-linguistically, languages differ with respect to whether multiple instances of $\bar{A}$ -movement can nest or cross, which can be accounted for by assuming that $\bar{A}$ -probes may be relativized to specific features (e.g., [*Wh*] for *wh*-movement, [*Rel*] for relativization, [*Top*] for topicalization, [*Foc*] for focalization, etc.), or be flat: while a relativized $\bar{A}$ -probe must be satisfied by a goal that has a specific $\bar{A}$ -feature that matches with the specific feature on the probe, a flat $\bar{A}$ -probe can be satisfied by any $\bar{A}$ -feature on the goal (see Rizzi 1997, 2004; Abels 2012; a.o.). Concretely, in Italian, multiple instances of $\bar{A}$ -movement can proceed in either a nested or a crossed fashion, as seen in (51); this can be accounted for by assuming that in Italian, $\bar{A}$ -probes are relativized to specific features (see Rizzi 1997, 2004; Abels 2012; a.o.).

(51) *Italian multiple $\bar{A}$ -dependencies*

a. *Nested dependencies*

?Mi domando, [_{CP} a chi_i, il premio Nobel_j, lo potrebbero dare _____j
 I wonder to whom the prize Nobel it potrebbero could give
 _____i].

‘I wonder to whom, the Nobel Prize, they could give it.’ (Rizzi 1997: 14b)

b. *Crossed dependencies*

Mi domando, [_{CP} il premio Nobel_i, a chi_j lo potrebbero dare _____i
 I wonder the prize Nobel to whom it potrebbero could give
 _____j].

‘I wonder, the Nobel Prize, to whom they could give it.’ (Rizzi 1997: 14a)

By contrast, in English, multiple instances of $\bar{A}$ -movement (of any kinds) must form nested dependencies, as seen in (52) and (53) (see e.g., Pesetsky 1982). This can be accounted for by assuming that in English, $\bar{A}$ -probes are flat, which can be satisfied by any $\bar{A}$ -feature on the goal (see Rizzi 1997, 2004; Abels 2012; a.o.).²⁴

- (52) *English multiple $\bar{A}$ -dependencies*
- a. *Nested dependencies*
This problem_i, Mary knows who_j to consult ___j about ___i. (Pesetsky 1982: 269, ex. 36a)
 - b. *Crossed dependencies*
*This specialist_i, Mary knows what problems_j to consult ___i about ___j. (Pesetsky 1982: 269, ex. 36b)
- (53) *English multiple $\bar{A}$ -dependencies*
- a. *Nested dependencies*
What balalaika_i are these partitas_j easy to play ___j on ___i. (Pesetsky 1982: 269, ex. 38a)
 - b. *Crossed dependencies*
*What partitas_i is this balalaika_j easy to play ___i on ___j. (Pesetsky 1982: 269, ex. 38b)

Mandarin, like Italian and unlike English, allows for multiple instances of $\bar{A}$ -movement to form either nested or crossed dependencies (see e.g., Xu 2000; Kuo 2009; a.o.). Concretely, the examples in (54) involve the indirect object and the direct object of a ditransitive verb undergoing IP-external topicalization, in either a nested or a crossed fashion.

- (54) *Mandarin multiple $\bar{A}$ -dependencies*
- a. *Nested dependencies*
Zhe-jian shi_i (ne), (ta shuo) you-xie ren_j (a), ta mei gaosu
this-CL matter TOP 3SG say exist-PCL person TOP 3SG not tell
___j ___i.

‘This matter, (he said that) some people, he didn’t tell (them) (about it).’ (Based on Xu 2000: ex. 18)
 - b. *Crossed dependencies*

²⁴It is worth mentioning here that the exact analysis of English *tough*-movement by Longenbaugh (2017) assumes that the composite probe on the Voice head consists of a ϕ -feature and a relativized $\bar{A}$ -feature, [AT] (for ‘aboutness topic’) (see Longenbaugh 2017: 21-22); such an assumption is made in order to prevent generalized composite A/ $\bar{A}$ -movement in English (see Longenbaugh 2017: 26-28), but is problematic, because it would allow for crossed dependencies as in (53-b). In order to account for the requirement for nested dependencies as in (53-a), it must be assumed that the composite probe on the Voice head consists of a ϕ -feature and a flat $\bar{A}$ -feature. Hence, one must restrict composite probing in English by positing that only the Voice head(s) involved in the path of *tough*-movement can host the composite probe [$\phi + \bar{A}$].

You-xie ren_i (a), (ta shuo) zhe-jian shi_j (ne), ta mei gaosu
 exist-PCL person TOP 3SG say this-CL matter TOP 3SG not tell
 ___i ___j.

‘Some people, (he said that) this matter, he didn’t tell (them) (about it).’ (Based on Xu 2000: ex. 19)

The examples in (55) involve the matrix object and the embedded object undergoing IP-external topicalization and relativization, in either a nested or a crossed fashion.

(55) *Mandarin multiple $\bar{A}$ -dependencies*

a. *Nested dependencies*

Zhang_i (zhe-ge haizi), [NOP_i wo mingling ___j zhaogu ___i] de
 Zhang this-CL child 1SG order take care of REL
 laoshi_j shi Li.
 police be Li
 ‘Zhang_i (this child), the teacher that [I ordered to take care of (him_i)] is Li.’

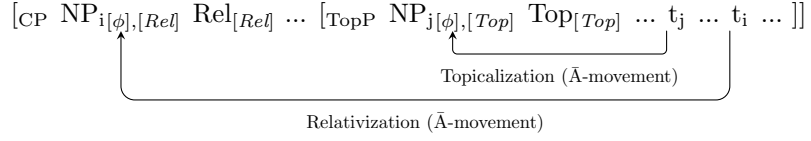
b. *Crossed dependencies*

Zhang_i (zhe-ge laoshi), [NOP_i wo mingling ___i zhaogu ___j] de
 Zhang this-CL teacher 1SG order take care of REL
 haizi_j shi Li.
 child be Li
 ‘Zhang_i (this teacher), the child that [I ordered (him_i) to take care of] is Li.’

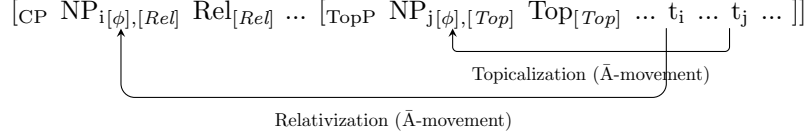
The possibility of either nested or crossed dependencies with multiple instances of $\bar{A}$ -movement in Mandarin suggests that pure $\bar{A}$ -probes in Mandarin are relativized to specific features (e.g., [*Wh*] for *wh*-movement, [*Rel*] for relativization, [*Top*] for topicalization, [*Foc*] for focalization, etc.). Specifically, nested dependencies are formed when the structurally higher NP has the specific $\bar{A}$ -feature of the structurally lower probe and the structurally lower NP has the specific $\bar{A}$ -feature of the structurally higher probe, as illustrated in (56-a), while crossed dependencies are formed when the structurally higher NP has the specific $\bar{A}$ -feature of the structurally higher probe and the structurally lower NP has the specific $\bar{A}$ -feature of the structurally lower probe, as illustrated in (56-b). Note that a detail not illustrated in (56) is successive-cyclic movement – assuming that in the active voice the Voice head heads a phase (Chomsky 2001), topicalization and relativization should proceed successive-cyclically via Spec, VoiceP. Accordingly, the Voice head must also host the specific $\bar{A}$ -features (e.g., [*Wh*] for *wh*-movement, [*Rel*] for relativization, [*Top*] for topicalization, [*Foc*] for focalization, etc.).

(56) *Mandarin multiple $\bar{A}$ -movement*

a. *Nested dependencies*



b. *Crossed dependencies*



However, when two NPs with both ϕ - and $\bar{A}$ -features move from BEI's complement, only the NP closer to BEI can be the subject of BEI. Under the proposed analysis of the BEI-construction where composite $\bar{A}$ /A-movement is triggered by the passive head/BEI, this follows if the $\bar{A}$ -feature on BEI is flat, which can be satisfied by any $\bar{A}$ -feature on the goal (see Rizzi 1997, 2004; Abels 2012; a.o.).²⁵ To set the stage, the examples in (57) show that either the indirect object or the direct object of a ditransitive verb can be the subject in the BEI-construction. In particular, the use of *mei-you ren* 'nobody' and *mei-you shi* 'nothing' in (57) is to ensure that these phrases remain as the subject of BEI, for they resist topicalization, as seen in (58).

- (57) a. {Zhe-ge ren_i, Mei-you ren_i} bei (wo) gaozhi-guo ___i zhe-jian
 this-CL person not-have person BEI 1SG inform-EXP this-CL
 shi.
 matter
 'This person/Nobody was informed about this matter (by me).'
- b. {Zhe-jian shi_i, Mei-you shi_i} bei (wo) gaozhi-guo zhe-ge ren
 this-CL matter not-have matter BEI 1SG inform-EXP this-CL person
 ___i.
 'This matter/Nothing was informed this person (by me).'
- (58) a. {Zhe-ge ren_i, *Mei-you ren_i} (a), wo gaozhi-guo ___i zhe-jian
 this-CL person not-have person TOP 1SG inform-EXP this-CL
 shi.
 matter
 'This person/*Nobody, I informed (them) about this matter.'

²⁵By contrast, the various proposals under the alternative approach to the BEI-construction involving base-generation of the subject of BEI and NOP movement in BEI's complement do not specify that the NOP movement is triggered by a specific head (in BEI's complement) (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.), with the exception of Ngui (2024), who makes an explicit assumption that NOP movement targets Spec, BEIP.

- b. {Zhe-jian shi_i, *Mei-you shi_i} (ne), wo gaozhi-guo zhe-ge ren
 this-CL matter not-have matter TOP 1SG inform-EXP this-CL person
 __i.

‘This matter/*Nothing, I informed this person (about it).’

The well-formed example in (59-a) is derived from (57-a) via topicalization of the direct object (*zhe-jian shi* ‘this matter’) from BEI’s complement. In this case, the subject of BEI (*mei-you ren* ‘nobody’, which remains as the subject of BEI for it resists topicalization) is linked to the indirect object gap, and nested dependencies are formed. By contrast, the ill-formed example in (59-b) is derived from (57-b) via topicalization of the indirect object (*Lisi*) from BEI’s complement. In this case, the subject of BEI (*mei-you shi* ‘nothing’, which remains as the subject of BEI for it resists topicalization) is linked to the direct object gap, and crossed dependencies are formed. Note that (59-c), which has the same derivation as (59-a) except that the subject of BEI (*zhe-ge ren* ‘this person’) undergoes further topicalization, is possible.²⁶

(59) BEI-construction: only nested dependencies

- a. Zhe-jian shi_i (ne), mei-you ren_j bei (wo) gaozhi-guo __j __i.
 this-CL matter TOP not-have person BEI 1SG inform-EXP
 ‘This matter, nobody was informed about (it) (by me).’
- b. *Zhe-ge ren_i (a), mei-you shi_j bei (wo) gaozhi-guo __i __j.
 this-CL person TOP not-have matter BEI 1SG inform-EXP
 Int. ‘This person, nothing was informed (him) about (by me).’
- c. Zhe-ge ren_j (a), zhe-jian shi_i (ne), __j bei (wo) gaozhi-guo
 this-CL person TOP this-CL matter TOP BEI 1SG inform-EXP
 __j __i.

‘This person, this matter, (he) was informed about (it) (by me).’

Similar to the examples in (58), the examples in (60) show that either the matrix object or the embedded object can be the subject in the BEI-construction.

²⁶Holmberg, Sheehan, & Van der Wal (2019) (see also Newman 2021) observe that in North-West English and a number of other languages (e.g., Norwegian, Zulu, and Lubukusu) where either object in a double-object construction is free to undergo passivization and *wh*-movement, while it is possible to passivize the indirect object and *wh*-move the direct object, as seen in (i-a), it is impossible to passivize the direct object and *wh*-move the indirect object, as seen in (i-b).

(i) *North-West English*

- a. Which book_i was John_j given/sent/handed __j __i? (Holmberg, Sheehan, & Van der Wal 2019: ex. 9b)
- b. *Who_i was the book_j given/sent/handed __i __j (by Mary)? (Holmberg, Sheehan, & Van der Wal 2019: ex. 9d)

Note that the contrast in (i) can also be viewed in terms of a contrast between (well-formed) nested and (ill-formed) crossed dependencies.

- (60) a. Li_i (zhe-ge laoshi) ceng bei wo mingling ___i zhaogu-guo Zhang
 Li this-CL teacher once BEI 1SG order take care of-EXP Zhang
 (zhe-ge haizi).
 this-CL child
 ‘Li (this teacher) was once ordered to take care of Zhang (this child) by me.’
- b. Zhang_i (zhe-ge haizi) ceng bei wo mingling Li (zhe-ge laoshi)
 Zhang this-CL child once BEI 1SG order Li this-CL teacher
 zhaogu-guo ___i.
 take care of-EXP
 Lit. ‘Zhang (this child) was once ordered Li (this teacher) to take care of by me.’

The examples in (61) involve both the matrix object and the embedded object moving from BEI’s complement – the matrix and embedded object gaps are linked to the subject of BEI and the head of the relative clause. In all of the examples, *Zhang*, *mei-ge ren* ‘everyone’, or *zhe-ge ren* ‘this person’ is forced to be the subject of BEI, by means of binding the (compound) reflexive *(ta-)ziji* ‘3SG-self’, as in (61-a), or co-reference with the pronominal possessor without incurring weak crossover effects, as in (61-b), or co-reference with the pronoun without incurring reconstruction effects for Principle C, as in (61-c).²⁷ Unlike (60), where the subject of BEI can be linked to either the matrix or embedded object gap, the subject of BEI in the examples in (61) can only be linked to the matrix object gap (e.g., be interpreted as the care taker, and crucially not the person being taken care of).

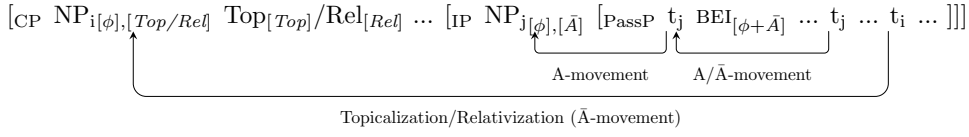
- (61) BEI-construction: only nested dependencies
- a. [Zhang_i bei (ta-)ziji_i-de pengyou mingling ___i/*_j zhaogu ___j/*_i]
 Zhang BEI 3SG-self’s friend order take care of
 de ren_j shi Li.
 REL person be Li
 ‘The person x that [Zhang_i was ordered to take care of x by his_i friend] is Li.’ (Not ‘The person x that [Zhang_i was ordered x to take care of by his_i friend] is Li.’)
- b. [Mei-ge ren_i dou bei ta_i-de pengyou mingling ___i/*_j zhaogu
 every-CL person DIST BEI 3SG’s friend order take care of
 ___j/*_i] de ren_j shi Li.
 REL person be Li
 ‘The person x that [everyone_i was ordered to take care of x by his_i friend] is Li.’ (Not ‘The person x that [everyone_i was ordered x to take care of by his_i friend] is Li.’)

²⁷Recall that the BEI-construction but not $\bar{A}$ -movement (i) creates new antecedents for anaphor binding; (ii) is immune to weak crossover; and (iii) does not show reconstruction effects for Principle C. These facts have been established in sections 3.3.1 and 3.3.2.

- c. [Zhe-ge ren_i-de pengyou_j bei ta_i mingling ___j/*_k zhaogu
 this-CL person's friend BEI 3SG order take care of
 ___k/*_j] de ren_k shi Li.
 REL person be Li
 'The person x that [this person_i's friend_j was ordered to take care of x
 by him_i] is Li.' (Not 'The person that [this person_i's friend_j was ordered
 x to take care of by him_i] is Li.')

To summarize, both (59) and (61) show that when two NPs with both ϕ - and $\bar{A}$ -features move from BEI's complement, only the NP closer to BEI can be the subject of BEI. Under the proposed analysis, this follows if the composite probe on the passive head/BEI consists of a ϕ -feature and a flat $\bar{A}$ -feature, which can be satisfied by any $\bar{A}$ -feature on the goal (see Rizzi 1997, 2004; Abels 2012; a.o.), as illustrated in (62).

(62) *Mandarin passivization and topicalization/relativization: only nested dependencies*



Importantly, the contrast between (54), (55) and (59), (61) suggests a difference between the active voice and the BEI-construction, which, under the proposed analysis, is a passive construction. Specifically, the difference is that the $\bar{A}$ -feature on the Voice head must either be relativized to specific features (e.g., [*Wh*] for *wh*-movement, [*Rel*] for relativization, [*Top*] for topicalization, [*Foc*] for focalization, etc.) or be flat (for purposes of successive-cyclic composite A/ $\bar{A}$ -movement in the BEI-construction), while the $\bar{A}$ -feature on the passive head/BEI can only be flat.

Finally, because in Mandarin, topicalization is triggered by a specific [*Top*] feature, while the (composite A/ $\bar{A}$ -)movement involved in the BEI-construction is triggered by (a ϕ -feature and) a flat $\bar{A}$ -feature on BEI, it is expected that restrictions imposed specifically on a topic should not be imposed on the subject of BEI. As seen previously in (57) and (58), while phrases like *mei-you ren* 'nobody' and *mei-you shi* 'nothing' resist topicalization, as seen in (58), they are fully acceptable as the subject of BEI, as seen in (57). A further difference between topicalization and the BEI-construction lies in their information-structural effects: answers to questions such as *what happened* consist solely of new information – in such cases, topicalization is infelicitous, while the use of a BEI-construction is felicitous, as seen in (63).

(63) *Context: What happened?*

- a. #Lisi_i, jingcha zhuazou-le ___i.
 Lisi police arrest-PRF
 'Lisi, the police arrested (him).'
- b. Lisi_i bei (jingcha) zhuazou-le ___i.
 Lisi BEI police arrest-PRF
 'Lisi was arrested (by the poice).'

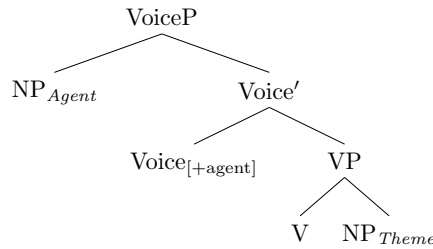
4 Proposed analysis

4.1 Bei as a passive head

In a nutshell, I propose to analyze the BEI-construction as a passive construction where BEI spells out a passive head, which selects a projection of the agent/external-argument-introducing Voice head (Kratzer 1996), in the sense of Bruening (2013). In overt-agent overt-agent BEI-constructions, which might be dubbed non-canonical passive constructions, the passive head/BEI selects a projection of a $\text{Voice}_{[+\text{agent}]}$ head (with the $[+\text{agent}]$ feature borrowed from Erlewine & Smith 2024) and is responsible for case-licensing the overtly expressed agent/external argument of the matrix verb introduced in Spec, VoiceP, albeit being semantically vacuous. In agent-less BEI-constructions, which might be dubbed canonical passive constructions, the passive head/BEI selects a projection of a $\text{Voice}_{[-\text{agent}]}$ head (with the $[-\text{agent}]$ feature borrowed from Erlewine & Smith 2024) and is responsible for existentially binding the agent/external argument (following Bruening 2013; see also Bach 1980; Keenan 1980, 1985; Williams 1987; a.o.).

Specifically, I assume, following Kratzer (1996), that a simple transitive construction (in the active voice) has the structure in (64), where the $\text{Voice}_{[+\text{agent}]}$ head (with the $[+\text{agent}]$ feature borrowed from Erlewine & Smith 2024) is responsible for introducing the agent/external argument of the matrix verb in Spec, VoiceP.

(64) *Simple transitive (active voice)*



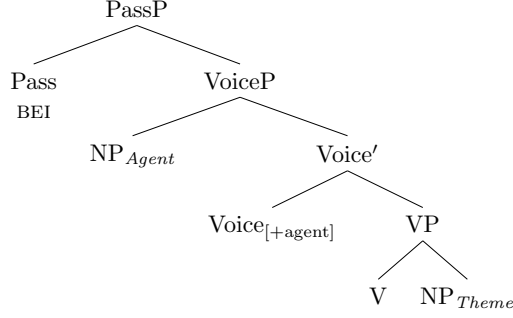
- a. *Definition of agent/external-argument-introducing Voice head (Kratzer 1996)*
 $\text{Voice}: \lambda x. \lambda e. \text{Agent}(e, x)$
- b. $\text{VP}_{\langle s, t \rangle}: \lambda e. \text{V}(e, \text{NP}_{\text{Theme}})$
- c. $\text{Voice}_{\langle e, \langle s, t \rangle \rangle}: \lambda x. \lambda e. \text{Agent}(e, x)$
- d. $\text{Voice}'_{\langle e, \langle s, t \rangle \rangle}: \lambda x. \lambda e. \text{V}(e, \text{NP}_{\text{Theme}}) \ \& \ \text{Agent}(e, x)^{28}$
- e. $\text{VoiceP}_{\langle s, t \rangle}: \lambda e. \text{V}(e, \text{NP}_{\text{Theme}}) \ \& \ \text{Agent}(e, \text{NP}_{\text{Agent}})$

I propose to analyze the BEI-construction as a passive construction where BEI spells out a passive head, which selects a projection of the Voice head, in the sense of Bruening (2013). Specifically, overt-agent BEI-constructions, which might be dubbed non-canonical passive constructions, have the structure in (65). The passive head/BEI selects a projection of a $\text{Voice}_{[+\text{agent}]}$ head (with the $[+\text{agent}]$ feature borrowed

²⁸The Voice head and the VP combine via *Event Identification* (Kratzer 1996).

from Erlewine & Smith 2024), which introduces the agent/external argument of the matrix verb in Spec, VoiceP. In this case, the VoiceP remains both syntactically and semantically identical to that in the active voice. The passive head/BEI is responsible for case-licensing the overtly expressed agent/external argument of the matrix verb introduced in Spec, VoiceP, albeit being semantically vacuous (in that it denotes an identity function).

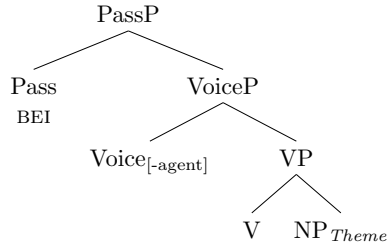
(65) *Overt-agent BEI-construction as (non-canonical) passive construction*



- a. *Definition of passive head (Bruening 2013)*
Pass (with overt agent): $\lambda f_{\langle st \rangle}. \lambda e. f(e)$
- b. VoiceP (with overt agent): $\lambda e. V(e, NP_{Theme}) \ \& \ Agent(e, NP_{Agent})$
- c. PassP (with overt agent): $\lambda e. V(e, NP_{Theme}) \ \& \ Agent(e, NP_{Agent})$

Agent-less BEI-constructions, which might be dubbed canonical passive constructions, have the structure in (66). The passive head/BEI selects a projection of a Voice_[-agent] head (with the [-agent] feature borrowed from Erlewine & Smith 2024). In this case, the VoiceP syntactically lacks a specifier and semantically has an “unsaturated” agent/external argument (in Bruening’s 2013 terms). The passive head/BEI is responsible for existentially binding the agent/external argument (following Bruening 2013; see also Bach 1980; Keenan 1980, 1985; Williams 1987; a.o.).

(66) *Agent-less BEI-construction as (canonical) passive construction*



- a. *Definition of passive head (Bruening 2013)*
Pass (agent-less): $\lambda f_{\langle e, st \rangle}. \lambda e. \exists x : f(x, e)$
- b. VoiceP (agent-less): $\lambda x. \lambda e. V(e, NP_{Theme}) \ \& \ Agent(e, x)$

c. PassP (agent-less): $\lambda e. V(e, NP_{Theme}) \ \& \ \exists x : Agent(e, x)$

The proposed analysis of the BEI-construction straightforwardly incorporates the syntactic properties of the the BEI-construction presented in section 2 – specifically, that in overt-agent the BEI-constructions the agent/external argument of the matrix verb is introduced in Spec, VoiceP; that in agent-less BEI-constructions, the the agent/external argument of the matrix verb is existentially bound; that BEI is best analyzed as a predicate selecting a VoiceP complement; and that BEI has been fully grammaticalized as a functional category in modern Mandarin.

In addition, I assume Burzio’s generalization (Burzio 1986), which states that all and only the verbs that can assign a theta-role to the (logical) subject can assign accusative case to an object.²⁹ Hence, in overt-agent BEI-constructions, the matrix Voice head not only assigns an agent theta-role to the external argument of the matrix verb but also assigns (accusative) case; in contrast, in agent-less BEI-constructions, the matrix Voice head does not assign a theta-role, nor does it assign case. Consequently, in agent-less BEI-constructions, when there is an overt NP that cannot be assigned case by the matrix Voice head, that NP must become the subject of BEI, where it can receive case from Infl; in such cases, it is predicted that long-distance dependencies between the subject of BEI and a deeply embedded gap in BEI’s complement is impossible (see section 5.1).

Before proceeding, a few more words are in order on some implementation details of the passive voice. To provide context, at the core of Bruening’s (2013) analysis of the passive construction, which is based on the English *be*-passive, is a (syntactic and semantic) distinction between the passive head and the agent/external-argument-introducing Voice head, which has two consequences: syntactically, in the English *be*-passive, the passive head must select a VoiceP with an “unsaturated” agent/external argument (that is, a VoiceP headed by a Voice_[-agent], in the sense of Erlewine & Smith 2024), regardless of whether an agent/external argument is overtly expressed or non-overt; semantically, the passive head exhibits distinct semantics depending on whether an agent/external argument is overtly expressed, with vacuous semantics when the agent/external argument is non-overt. The proposed analysis of the BEI-construction builds on Bruening’s (2013) analysis of the passive construction, but crucially allows the passive head/BEI to also select a VoiceP with a “saturated” agent/external argument (that is, a VoiceP headed by a Voice_[+agent], in the sense of Erlewine & Smith 2024). Consequently, the passive head/BEI exhibits distinct syntactic and semantic properties in overt-agent BEI-constructions, which might be dubbed non-canonical passive constructions, and agent-less BEI-constructions, which might be dubbed canonical passive constructions: syntactically, the passive head/BEI selects a projection of Voice_[+agent] in overt-agent BEI-constructions and is responsible for case-licensing the overtly expressed agent/external argument of the matrix verb introduced in Spec, VoiceP, but it selects a projection of Voice_[-agent] in agent-less BEI-constructions; semantically, the passive head/BEI is vacuous (in that it denotes an

²⁹It is worth noting that recent studies on various languages have argued against the universality of Burzio’s generalization (see, e.g., Legate 2014 on grammatical object passives in Icelandic and certain Slavic and Celtic languages; Bruening & Tran 2015 on the Vietnamese *bi*-construction; Šereikaite 2021 on Lithuanian active existentials; Akkus 2022 on Sason Arabic causatives; a.o.).

identity function) in overt-agent BEI-constructions, but is responsible for existentially binding the agent/external argument in agent-less BEI-constructions.

Alternatively, Legate, Akkuş & Šereikaitė (2020), building on Legate (2012, 2014), propose an analysis where the passive construction involves a single passive Voice head, treated as a variant of the Voice head. In the English *be*-passive, the passive Voice head differs from the active Voice head in two respects: syntactically, it introduces but does not project the agent/external argument in Spec, VoiceP, allowing it to be specified by a *by*-phrase; semantically, it allows the agent/external argument to either be specified by a *by*-phrase or be existentially bound. If one were to pursue an alternative analysis of the BEI-construction along the lines of Legate, Akkuş & Šereikaitė (2020), BEI might be treated as a predicate selecting either an active VoiceP or a passive VoiceP, similar to Bruening & Tran (2015) and Ngui (2024), which I will review in section 6.2. However, under such an analysis, the precise nature of BEI would remain unclear.

Ultimately, the goal of this paper is not to settle the debate between the two major theories of the passive but to establish a foundation for analyzing the BEI-construction as a passive construction. I leave it to future research to refine the implementation of the proposed analysis and to further investigate its implications for the broader theory of the passive.

4.2 Composite A/ $\bar{A}$ -movement

To account for the possibility of long-distance dependencies in the BEI-construction and, more generally, the mixed A/ $\bar{A}$ -properties associated with the BEI-construction, which I have considered to be direct consequences of (successive-cyclic) composite A/ $\bar{A}$ -movement following Van Urk (2015) and Longenbaugh (2017), I propose that the passive head/BEI hosts a composite probe $[\phi + \bar{A}]$, which attracts the closest NP with both a matching ϕ -feature and a matching $\bar{A}$ -feature – in particular, the $\bar{A}$ -feature on the passive head/BEI is flat, which can be satisfied by any $\bar{A}$ -feature on the goal (see Rizzi 1997, 2004; Abels 2012; a.o.); the subject in the BEI-construction undergoes (successive-cyclic) composite A/ $\bar{A}$ -movement to Spec, PassP, triggered by the composite probe $[\phi + \bar{A}]$ on the passive head/BEI (as well as the Voice head(s) involved in the successive-cyclic chain), as illustrated in (67-a), followed by a terminating step of A-movement, as illustrated in (67-b), which is akin to the analysis proposed by Longenbaugh (2017) for English *tough*-movement (see section 3.2).

(67) *Proposed analysis of BEI-construction*

a. *Step 1: (Successive-cyclic) composite movement to Spec, PassP*

$$[\text{PassP } \text{NP}_{[\phi], [\bar{A}]} \text{ BEI}_{[\phi + \bar{A}]} [\text{VoiceP } (\text{NP}_{[\phi]}) \text{ V } (\dots \text{ V } \dots) \text{ t } (\dots)]]$$

A/ $\bar{A}$ -movement

b. *Step 2: A-movement to Spec, IP*

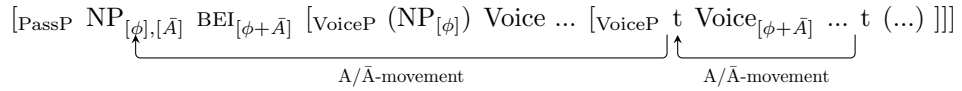
$$[\text{IP } \text{NP}_{[\phi], [\bar{A}]} \text{ Infl}_{[\phi]} [\text{PassP } \text{t } \text{ BEI } \dots \text{ t } (\dots)]]$$

A-movement

A few clarifications are in order: First, under the proposed analysis of the BEI-construction, the composite probe $[\phi + \bar{A}]$ on the passive head/BEI attracts the closest NP with both a matching ϕ -feature and a matching $\bar{A}$ -feature; other NPs between the passive head/BEI which hosts the composite probe $[\phi + \bar{A}]$ and the closest NP with both ϕ - and $\bar{A}$ -features are not interveners if they lack an $\bar{A}$ -feature. In overt-agent BEI-constructions, such NPs include the agent/external argument of the matrix verb, which is introduced in Spec, VoiceP.³⁰ Also recall that, in section 3.3, I have shown that when two NPs with both ϕ - and $\bar{A}$ -features move from BEI's complement, only the NP closer to BEI can be the subject of BEI. In such cases, the NP closer to BEI is an intervener for the other NP.

Second, a few more words are in order on successive-cyclic composite A/ $\bar{A}$ -movement involved in (step 1 of) the derivation of the BEI-construction. I assume that in the passive voice, it is the passive head (instead of the Voice head immediately below it) that heads a phase (see e.g., Collins 2005: 98), while in the active voice, the agent/external-argument-introducing Voice head heads a phase (Chomsky 2001). Hence, in addition to the passive head/BEI, which hosts a composite probe $[\phi + \bar{A}]$, the Voice head, when it heads a phase, must also host a composite probe $[\phi + \bar{A}]$, for purposes of successive-cyclic composite A/ $\bar{A}$ -movement. Concretely, I propose that when the BEI-construction involves multiple verbal projections and a long-distance dependency between the subject of BEI and a deeply embedded gap in BEI's complement, the subject of BEI is derived via successive-cyclic composite A/ $\bar{A}$ -movement through the specifiers of successive VoicePs, terminating at Spec, PassP, as illustrated in (68).

(68) *Successive-cyclic composite movement in BEI-construction*



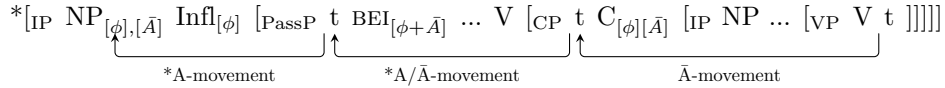
³⁰It is worth mentioning here that the agent/external argument of the matrix verb cannot become the subject of BEI, as seen in (i-a), and resists extraction in general, as seen in (i-b-c).

- (i) a. *Jingcha_i bei ___i zhuazou-le Lisi.
 police BEI arrest-PRF Lisi
 Int. 'The police was arrested Lisi (by him).'
- b. *Jingcha_i, Lisi_j bei ___i zhuazou-le ___j.
 police Lisi BEI arrest-PRF
 Int. 'The police, Lisi was arrested (by him).'
- c. *[Lisi_i bei ___j zhuazou-le ___i] de jingcha_j
 Lisi BEI scold-PRF REL police
 Int. 'the police by whom Lisi was arrested'

The ungrammaticality of the examples in (i) could be due to several reasons, which I will leave for further investigation in future research: (i) (composite A/ $\bar{A}$ -)movement from Spec, VoiceP to Spec, BEIP would violate Spec-to-Spec anti-locality in the sense of Erlewine (2016, 2020) (which is also suggested by a reviewer); (ii) extracting the agent/external argument of the matrix verb stranding BEI might be ruled out due to a ban on *predicate* stranding in certain contexts in Mandarin, which is also observed in cases of matrix object extraction from object control constructions with a monosyllabic matrix verb (see e.g., Tang 2002) and in the extraction of an NP located between two verbs in (purposive) serial verb constructions (see e.g., Liao & Lin 2019) (which aligns better with my native speaker intuition). Crucially, the examples in (i) should not be taken as evidence that the agent/external argument of the matrix verb is introduced in a PP adjunct (contra Ngui 2024).

Finally, I assume, following Longenbaugh (2017), that in (step 2 of) the derivation of the BEI-construction, (successive-cyclic) composite A/ $\bar{A}$ -movement (to Spec, PassP) can be followed by A-movement (to Spec, IP), without violating the ban on improper (A- after $\bar{A}$ -) movement (see 14). I also assume, again following Longenbaugh (2017), a ban on composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement, which extends from the ban on improper A-after- $\bar{A}$ -movement (see 15); in particular, if a finite clause object undergoes $\bar{A}$ -movement to Spec, CP (because it crosses over the finite clause subject, which rules out the possibility of it undergoing A-movement to Spec, CP, and under the assumption that the Mandarin C does not host a composite probe $[\phi + \bar{A}]$, which rules out the possibility of the finite clause object undergoing composite A/ $\bar{A}$ -movement to Spec, CP), then it cannot undergo further composite A/ $\bar{A}$ -movement (and A-movement), as illustrated in (69) (see section 5.2).

(69) *No improper composite-after- $\bar{A}$ -movement in BEI-construction*



4.3 Overview of alternative analyses

As mentioned previously, a widely accepted alternative approach that derives the dependency involved in the BEI-construction via base-generation of the subject of BEI as an argument of BEI and $\bar{A}$ -movement of a NOP in BEI's complement, on a par with Chomsky's (1977, 1981) analysis of English *tough*-movement (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.). Hence, like Chomsky's (1977, 1981) analysis of English *tough*-movement, the alternative approach to the BEI-construction is bipartite, in the sense that it attempts to derive the A-properties associated with the BEI-construction by base-generating the subject of BEI as an argument of BEI, and derives the $\bar{A}$ -properties associated with the BEI-construction via $\bar{A}$ -movement of a NOP in BEI's complement.

More specifically, these various proposals differ in their specific analyses of overt-agent and agent-less BEI-constructions. For Huang, Li & Li (2009) (see also Ting 1995, 1998; Huang 1999; a.o.), while the subject of BEI is always base-generated, only overt-agent BEI-constructions involve NOP movement within BEI's complement. By contrast, agent-less BEI-constructions involve A-movement of a PRO, which is controlled by the subject of BEI. But for Bruening & Tran (2015) and Ngui (2024), both overt-agent and agent-less BEI-constructions uniformly involve base-generation of the subject of BEI and NOP movement in BEI's complement. In addition, building on Huang, Li & Li (2009) (see also Ting 1995, 1998; Huang 1999; a.o.), Liu & Huang (2016) consider the possibility of both a base-generation analysis and a raising analysis of the subject in the BEI-construction, but only for (overt-agent and agent-less) BEI-constructions involving one, rather than multiple, (extended) verbal projections, in BEI's complement.

In section 5, I will argue that the two restrictions on long-distance dependencies in the BEI-construction – namely, the requirement that no overt, case-less NPs should intervene between the subject of BEI and the gap in agent-less BEI-constructions,

and the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI – provide evidence for the proposed analysis of the BEI-construction as a passive construction where the subject in the BEI-construction is derived via A-movement after (successive-cyclic) composite A/ $\bar{A}$ -movement and, crucially, are not accounted for by the various proposals under the alternative approach to the BEI-construction involving base-generation of the subject of BEI and NOP movement in BEI’s complement. Then, in section 6, I will review the alternative analyses in more detail and recapitulate their problems.

5 On the restricted long-distance dependencies in the bei-construction

In this section, I will account for the two restrictions on long-distance dependencies in the BEI-construction presented previously in section 2.4. Specifically, in section 5.1, I will account for the ban on any overt, case-less NPs intervening between the subject of BEI and the gap in agent-less BEI-constructions, which follows from the proposed analysis of the BEI-construction as a passive construction and Burzio’s generalization (Burzio 1986); in section 5.2, I will account for the contrast when the BEI-construction involves a cross-finite-clause dependency between the subject of BEI and a subject vs. object gap, which crucially relies on the proposed analysis of the BEI-construction where the subject in the BEI-construction is derived via (successive-cyclic) composite A/ $\bar{A}$ -movement, followed by a terminating step of A-movement.

Before proceeding, a few words are in order about the finite vs. non-finite distinction in Mandarin, which has implications for clause size (see e.g., Huang 1982, 1989, 2022; Li 1990; Tang 2000; Lin 2011, 2012, 2015; a.o.). Specifically, as is commonly assumed, a distinction between finite and non-finite clauses in Mandarin “may be made on the basis of the potential occurrence of any element of the auxiliary category (such as an aspect marker or a modal)” (Huang 1989: 189). Both object control verbs like *bi(po)* ‘force’ (also *pai* ‘send’, *jiao* ‘order’, *qing(qiu)* ‘ask’, *(bai)tu* ‘entrust’) and subject control verbs like *shefa* ‘manage (Lit. find a way)’ (also *changshi* ‘try’, *qitu* ‘attempt’) take a non-finite clausal complement, because their complement cannot contain a modal verb (*hui* ‘will’, *yinggai* ‘should’, *neng* ‘can (be able)’, *keyi* ‘can (be permissible)’) or the progressive aspect (*zhengzai*), as seen in (70).

(70) *Non-finite clause: incompatible with aspect and modal*

a. *Object control*

Wo bi(po)/pai/jiao/qing(qiu)/(bai)tu Lisi_i [PRO_i
 1SG force/send/order/ask/entrust Lisi
 (*hui/*yinggai/*neng/*zhengzai/*zai) guolai].
 will/should/can/PROG/PROG come over
 ‘I forced/sent/ordered/asked/entrusted Lisi to (*will/*should/*can)
 come over(/*be coming over).’ (Based on Huang 1989: 189, ex. 8b)

b. *Subject control*

Lisi_i shefa/changshi/qitu [PRO_i *hui/*neng/*keyi/*zhengzai/*zai
 Lisi manage/try/attempt will/can/can/PROG/PROG
 guolai].
 come over
 ‘Lisi managed/tried/attempted to (*will/*can/*can) come over(/*be
 coming over).’ (Based on Huang 1989: 189, ex. 9b)

By contrast, verbs like *shuo* ‘say’ (also *renwei* ‘think’, *huaiyi* ‘suspect’, *xiangxin* ‘believe’) take a finite CP complement, which can contain a modal verb as well as the progressive aspect (*zheng*)*zai*, as seen in (71).

- (71) *Finite clause: compatible with aspect and modal*
 Lisi_i shuo/renwei/huaiyi/xiangxin [_{CP} ta_{i/j}
 Lisi say/think/suspect/believe 3SG
 hui/yinggai/neng/keyi/(zheng)zai guolai].
 will/should/can/can/PROG come over
 ‘Lisi_i said/thought/suspected/believed that he_{i/j} will/should/can/can come
 over/is coming over.’

Importantly, based on the contrast between (70) and (71), it is commonly assumed that infinitival complements of object control and subject control verbs are non-finite clauses that lack a CP projection (see e.g., Huang 1982, 1989, 2022; Li 1990; Tang 2000; Lin 2011, 2012, 2015; see also e.g., Wurmbrand 2014 for English). The absence of the C head in the embedded clause(s) will be necessary for the analysis of the requirement that no overt, case-less NPs should intervene between the subject of BEI and the gap in agent-less BEI-constructions, which I will present in section 5.1. Similarly, the presence of the C head in the embedded clause will also be crucial to the analysis of the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI, which I will present in section 5.2.

5.1 Long-distance dependencies in agent-less bei-constructions

In this section, I will account for the ban on any overt, case-less NPs intervening between the subject of BEI and the gap in agent-less BEI-constructions, which follows from the proposed analysis of the BEI-construction as a passive construction and Burzio’s generalization (Burzio 1986), which states that all and only the verbs that can assign a theta-role to the (logical) subject can assign accusative case to an object.

Recall that, under the proposed analysis of the BEI-construction as a passive construction, in overt-agent BEI-constructions, the passive head/BEI selects a projection of a Voice_[+agent] head (with the [+agent] feature borrowed from Erlewine & Smith 2024), which introduces the agent/external argument of the matrix verb in Spec, VoiceP, while in agent-less BEI-constructions, the passive head/BEI selects a projection of a Voice_[-agent] head (with the [-agent] feature borrowed from Erlewine & Smith 2024), which lacks a specifier. Furthermore, according to Burzio’s generalization, in

overt-agent BEI-constructions, the matrix Voice head not only assigns an agent theta-role to the external argument of the matrix verb but also assigns (accusative) case; in contrast, in agent-less BEI-constructions, the matrix Voice head does not assign a theta-role, nor does it assign case. Consequently, in agent-less BEI-constructions, when there is an overt NP that cannot be assigned case by the matrix Voice head, that NP must become the subject of BEI, where it can receive case from Infl; in such cases, it is predicted that long-distance dependencies between the subject of BEI and a deeply embedded gap in BEI's complement is impossible.

In the remainder of this section, I will show that the possibility of long-distance dependencies in agent-less BEI-constructions with object control matrix verbs (*Case 1*, in section 5.1.1), subject control matrix verbs (*Case 2*, in section 5.1.2), and *exceptional case-marking* (ECM) matrix verbs (*Case 3*, in section 5.1.3), all depends on whether there is an overt NP – the thematic object of an object control verb, or an overt controllee in the case of subject control, or an overt NP that is underlying the subject of the infinitival complement to an ECM verb – that cannot be assigned case by the matrix Voice head. Lastly, in section 5.1.4, I will discuss the possibility of analyzing agent-less BEI-constructions which (apparently) involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI's complement across non-finite clause boundaries as involving *(voice) restructuring* in BEI's complement, on a par with Wurmbrand's (2001, 2007) analysis of the German long passive.

5.1.1 Case 1: object control

First, I consider the (im)possibility of long-distance dependencies in agent-less BEI-constructions with object control matrix verbs, which has long been recognized, but has been a controversy, in the literature (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Her 2009; Bruening & Tran 2015; a.o.). Recall that in the following BEI-constructions in (72), which involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI's complement across non-finite clause boundaries, the agent/external argument of the matrix verb must be overtly expressed and cannot be non-overt. All of the BEI-constructions in (72) involve an overt NP, the matrix object, between the subject of BEI and the deeply embedded object gap. Under the proposed analysis, the BEI-constructions in (72) cannot be agent-less, because the matrix object would not be assigned case by the matrix Voice head.

(72) *Long-distance dependency in BEI-construction (object control)*

- a. Zhangsan_i bei *(Lisi) bipo na-ge jingcha_j [PRO_j zhuzou-le ___i].
 Zhangsan BEI Lisi force that-CL police arrest-PRF
 Lit. 'Zhangsan was forced that police to arrest *(by Lisi).'
- b. Na-feng xin_i bei *(wo) jiao Li_j [PRO_j qing Wang_k [PRO_k tuo
 that-CL letter BEI 1SG order Li ask Wang entrust
 Zhang_i [PRO_i ji-chu-le ___i]].
 Zhang send-out-PRF

- Lit. ‘That letter was ordered Li to ask Wang to entrust Zhang to send out *(by me).’ (Based on Huang, Li & Li 2009: 132: ex. 47b, 48b)
- c. Na-feng xin_i bei *(Zhangsan) pai/jiao/qing(qiu)/(bai)tu_j Lisi_j [PRO_j that-CL letter BEI Zhangsan send/order/ask/entrust Lisi shefa [PRO_j ji-chu-le ___i]].
manage send-out-PRF
Lit. ‘That letter was send/order/ask/entrust Lisi to manage to send out *(by Zhangsan).’
- d. Wang_i bei *(wo) mingling Li qingqiu ___i baituo Zhang ji-chu-le
Wang BEI 1SG order Li ask entrust Zhang send-out-PRF
na-feng xin.
that-CL letter
Lit. ‘Wang was ordered Li to ask to entrust Zhang to send out that letter *(by me).’ (Based on Pan & Hu 2021: ex. 29b)
- e. Zhang_i bei *(wo) mingling Li qingqiu Wang baituo ___i ji-chu-le
Zhang BEI 1SG order Li ask Wang entrust send-out-PRF
na-feng xin.
that-CL letter
Lit. ‘Zhang was ordered Li to ask Wang to entrust to send out that letter *(by me).’ (Based on Pan & Hu 2021: ex. 29c)

A digression is necessary at this point. Her (2009) provides two apparent counterexamples found on the internet (the % signs are mine to indicate speaker variation on the judgements):

- (73) a. %Wo_i gang zou-jin gongyuan, *pro*_i jiu bei pai jingcha
1SG just walk-into park at once BEI send police
qing-jin-le ___i jin-le paichusuo.
ask-into-PRF police station
‘Just as I walked into the park, (I) was sent police to ask into the police station.’ (Her 2009: ex. 20a)
- b. %Tongdao_i dou yijing bei pai bing bashou ___i.
passage DIST already BEI send troop guard
‘All passages were already sent troops to guard.’ (Her 2009: ex. 20a)

While virtually every speaker I have consulted (many of whom offered to consult additional speakers who reported the same judgements) reported that these examples in (73) are entirely unacceptable (and not just ‘admittedly odd’, as Her 2009 describes them), two suggestions have been made to improve their acceptability. First, Yitong Luo (p.c.) notes that (73-a) could be made acceptable with a strong contextual setup. For instance, imagine Zhang has been sending police to escort anyone entering the park to the police station for interrogation. Previously, Li entered the park and was brought to the police station by some police sent by Zhang. Then, just as I walked into the park, I too was brought to the police station by some of Zhang’s police. Note that, with such a context, (73-a) in fact receives a null pronoun reading, where the null pronoun

coreferences with Zhang – that is to say, (73-a) is not truly agent-less. Second, most of the speakers, including myself, find (73-b) more acceptable than (73-a), possibly because in (73-b), *bing* ‘troop’ is more readily interpreted as an indefinite expression. In Mandarin, indefinite NPs arguably receive inherent case, allowing them to surface in case-less positions (e.g., the thematic object position of an unaccusative verb), in contrast to definite NPs, as seen in (74):³¹

- (74) (Zheli) si/lai-le bing/yi-xie bing/henduo bing/*na-xie
 here die/come-PRF troop/one-PCL troop/many troop/that-PCL
 bing/*Lisi.
 troop/Lisi
 ‘(Here) died/came troops/some troops/many troops/*those troops/*Lisi.’

Notably, the acceptability of (73) (for some but arguably not the majority of speakers, and potentially due to one of the two confounds discussed above) was taken to the extreme by Her (2009) – and subsequently by Bruening & Tran (2015) and Ngui (2024) – as evidence for the general possibility of long-distance dependencies in agent-less BEI-constructions, which, I contend, is wrong-headed.

Returning to the proposed analysis. It is also expected that the following BEI-constructions in (75) are well-formed when agent-less, because the matrix object becomes the subject of BEI, where it can receive case from Infl.

- (75) a. Jingcha_i bei (Lisi) bipo ___i [PRO_i zhuazou-le Zhangsan].
 police BEI Lisi force arrest-PRF Zhangsan
 ‘The police was forced to arrest Zhangsan (by Lisi).’
 b. Li_i bei (wo) mingling ___i qing Wang tuo Zhang ji-chu-le
 Li BEI 1SG order ask Wang entrust Zhang send-out-PRF
 na-feng xin.
 that-CL letter
 ‘Li was ordered to ask Wang to entrust Zhang to send out that letter (by me).’ (Adapted from Pan & Hu 2021: ex. 29a)

5.1.2 Case 2: subject control

Second, I consider the possibility of long-distance dependencies in agent-less BEI-constructions with subject control matrix verbs (see e.g., Her 2009; Bruening & Tran 2015; a.o.). Recall that in the following BEI-constructions in (76), which also involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI’s complement across non-finite clause boundaries, the agent/external argument of the matrix verb can be overtly expressed or non-overt (note that the original examples from Her 2009, where the agent/external argument of the matrix verb is non-overt, are found on the internet). In contrast to the BEI-constructions in (72), the BEI-constructions in (76) involve no overt NPs between the subject of BEI and the deeply embedded object gap. Under the proposed analysis, the BEI-constructions

³¹In light of this potential confound, the examples presented in this paper are carefully constructed to always use a definite NP in this position.

in (76) can be agent-less, because no NP needs to be assigned case by the matrix Voice head.³²

- (76) *Long-distance dependency in BEI-construction (subject control)*
- a. Gongsì-de wangluo_i ceng bei (heike_j) changshi/qitu [PRO_j ruqin ___i].
 company's network once BEI hacker try/attempt hack
 Lit. 'The company's network once was tried/attempted to hack (by the hacker).' (Based on Her 2009: ex. 21a)
- b. Ziliao_i bei (xiaotou_j) shefa [PRO_j kaobei-le ___i].
 document BEI thief manage copy-PRF
 Lit. 'The documents were managed to copy (by the thief).' (Based on Her 2009: ex. 21b)

In (77), there is an overt NP, *Lisi*, which is the object of the embedded verb (*pai/jiao/qing(qiu)/(bai)tuo* 'send/order/ask/entrust'), intervening between the subject of BEI and the deeply embedded object gap. In this case, the embedded object *Lisi* is assigned case by the embedded Voice head; hence, no case problem arises when the agent/external argument of the matrix verb is non-overt (in which case it is the matrix Voice head that cannot assign case) (see 9).

- (77) Na-feng xin_i bei %(Zhangsan_j) shefa [PRO_j
 that-CL letter BEI Zhangsan manage
 pai/jiao/qing(qiu)/(bai)tuo Lisi_k [PRO_k ji-chu-le ___i].
 send/order/ask/entrust Lisi send-out-PRF
 Lit. 'That letter was managed to send/order/ask/entrust Lisi to send out %(by Zhangsan).'

In Mandarin, certain subject control verbs, e.g., *jihua* 'plan', *jueding* 'decide', allow for an overt controllee in their complement, as seen in (78) (see e.g., Zhang 2016; Li 2024).

- (78) *Subject control construction*
- Pingguo-gongsì_i jihua/jueding [ta-men_{i/*j}/PRO_i daliang shengchan
 Apple-Inc. plan/decide 3-PL massively produce
 zhe-kuan shouji].
 this-type cellphone
 'Apple Inc. planned/decided that they massively produce this type of cellphones.'

Note that in (79-a), where the controllee is overt, the agent/external argument of the matrix verb must be overtly expressed – in this case, I suggest that the overt controllee needs to be assigned case by the matrix Voice head, hence the BEI-construction cannot be agent-less. By contrast, in (79-b), where the embedded subject is a PRO, the agent/external argument of the matrix verb can be overtly expressed or non-overt –

³²Here, I conveniently assume that PROs are case-less, but this assumption is not strictly required – even if PROs require a special null case, such a case would be assigned by the embedded Infl, and hence the PROs are expected to be insensitive to whether or not the matrix Voice head can assign case.

in this case, no NP needs to be assigned case by the matrix Voice head, hence the BEI-construction can be agent-less.³³

- (79) *Long-distance dependency in BEI-construction (subject control)*
- a. Zhe-kuan shouji bei *(Pingguo-gongsi_i) jihua/jueding [ta-men_i/*_j
 this-type cellphone BEI Apple-Inc. plan/decide 3-PL
 daliang shengchan __].
 massively produce
 Lit. ‘This type of cellphones was planned/decided that (they) massively produce *(by Apple Inc.).’
- b. Zhe-kuan shouji bei (Pingguo-gongsi_i) jihua/jueding [PRO_i
 this-type cellphone BEI Apple-Inc. plan/decide
 daliang shengchan __].
 massively produce
 Lit. ‘This type of cellphones was planned/decided to massively produce (by Apple Inc.).’

In (80), there is an overt NP, *Pingguo-gongsi* ‘Apple Inc.’, which is the object of the embedded verb (*pai/jiao/qing(qiu)/(bai)tu* ‘send/order/ask/entrust’), intervening between the subject of BEI and the deeply embedded object gap. In this case, the embedded object *Pingguo-gongsi* ‘Apple Inc.’ is assigned case by the embedded Voice head; hence, no case problem arises when the agent/external argument of the matrix verb is non-overt (in which case it is the matrix Voice head that cannot assign case).³⁴

- (80) Zhe-kuan shouji bei %(zhengfu_i) jihua/jueding [PRO_i
 this-type cellphone BEI government plan/decide
 pai/jiao/qing(qiu)/(bai)tu Pingguo-gongsi_j PRO_j daliang shengchan
 send/order/ask/entrust Apple-Inc. massively produce
 __]].

³³A plausible alternative explanation for this contrast is that overt controllees and PROs achieve coreference with their controllers in different ways: overt controllees, being pronouns, cannot, for independent reasons, be bound by a non-overt agent/external argument of the matrix verb; in contrast, PROs are subject to implicit control, a phenomenon that is cross-linguistically well-established (Julie Legate, p.c.). Here, I will not attempt to explore the binding possibilities of implicit arguments further, but to note that in numerous other context where there is an overt NP that cannot be assigned case by the matrix Voice head, long-distance dependencies between the subject of BEI and a deeply embedded gap in BEI’s complement is indeed impossible.

³⁴Similar to the case reported in 9, a few speakers I have consulted reported that (80) is not (entirely) acceptable when agent-less (with varying degree of degradedness reported); interestingly, the degradedness of (80) can be mitigated by embedding a future-oriented adverb such as *ming-nian* ‘next year’ within the infinitival complement of the subject control matrix verb, as in (i):

- (i) Zhe-kuan shouji bei (zhengfu_i) jihua/jueding [PRO_i ming-nian pai/jiao/qing(qiu)/(bai)tu
 this-type cellphone BEI government plan/decide next-year send/order/ask/entrust
 Pingguo-gongsi_j [PRO_j daliang shengchan __]].
 Apple-Inc. massively produce
 Lit. ‘This type of cellphones was planned/decided to send/order/ask/entrust Apple Inc. to massively produce next year (by the government).’

This piece of data will be accounted for in section 5.1.4.

Lit. ‘This type of cellphones was planned/decided to send/order/ask/entrust Apple Inc. to massively produce %(by the government).’

5.1.3 Case 3: exceptional case-marking

Lastly, I consider the possibility of long-distance dependencies in agent-less BEI-constructions with matrix verbs like *yunxu* ‘allow’, *jinzhi* ‘forbid’, *tongyi* ‘agree’, *fandui* ‘object’, which I analyze as ECM verbs (contra Li 1990, who refutes the existence of ECM verbs in Mandarin). Like object control verbs, these verbs take a non-finite clause complement that lacks a CP projection, which cannot contain a modal verb or the progressive aspect (*zheng*)*zai*, as seen in (81).

- (81) Wo yunxu/jinzhi/tongyi/fandui Lisi (*hui/*neng/*keyi/*zhengzai/*zai)
 1SG allow/forbid/agree/object Lisi will/can/can/PROG/PROG
 guolai.
 come over
 Lit. ‘I allowed/forbade/agreed/objected Lisi to (*will/*can/*can) come over(/*be coming over).’

But unlike object control verbs, these verbs allow for the apparent matrix object to be identified with the thematic object of the embedded verb, which can be achieved by means of embedding a BEI-construction, as seen in (82-a); this suggests that the apparent matrix object is underlyingly the embedded subject, which is not thematically related to the matrix verb.

- (82) a. Zhengfu yunxu/jinzhi/tongyi/fandui zhe-kuan shouji_i bei
 government allow/forbid/agree/object this-type cellphone BEI
 (Pingguo-gongsi) daliang shengchan ___i.
 Apple-Inc. massively produce
 Lit. ‘The government allowed/forbade/agreed/objected this type of cellphones to be produced massively (by Apple Inc.).’
 b. *Cf. Object control construction*
 *Zhengfu bi(po)/pai/jiao/qing(qiu)/(bai)tuozhe-kuan shouji_i bei
 government force/send/order/ask/entrust this-type cellphone BEI
 (Pingguo-gongsi) daliang shengchan ___i.
 Apple-Inc. massively produce
 Int. ‘The government forced/sent/ordered/asked/entrusted this type of cellphones to be produced massively (by Apple Inc.).’

Also unlike object control verbs, these verbs allow for the subject of the embedded clause to be an arbitrary PRO, as seen in (83-a).

- (83) a. Zhengfu yunxu/jinzhi/tongyi/fandui Pingguo-gongsi/PRO_{arb}
 government allow/forbid/agree/object Apple-Inc.
 daliang shengchan zhe-kuan shouji.
 massively produce this-type cellphone

Lit. ‘The government allowed/forbade/agreed/objected (Apple Inc.) to produce this type of cellphones massively.’

b. *Cf. Object control construction*

Zhengfu bi(po)/pai/jiao/qing(qiu)/(bai)tu
government force/send/order/ask/entrust

Pingguo-gongsi/*PRO_{arb} daliang shengchan zhe-kuan shouji.

Apple-Inc. massively produce this-type cellphone

‘The government forced/sent/ordered/asked/entrusted *(Apple Inc.) to produce this type of cellphones massively.’

Note that in (84-a), where the embedded subject is overt, the agent/external argument of the matrix verb must be overtly expressed – in this case, the overt embedded subject needs to be assigned case by the matrix Voice head, hence the BEI-construction cannot be agent-less. By contrast, in (84-b), where the embedded subject is an arbitrary PRO, the agent/external argument of the matrix verb can be overtly expressed or non-overt – in this case, no NP needs to be assigned case by the matrix Voice head, hence the BEI-construction can be agent-less.³⁵

(84) *Long-distance dependency in BEI-construction (ECM)*

- a. Zhe-kuan shouji_i bei *(zhengfu) yunxu/jinzhi/tongyi/fandui
this-type cellphone BEI government allow/forbid/agree/object

Pingguo-gongsi daliang shengchan ___i.

Apple-Inc. massively produce

Lit. ‘This type of cellphones was allowed/forbidden/agreed/objected Apple Inc. to massively produce *(by the government).’

- b. Zhe-kuan shouji_i bei (zhengfu) yunxu/jinzhi/tongyi/fandui
this-type cellphone BEI government allow/forbid/agree/object

PRO_{arb} daliang shengchan ___i.

massively produce

Lit. ‘This type of cellphones was allowed/forbidden/agreed/objected to massively produce (by the government).’

In (85), there is an overt NP, *Pingguo-gongsi* ‘Apple Inc.’, which is the object of the embedded verb (*pai/jiao/qing(qiu)/(bai)tu* ‘send/order/ask/entrust’), intervening between the subject of BEI and the deeply embedded object gap. In this case, the embedded object *Pingguo-gongsi* ‘Apple Inc.’ is assigned case by the embedded Voice head; hence, no case problem arises when the agent/external argument of the matrix verb is non-overt (in which case it is the matrix Voice head that cannot assign case).³⁶

³⁵As with controlled PROs, I conveniently assume that arbitrary PROs are also case-less, but this assumption is not strictly required – even if PROs require a special null case, such a case would be assigned by the embedded Infl, and hence the PROs are expected to be insensitive to whether or not the matrix Voice head can assign case.

³⁶Similar to the cases reported in 9 and 34, a few speakers I have consulted reported that (85) is not (entirely) acceptable when agent-less (with varying degree of degradedness reported); interestingly, the degradedness of (85) can be mitigated by embedding a future-oriented adverb such as *ming-nian* ‘next year’ within the infinitival complement of the ECM matrix verb, as in (i):

- (85) Zhe-kuan shouji_i bei %(zhengfu) yunxu/jinzhi/tongyi/fandui PRO_{arb}
 this-type cellphone BEI government allow/forbid/agree/object
 pai/jiao/qing(qiu)/(bai)tu Pingguo-gongsi_j [PRO_j daliang shengchan
 send/order/ask/entrust Apple-Inc. massively produce
 ___i].

Lit. ‘This type of cellphones was allowed/forbidden/agreed/objected to send/order/ask/entrust Apple Inc. to massively produce %(by the government).’

Also, the BEI-construction in (86) is well-formed, because the otherwise case-less NP (i.e., the embedded subject) becomes the subject of BEI, where it can receive case from Infl.

- (86) Pingguo-gongsi_i bei (zhengfu) yunxu/jinzhi/tongyi/fandui ___i daliang
 Apple-Inc. BEI government allow/forbid/agree/object massively
 shengchan zhe-kuan shouji.
 produce this-type cellphone
 ‘Apple Inc. was allowed/forbidden/agreed/objected to massively produce this
 type of cellphones (by the government).’

5.1.4 Not (necessarily) restructuring

To summarize, all of the three cases I have considered in sections 5.1.1, 5.1.2 and 5.1.3 suggest that long-distance dependencies in agent-less BEI-constructions are possible, but, crucially, are restricted by the requirement that no overt, case-less NPs should intervene between the subject of BEI and the gap in agent-less BEI-constructions. Under the proposed analysis of the BEI-construction as a passive construction and in light of Burzio’s generalization (Burzio 1986), this restriction arises because in agent-less BEI-constructions, when there is an overt NP that cannot be assigned case by the matrix Voice head, that NP must become the subject of BEI, where it can receive case from Infl. With this, it can be maintained that both overt-agent and agent-less BEI-constructions uniformly involve composite A/ $\bar{A}$ -movement.

As mentioned previously, the alternative proposals differ in their analysis of agent-less BEI-constructions: Huang, Li & Li (2009) (see also Ting 1995, 1998; Huang 1999; a.o.) propose that agent-less BEI-constructions involve A-movement of a PRO, which is controlled by the subject of BEI; in contrast, both Bruening & Tran (2015) and Ngui (2024) maintain that agent-less BEI-constructions still involve $\bar{A}$ -movement of a NOP. I contend that both sets of proposals are problematic: Huang, Li & Li’s (2009) (see also Ting 1995, 1998; Huang 1999; a.o.) analysis of agent-less BEI-constructions

-
- (i) Zhe-kuan shouji_i bei (zhengfu) yunxu/jinzhi/tongyi/fandui PRO_{arb} ming-nian
 this-type cellphone BEI government allow/forbid/agree/object next-year
 pai/jiao/qing(qiu)/(bai)tu Pingguo-gongsi_j [PRO_j daliang shengchan ___i].
 send/order/ask/entrust Apple-Inc. massively produce
 Lit. ‘This type of cellphones was allowed/forbidden/agreed/objected to send/order/ask/entrust
 Apple Inc. to massively produce next year (by the government).’

This piece of data will be accounted for in section 5.1.4.

fails to account for the well-formedness of certain agent-less BEI-constructions which (apparently) involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI's complement across non-finite clause boundaries (namely, Case 2, with a subject control matrix verb and a PRO subject in the infinitival complement; and Case 3, with an ECM matrix verb and an arbitrary PRO subject in the infinitival complement). By contrast, the analyses proposed by Bruening & Tran (2015) and Ngui (2024) encounter the opposite problem: they fail to account for the ill-formedness of certain agent-less BEI-constructions that involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI's complement across non-finite clause boundaries (namely, Case 1, with an object control matrix verb; and Case 2, with a subject control matrix verb and an overt controllee in the infinitival complement; and Case 3, with an ECM matrix verb and an overt subject in the infinitival complement).

Nevertheless, it is still worth considering the possibility of analyzing agent-less BEI-constructions which (apparently) involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI's complement across non-finite clause boundaries as involving *(voice) restructuring* in BEI's complement, on a par with Wurmbrand's (2001, 2007) analysis of the German long passive. In the remainder of this section, I will first highlight the differences between the German long passive and agent-less BEI-constructions which (apparently) involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI's complement across non-finite clause boundaries, and then show that it is indeed possible, but crucially not necessary, for (agent-less) BEI-constructions to involve (voice) restructuring.

To provide context, in German, the so-called long passive, which involves a long-distance dependency between the passive-subject and a deeply embedded gap, is licensed by a so-called restructuring verb (e.g., *versuchen* 'to try'), as seen in (87-a); crucially, the long passive cannot be licensed by a so-called non-restructuring verb (e.g., *planen* 'to plan' and *beschliessen* 'to decide'), as seen in (87-b) (Wurmbrand 2001).

- (87) a. *German long passive: licensed by restructuring verb*
dass der Traktor_i [___i zu reparieren] versucht wurde
that the tractor-NOM to repair tried was
'that the tractor was tried to repair' (Wurmbrand 2001: 19, ex. 6a)
- b. *German long passive: not licensed by non-restructuring verb*
*dass der Traktor zu reparieren geplant/beschlossen wurde
that the tractor-NOM to repair plan/decide was
Int. 'that the tractor was planned/decided to repair' (Wurmbrand 2001: 267, ex. 214b-c)

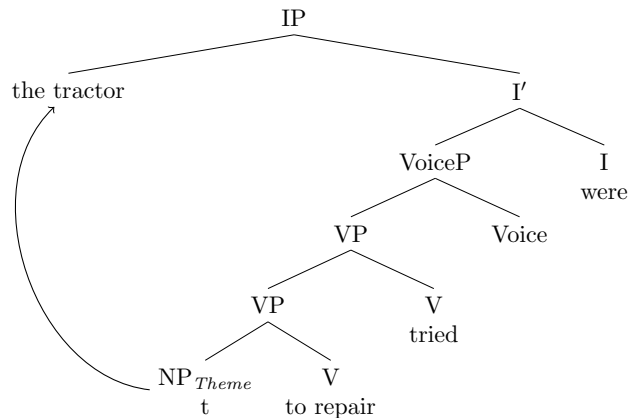
Furthermore, in German, a distinction between restructuring verbs like *versuchen* 'to try' and non-restructuring verbs like *planen* 'to plan' and *beschliessen* 'to decide' can be made on the basis of the possibility of embedded tense (and negation): restructuring verbs do not allow future adverbials, which introduce independent tense, in their

infinitival complement, as seen in (88-a), which indicates that their complement is structurally smaller than an IP; in contrast, non-restructuring verbs allow future adverbials in their infinitival complement, as seen in (88-b), which indicates that their complement is structurally as large as an IP.

- (88) a. *German restructuring verb: no embedded tense*
 Hans hat versucht (*morgen) zu verreisen.
 John has try tomorrow to travel
 ‘John tried to travel (*tomorrow).’ (Wurmbrand 2001: 78, ex. 61b)
- b. *German non-restructuring verb: embedded tense*
 Hans hat geplant/beschlossen (morgen) zu verreisen.
 John has plan/decide tomorrow to travel
 ‘John planned/decided to travel tomorrow.’ (Wurmbrand 2001: 78, ex. 61a)

Wurmbrand (2001, 2007) proposes that in German, (voice) restructuring is necessary to license a long passive, because apparent long-distance dependencies in the German long passive are nevertheless derived via A-movement, with the restructuring verb taking a VoiceP(-less) infinitival complement (see Wurmbrand 2016 and Wurmbrand & Shimamura 2017 for a more fine-grained analysis of voice restructuring involving a special Voice_R head), as illustrated in (89).

- (89) *German long passive as A-movement*



To support the restructuring analysis of the German long passive, it is further shown that, in the German long passive, which is only possible with restructuring verb, it is also the case that neither future adverbials nor sentential negation can be embedded within the restructuring infinitival complement (Wurmbrand 2001).

- (90) a. *German long passive: no embedded tense*

dass der Wagen (*morgen) über die Grenze zu schmuggeln
 that the car-NOM (*tomorrow) across the border to smuggle
 versucht wurde
 try was

Lit. ‘that the car was tried to smuggle across the border (*tomorrow)’
 (Wurmbrand 2001: 84, ex. 66a)

b. *German long passive: no embedded negation*

*... weil der Kuchen nicht zu essen versucht wurde
 since the cake not to eat try was

Int. ‘... since the cake was tried not to eat’ (Wurmbrand 2001: 118, ex. 91b)

In Mandarin, (subject control) verbs like *shefa* ‘manage (Lit. find a way)’ (which might be a candidate of a restructuring verb), (subject control) verbs like *jihua* ‘plan’, *jueding* ‘decide’ (which might be candidates of non-restructuring verbs), as well as (ECM) verbs like *yunxu* ‘allow’, *tongyi* ‘agree’ exhibit no distinction in their ability to both allow for embedded tense and negation in their infinitival complements and license long-distance dependencies in the BEI-construction. Hence, in contrast to German, there is no evidence that Mandarin makes a distinction between restructuring and non-restructuring verbs. More importantly, in contrast to the German long passive, it is easy to show that (voice) restructuring is not required to license long-distance dependencies in agent-less BEI-constructions: as seen in (91), both future adverbs and sentential negation can occur in the infinitival complement of the subject control or ECM verb, suggesting that the infinitival complement of these verbs can be structurally as large as an IP.³⁷

(91) *Long-distance dependency in BEI-construction*

a. *Embedded tense*

Zhe-feng youjian_i bei (Lisi_j) shefa [PRO_j ming-tian fa-chu ___i].
 this-CL email BEI Lisi manage next-day send-out
 Lit. ‘This email was managed to send tomorrow (by Lisi).’

b. *Embedded negation*

Zhe-ge xiaoxi_i bei (ta-men_j) shefa [PRO_j bu rang Lisi zhidao ___i].
 this-CL news BEI 3-PL manage not let Lisi know
 Lit. ‘This news was managed to not let Lisi know (by them).’

c. Zhe-kuan shouji_i bei (Pingguo-gongsi_j) jihua/jueding [PRO_j
 this-type cellphone BEI Apple-Inc. plan/decide
 (ming-nian) (bu) daliang shengchan ___i].

next-year not massively produce

Lit. ‘This type of cellphones was planned/decided to (not) massively produce (next year) (by Apple-Inc.).’

³⁷This should not be confused with the evidence that BEI’s complement is structurally smaller than an IP/AspP, presented in section 2.3.

- d. Zhe-ge xiaoxi_i bei (jingcha) yunxu/tongyi [PRO_{arb} (ming-tian) (bu)
 this-CL news BEI police allow/agree next-day not
 gongbu ___i].
 publish
 Lit. ‘This news was allowed/agreed to (not) publish (tomorrow) (by the
 police).’

Nevertheless, it should be noted that the ability for these subject control or ECM verbs to take an IP complement does not rule out their possibility of taking a VoiceP(-less) complement. As noted in 9, 34, and 36, a few speakers I have consulted reported that the examples in (92) (which are fully acceptable when agent-less, but crucially with embedded tense) would not be entirely acceptable when agent-less (with varying degree of degradedness reported), without the future-oriented adverb embedded within the infinitival complement of the subject control matrix verb:³⁸

- (92) a. Na-feng xin_i bei (Zhangsan_j) shefa [PRO_j ming-tian
 that-CL letter BEI Zhangsan manage next-day
 pai/jiao/qing(qiu)/(bai)tu Lisi_k [PRO_k ji-chu ___i].
 send/order/ask/entrust Lisi send-out-PRF
 Lit. ‘That letter was managed to send/order/ask/entrust Lisi to send out
 tomorrow (by Zhangsan).’
- b. Zhe-kuan shouji bei (zhengfu_i) jihua/jueding [PRO_i ming-nian
 this-type cellphone BEI government plan/decide next-year
 pai/jiao/qing(qiu)/(bai)tu Pingguo-gongsi_j PRO_j daliang
 send/order/ask/entrust Apple-Inc. massively
 shengchan ___j].
 produce
 Lit. ‘This type of cellphones was planned/decided to
 send/order/ask/entrust Apple Inc. to massively produce next year (by
 the government).’
- c. Zhe-kuan shouji_i bei (zhengfu) yunxu/jinzhì/tongyi/fandui
 this-type cellphone BEI government allow/forbid/agree/object
 PRO_{arb} ming-nian pai/jiao/qing(qiu)/(bai)tu Pingguo-gongsi_j [PRO_j
 next-year send/order/ask/entrust Apple-Inc.
 daliang shengchan ___i].
 massively produce
 Lit. ‘This type of cellphones was allowed/forbidden/agreed/objected to
 send/order/ask/entrust Apple Inc. to massively produce next year (by
 the government).’

³⁸In particular, Danfeng Wu (p.c.) notes that for her, while (92-a) and (92-c) are degraded when agent-less without the future-oriented adverb, (92-b), which involves *jihua/jueding* ‘plan/decide’, is fully acceptable when agent-less without the need of a future-oriented adverb. This indicates that, despite the lack of contrast in (91), a restructuring vs. non-restructuring distinction could still be present in Mandarin, in the sense that restructuring verbs are assumed to take a VoiceP-less complement by default (although they can also take an IP complement), while non-restructuring verbs are assumed to take an IP complement by default.

This, I suggest, is because for some speakers, without the presence of the future-oriented adverb, restructuring indeed occurs in the above examples, which renders the embedded infinitival complement(s) VoiceP-less; as a result, even an embedded object (*Lisi* or *Pingguo-gongsi* ‘Apple-Inc.’) would require case from the matrix Voice head, which is unavailable when the examples are agent-less. However, with the presence of a future-oriented adverb, a non-restructuring analysis is enforced, in which case the embedded object (*Lisi* or *Pingguo-gongsi* ‘Apple-Inc.’) is assigned case by the embedded Voice head; in such a case, no case problem arises when the examples are agent-less.

5.2 Long-distance dependency across finite clause boundary

Recall that another restriction on long-distance dependencies in the BEI-construction lies in a contrast when the BEI-construction involves a cross-finite-clause dependency between the subject of BEI and a subject vs. object gap. Specifically, the BEI-construction does not allow for a cross-finite-clause dependency between the subject of BEI and an object gap, as seen in (93-b) (see e.g., Ting 1995, 1998; a.o.), but allows for a cross-finite-clause dependency between the subject of BEI and a subject gap, as seen in (93-c) (see e.g., Her 2009).

- (93) a. *Cross-finite-clause dependency in BEI-construction (finite clause object gap)*
 *Lisi_i bei (jingcha) renwei/huaiyi/xiangxin [_{CP} Zhangsan hui mousha
 Lisi BEI police think/suspect/believe Zhangsan will murder
 ___i].
 Int. ‘Lisi was thought/suspected/believed that Zhangsan will murder (him) (by the police).’ (Based on Ting 1998: ex. 28c)
- b. *Cross-finite-clause dependency in BEI-construction (finite clause subject gap)*
 Zhangsan_i bei (jingcha) renwei/huaiyi/xiangxin [_{CP} ___i hui mousha
 Zhangsan BEI police think/suspect/believe will murder
 Lisi].
 Lisi
 Lit. ‘Zhangsan was thought/suspected/believed that (he) will murder Lisi (by the police).’ (Based on Her 2009: ex. 25a)

In this section, I will account for this subject/object contrast, which crucially relies on the proposed analysis of the BEI-construction where the subject in the BEI-construction is derived via (successive-cyclic) composite A/ $\bar{A}$ -movement, followed by a terminating step of A-movement. In section 5.2.1, I will propose that the possibility of cross-finite-clause dependency between the subject of BEI and a subject gap follows from the possibility of raising to subject via A-movement to Spec, CP, or hyper-raising to subject (see e.g., Fong 2019; Wurmbrand 2019; Lohninger, Kovač & Wurmbrand 2022; a.o.). In section 5.2.2, I will propose that the impossibility of cross-finite-clause dependency between the subject of BEI and an object gap follows from the

ban on improper $\bar{A}$ -movement to Spec, CP followed by composite A/ $\bar{A}$ -movement (see Longenbaugh 2017).³⁹ The proposed analysis of the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI has implications for the feature composition of the probe on the Mandarin C head, which I will discuss in section 5.2.3.

Importantly, this subject/object contrast does not receive a straightforward explanation under the alternative approach to the BEI-construction involving base-generation of the subject of BEI and NOP movement in BEI's complement (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.). Specifically, if the dependency involved in the BEI-construction is derived via $\bar{A}$ -movement of a NOP, then cross-finite-clause dependencies between the subject of BEI and either a finite subject gap or a finite object gap should be possible (or impossible, depending on independent assumptions/evidence regarding whether or not NOP movement can cross a finite CP boundary).⁴⁰

5.2.1 Case 1: finite subject gap

I propose that in the BEI-construction, cross-finite-clause dependencies between the subject of BEI and a subject gap boundary are possible, as seen in (93-b), as the result of raising to subject via A-movement to Spec, CP, or hyper-raising to subject (see e.g., Fong 2019; Wurmbrand 2019; Lohninger, Kovač & Wurmbrand 2022; a.o.): if the finite clause subject can undergo A-movement to Spec, CP, triggered by a pure ϕ -probe on the C head, then it can undergo further composite A/ $\bar{A}$ -movement to Spec, PassP and A-movement to Spec, IP, as illustrated in (94).

$$(94) \quad \textit{Hyper-raising to subject in BEI-construction}$$

$$\begin{array}{c} \text{[IP NP}_{[\phi],[\bar{A}]} \text{ Infl}_{[\phi]} \text{ [PassP t BEI}_{[\phi+\bar{A}]} \dots \text{ V [CP t C}_{[\phi],[\bar{A}]} \text{ [IP t Infl}_{[\phi]} \dots]]]] \\ \quad \quad \quad \underbrace{\hspace{10em}}_{\text{A-movement}} \quad \quad \underbrace{\hspace{10em}}_{\text{A}/\bar{\text{A}}\text{-movement}} \quad \quad \underbrace{\hspace{10em}}_{\text{A-movement}} \end{array}$$

In (94), the assumption that the Mandarin C head hosts a pure ϕ -probe, which licenses hyper-raising to subject, is important, which, I argue, is supported by the *general possibility of hyper-raising to subject in Mandarin*.

To provide context, Lee & Yip (2024) observe that in Cantonese and Vietnamese, some CP-selecting verbs, e.g., *gamgok* 'feel like (Cantonese)', *tengman* 'hear (Cantonese)', *cảm giác* 'feel like (Vietnamese)', *nghe nói* 'hear (Vietnamese)', but crucially not some other CP-selecting verbs, e.g., *gokdak/jingwai* 'think (Cantonese)',

³⁹Recall that English *tough*-movement shows the *opposite* pattern: it is degraded for non-subjects and impossible for subjects across a phasal CP-projection (Longenbaugh 2017; see also Postal 1971; Bresnan 1972; Chomsky 1973; Lasnik & Fiengo 1974; Browning 1987; Rezac 2006) (see section 3.2). To account for the English *tough*-movement pattern, Longenbaugh (2017: 10) suggests that degraded (but possible) instances of *tough*-movement, as seen in (34), are derived without an intermediate step of $\bar{A}$ -movement to Spec, CP (see 16). In contrast, to account for the ill-formedness of (93-a), it must be assumed that an intermediate step of $\bar{A}$ -movement to Spec, CP is strictly required in Mandarin.

⁴⁰As mentioned previously in section 3.2, Chomsky's (1977, 1981) analysis of English *tough*-movement involving base-generation of the *tough*-subject and NOP movement in the *tough*-predicate's complement faces similar problems in accounting for the (opposite) subject/non-subject contrast.

cho/ngĩ ‘think (Vietnamese)’, can license hyper-raising to subject, as seen in (95) and (96).

(95) *Hyper-raising to subject in Cantonese*

a. *Licit with gamgok ‘feel like’, tengman ‘hear’*

Coeng jyui gamgok/tengman [CP waa __i m-wui ting].
CL rain feel like/hear C not-will stop

Lit. ‘The rain is felt/heard that (it) will not stop.’ (Lee & Yip 2024: ex. 1a)

b. *Illicit with gokdak/jingwai ‘think’*

*[Mongkau ni loeng go zi]i gokdak/jingwai [CP __i jiging
online shopping this two CL word think/think already
singwai Hoenggongjan ge jatsoeng sangwut].
become Hong Konger MOD daily life

Int. ‘The two words *online shopping* are thought that (it) has become the daily life of Hong Kongers.’ (Lee & Yip 2024: ex. 7c)

(96) *Hyper-raising to subject in Vietnamese*

a. *Licit with cảm giác ‘feel like’, nghe nói ‘hear’*

Cơn mưa nàyi cảm giác/nghe nói [CP rằng/là __i sẽ không dừng].
CL rain this feel like/hear C FUT not stop

Lit. ‘The rain is felt/heard that (it) will not stop.’ (Lee & Yip 2024: ex. 1b)

b. *Illicit with cho/ngĩ ‘think’*

*Vậy thì [cả đời nàyi] cho/ngĩ [CP rằng __i chẳng còn cơ hội
then whole life this think/think C not left chance
nữa]!
more

Int. ‘Then, this whole life there is thought that (it) left no more chance.’
(Based on Lee & Yip 2024: ex. 8c)

Lee & Yip (2024) propose that a CP-selecting verb licenses hyper-raising to subject if it lexically encodes *indirect evidence* (in the sense that “the source of the speaker’s information is of a secondary nature, e.g., reportative and inferential, and the information does not settle the truth of the associating proposition”) and not *direct evidence* (in the sense that “the source of the speaker’s information is of a primary nature and the information settles the truth of the associating proposition”), based on the following contrasts in Cantonese and Vietnamese:

(97) a. *Hyper-raising predicate is compatible with indirect evidence in Cantonese*

Context: Your friend told you that Ming is playing piano at his home.

Aaming tengman taan-gan kam.

Ming hear play-PROG piano

Lit. ‘Ming is heard that (he) is playing piano.’ (Lee & Yip 2024: ex. 10b)

- b. *Hyper-raising predicate is incompatible with direct evidence in Cantonese*
Context: You live next to Ming and heard him playing piano at his home.
 #Aaming tengman taan-gan kam.
 Ming hear play-PROG piano
 Int. ‘Ming is heard that (he) is playing piano.’ (Lee & Yip 2024: ex. 12b)
- (98) a. *Hyper-raising predicate is compatible with indirect evidence in Vietnamese*
Context: On a winter day, John saw through a window that people are shivering outside. He said:
 Bên ngoài cảm giác rất lạnh.
 outside feel like very cold
 Lit. ‘The outside is felt that (it) is very cold.’ (Lee & Yip 2024: ex. 11b)
- b. *Hyper-raising predicate is incompatible with direct evidence in Vietnamese*
Context: On a winter day, John went out without wearing a coat. Shivering, he said:
 #Bên ngoài cảm giác rất lạnh.
 outside feel like very cold
 Int. ‘The outside is felt that (it) is very cold.’ (Lee & Yip 2024: ex. 13b)

In Mandarin, it is also the case that CP-selecting verbs like *ganjue* ‘feel like’ and *tingshuo* ‘hear’ license hyper-raising to subject, as seen in (99-a), but CP-selecting verbs like *renwei* ‘think’, *huaiyi* ‘suspect’, *xiangxin* ‘believe’ do not license hyper-raising to subject, as seen in (99-b) (Ka Fai Yip, p.c.).

- (99) *Hyper-raising to subject in Mandarin*
- a. *Licit with ganjue ‘feel like’, tingshuo ‘hear’*
 Beijing_i ganjue/tingshuo [CP ___i hui xia baoyu].
 Beijing feel like/hear will rain(V) rainstorm
 Lit. ‘Beijing is felt/heard that (it) will have a rainstorm.’
- b. *Illicit with renwei ‘think’, huaiyi ‘suspect’, xiangxin ‘believe’*
 *Beijing_i renwei/huaiyi/xiangxin [CP ___i hui xia baoyu].
 Beijing think/suspect/believe will rain(V) rainstorm
 Int. ‘Beijing is thought/suspected/believed that (it) will have a rainstorm.’

In addition, it is also the case that verbs that can license hyper-raising to subject are compatible with indirect evidence but not direct evidence, as seen in (100) and (101).

- (100) a. *Hyper-raising predicate is compatible with indirect evidence in Mandarin*
Context: Your friend told you that Lisi is playing piano at his home.
 Lisi_i tingshuo ___i (zheng)zai tan gangqin.
 Lisi hear PROG play piano
 Lit. ‘Lisi is heard that (he) is playing piano.’

- b. *Hyper-raising predicate is incompatible with direct evidence in Mandarin*
Context: You live next to Lisi and heard him playing piano at his home.
 #Lisi_i tingshuo ___i (zheng)zai tan gangqin.
 Lisi hear PROG play piano
 Int. ‘Lisi is heard that (he) is playing piano.’
- (101) a. *Hyper-raising predicate is compatible with indirect evidence in Mandarin*
Context: On a winter day, John saw through a window that people are huddling under umbrellas. He said:
 Waimian_i ganjue ___i (zheng)zai xia xue.
 outside feel like PROG snow(V) snow
 Lit. ‘The outside is felt that (it) is snowing.’
- b. *Hyper-raising predicate is incompatible with direct evidence in Mandarin*
Context: On a winter day, John stepped outside without an umbrella. Being hit by cold snowflakes, he said:
 #Waimian_i ganjue ___i (zheng)zai xia xue.
 outside feel like PROG snow(V) snow
 Int. ‘The outside is felt that (it) is snowing.’

Importantly, BEI is incompatible with verbs that can license hyper-raising to subject in (99-a), as seen in (102-a) (Ka Fai Yip p.c.), while BEI is compatible with verbs that cannot license hyper-raising to subject in (99-b), as seen previously in (93-b) and repeated in (102-b).

- (102) a. BEI is incompatible with *ganjue* ‘feel like’, *tingshuo* ‘hear’
 *Zhangsan_i bei ganjue/tingshuo [CP ___i xiang mousha Lisi].
 Zhangsan BEI feel like/hear want murder Lisi
 Int. ‘Zhangsan is felt/heard that (he) wants to murder Lisi.’
- b. BEI is compatible with *renwei* ‘think’, *huaiyi* ‘suspect’, *xiangxin* ‘believe’
 Zhangsan_i bei (jingcha) renwei/huaiyi/xiangxin [CP ___i hui mousha
 Zhangsan BEI police think/suspect/believe will murder
 Lisi].
 Lisi
 Lit. ‘Zhangsan was thought/suspected/believed that (he) will murder Lisi (by the police).’

I propose that hyper-raising is generally possible in Mandarin, as a result of the Mandarin C head generally hosting a pure ϕ -probe which triggers A-movement to Spec, CP, which can feed further A-movement to Spec, IP in the active voice, and can feed further composite A/ $\bar{A}$ -movement to Spec, PassP and A-movement to Spec, IP in the BEI-construction, as illustrated previously in (94). Furthermore, I propose that verbs like *ganjue* ‘feel like’ and *tingshuo* ‘hear’, which have been shown to encode indirect evidence, *lack a thematic subject*, hence can license hyper-raising to subject in the active voice, as seen in (99-a). By contrast, verbs like *renwei* ‘think’, *huaiyi* ‘suspect’, *xiangxin* ‘believe’ necessarily have a thematic subject (introduced in Spec, VoiceP); these verb cannot license hyper-raising, because their thematic subject will

undergo A-movement to Spec, IP in the active voice, effectively blocking the embedded subject from hyper-raising into the matrix clause. Importantly, under the proposed analysis of the BEI-construction as a passive construction, it is expected that verbs (that encode indirect evidence) that lack a thematic subject not only license hyper-raising to subject in the active voice but also resist passivization (cf. English: **Mary was appeared/seemed to be smart.*). Hence the ill-formedness of (102-a). By contrast, it is expected that verbs that cannot license hyper-raising to subject in the active voice (because it is blocked by their thematic subject), when passivized – with their thematic subject embedded under and case-licensed by the passive head/BEI, if overtly expressed, or being existentially bound by the passive head/BEI, if non-overt – can license hyper-raising to subject. This, I suggest, is the case of (102-b).⁴¹

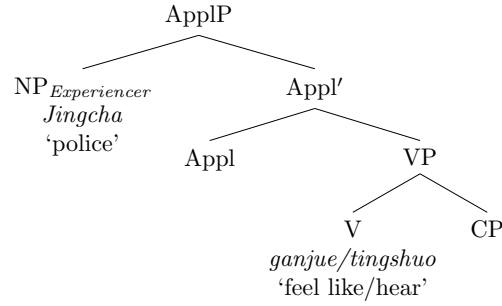
It is worth mentioning that in Mandarin, hyper-raising predicates like *ganjue* ‘feel like’ and *tingshuo* ‘hear’ also have apparently transitive uses, as seen in (103-a), and yet the BEI-construction in (103-b) is still ill-formed (Ka Fai Yip, p.c.).

- (103) a. Jingcha ganjue/tingshuo [_{CP} Zhangsan xiang mousha Lisi].
 police feel like/hear Zhangsan want murder Lisi
 ‘The police felt/heard that Zhangsan wants to murder Lisi.’
- b. *Zhangsan_i bei jingcha ganjue/tingshuo [_{CP} ___i xiang mousha Lisi].
 Zhangsan BEI police feel like/hear want murder Lisi
 Int. ‘Zhangsan is felt/heard that (he) wants to murder Lisi by the police.’

I suggest that in (103-a) the grammatical subject of hyper-raising predicates like *ganjue* ‘feel like’ and *tingshuo* ‘hear’ is not a thematic subject introduced by the Voice head but an experiencer indirect object introduced by an applicative head (cf. English, where experiencer arguments of raising predicates are introduced in PPs: *Mary seems/appears [_{PP} to John] to be smart.*), as illustrated in (104). Under the proposed analysis of the BEI-construction as a passive construction where the passive head/BEI selects a VoiceP, it is expected that (103-a), which lacks a thematic subject and effectively a VoiceP, cannot be passivized with BEI, hence the ill-formedness of (103-b) (see also section 2.3 for the incompatibility of BEI and an unaccusative verb).

⁴¹Note that the contrast between (102-a) and (102-b) does not receive a straightforward explanation under the alternative approach to the BEI-construction involving base-generation of the subject of BEI and NOP movement in BEI’s complement (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.). More generally, it is only under the proposed analysis of the BEI-construction as a passive construction that the (im)possibility of passivization with BEI diagnoses *agentivity*: the passive head/BEI requires the matrix verb to have a thematic subject, either assigning case to it (if it is overt) or existentially closing it (if it is non-overt).

- (104) *Experiencer subject as indirect object/applicative argument*



One piece of evidence supporting the analysis in (104) is that the grammatical subject of hyper-raising predicates like *ganjue* ‘feel like’ and *tingshuo* ‘hear’ in their apparently transitive use cannot be modified by a ‘deliberately’-type adverb, as seen in (105-a); in contrast, the grammatical subject of non-hyper-raising predicates like *renwei* ‘think’, *huaiyi* ‘suspect’, *xiangxin* ‘believe’ can be modified by a ‘deliberately’-type adverb, as seen in (105-b) (Ka Fai Yip, p.c.) (see section 7.1 for a discussion on the distribution and interpretations of ‘deliberately’-type adverbs).

- (105) a. Jingcha (*guyi) ganjue/tingshuo [_{CP} Zhangsan mousha-le Lisi].
 police deliberately feel like/hear Zhangsan murder-PRF Lisi
 ‘The police (*deliberately) felt/heard that Zhangsan wants to murder Lisi.’
- b. Jingcha guyi renwei/huaiyi/xiangxin [_{CP} Zhangsan
 police deliberately think/suspect/believe Zhangsan
 mousha-le Lisi].
 murder-PRF Lisi
 ‘The police deliberately thought/suspected/believed that Zhangsan murdered Lisi.’

5.2.2 Case 2: finite object gap

I propose that in the BEI-construction, cross-finite-clause dependency between the subject of BEI and an object gap is impossible, as seen in (93-b), due to the ban on improper $\bar{A}$ -movement to Spec, CP followed by composite A/ $\bar{A}$ -movement (Longenbaugh 2017): if a finite clause object undergoes $\bar{A}$ -movement to Spec, CP (because it crosses over the finite clause subject, which rules out the possibility of it undergoing A-movement to Spec, CP, and under the assumption that the Mandarin C does not host a composite probe $[\phi + \bar{A}]$, which rules out the possibility of the finite clause object undergoing composite A/ $\bar{A}$ -movement to Spec, CP), then it cannot undergo further composite A/ $\bar{A}$ -movement (and A-movement), as illustrated in (106) (see also section 4.2).

(106) *No improper composite-after- $\bar{A}$ -movement in BEI-construction*

$$\begin{array}{c}
*[\text{IP NP}_{[\phi],[\bar{A}]} \text{Infl}_{[\phi]} [\text{PassP t BEI}_{[\phi+\bar{A}]} \dots \text{V}_{[\text{CP t C}_{[\phi][\bar{A}]} [\text{IP NP} \dots [\text{VP V t}]]]}] \\
\begin{array}{ccc}
\text{---} & \text{---} & \text{---} \\
\text{*A-movement} & \text{*A/\bar{A}-movement} & \text{\bar{A}-movement}
\end{array}
\end{array}$$

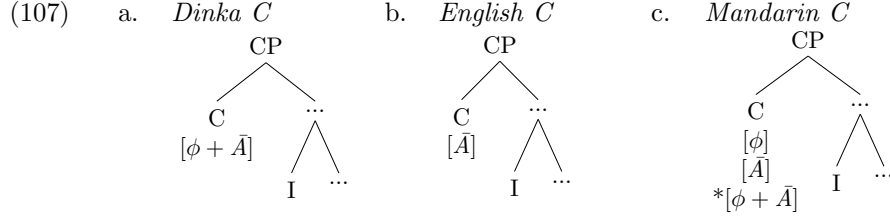
In (106), the assumption that the Mandarin C head does not host a composite probe $[\phi + \bar{A}]$, albeit housing a pure ϕ -probe (which licenses hyper-raising to subject, as seen in section 5.2.1), as well as pure $\bar{A}$ -probes (which are relativized to specific features, as previously discussed in section 3.3.3), is important, which, I argue, is supported by the *general clause-boundedness of composite A/ $\bar{A}$ -movement movement in Mandarin*. To provide context, in Mandarin, in addition to the BEI-construction, there are several other constructions – IP-internal topicalization and IP-internal focalization constructions – that both exhibit mixed A/ $\bar{A}$ -properties and are clause-bound (see e.g., Qu 1994; Shyu 1995; Ting 1995; Paul 2002, 2005; Kuo 2009; see also 18). Under the assumption that mixed A/ $\bar{A}$ -properties are direct consequences of (successive-cyclic) composite A/ $\bar{A}$ -movement and in light of the proposed analysis of the BEI-construction as a passive construction involving composite A/ $\bar{A}$ -movement, triggered by a composite probe $[\phi + \bar{A}]$ on the passive head/BEI, the IP-internal topicalization and IP-internal focalization constructions may be analyzed as also involving (successive-cyclic) composite A/ $\bar{A}$ -movement, triggered by a composite probe $[\phi + \text{Top}]$ on the IP-internal Topic head and a composite probe $[\phi + \text{Foc}]$ on the IP-internal Focus head, respectively. Under such an analysis, the clause-boundedness of all of these constructions would follow from the assumption that Mandarin C head does not host a composite probe $[\phi + \bar{A}]$. This line of analysis has been developed further in Chen (to appear), which I will not go into details here.

5.2.3 Feature composition of probe on C

As mentioned previously in section 3.2, Longenbaugh (2017) proposes that the distribution of composite probes can be different in different languages: In Dinka, both the C head and the Voice head host a composite probe $[\phi + \bar{A}]$; hence, composite A/ $\bar{A}$ -movement can cross finite clause boundaries (Van Urk 2015). In English, only the Voice head (involved in the path of *tough*-movement) hosts a composite probe $[\phi + \bar{A}]$ while the C head only hosts a pure $\bar{A}$ -probe; hence, composite A/ $\bar{A}$ -movement can proceed successive-cyclically through the specifiers of successive VoicePs, but cannot proceed from Spec, CP, i.e., following a step of $\bar{A}$ -movement to Spec, CP triggered by the pure $\bar{A}$ -probe on the C head, due to the ban on improper composite A/ $\bar{A}$ -movement after $\bar{A}$ -movement.

Under the proposed analysis of the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI, the Mandarin C head is not only unlike the Dinka C head in that it does not host a composite probe $[\phi + \bar{A}]$ and like the English C head in that it hosts pure $\bar{A}$ -probes, but also unlike the English C head in that it also generally hosts a pure ϕ -probe, which triggers A-movement to Spec, CP in the case of hyper-raising to subject. This

suggests a three-way difference in the feature composition of the probe on the C head across the three languages, as illustrated in (107).⁴²



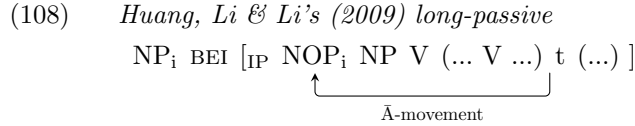
6 Remarks on alternative analyses

As mentioned previously, the various proposals under the alternative approach to the BEI-construction involving base-generation of the subject of BEI and NOP movement in BEI's complement differ in their specific analyses of overt-agent and agent-less BEI-constructions. In this section, I will review the alternative analyses in more detail and recapitulate their problems.

6.1 Huang, Li & Li (2009)

Huang, Li & Li (2009) (see also Ting 1995, 1998; Huang 1999; a.o.) analyze BEI as a two-place predicate (meaning ‘undergo’ or ‘experience’), both introducing the subject of BEI (as an experiencer argument of BEI) and selecting a secondary predicate of the subject of BEI. Additionally, they assume that different types of dependencies are involved in overt-agent BEI-constructions, which they dub *long-passives*, and agent-less BEI-constructions, which they dub *short-passives*.⁴³

Specifically, in their analysis of long-passives/overt-agent BEI-constructions, as illustrated in (108), BEI's complement is an IP, which contains an $\bar{A}$ -moved NOP, which is co-indexed with the subject of BEI; at LF, the NOP serves as a lambda operator, which turns a proposition into a predicate via lambda abstraction.⁴⁴



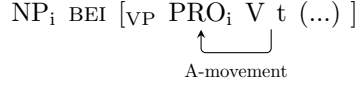
⁴²Lohninger, Kovač & Wurmbbrand (2022) arrive at a similar conclusion that ϕ - and $\bar{A}$ -features present on the same head may trigger movement together (which is the case of the Dinka C head) or must be satisfied independently (which is the case of the Mandarin C head).

⁴³As is pointed out by Bruening & Tran (2015), their naming of overt-agent and agent-less BEI-constructions as long- and short-passives is misleading, as there is nothing “passive” – neither object promotion nor agent/external argument demotion – in their analysis of the BEI-construction.

⁴⁴As is pointed out by Ernst (2010), an apparent problem with Huang, Li & Li's (2009) analysis of long-passives/overt-agent BEI-constructions is that BEI's complement cannot be as large as an IP/AspP (see section 2.3); such a problem can be avoided by simply reanalyzing BEI's complement as an extended verbal projection while maintaining other components of the analysis, i.e., the base-generation of the subject of BEI and NOP movement in BEI's complement.

In their analysis of short-passives/agent-less BEI-constructions, BEI's complement is a VP, which contains an A-moved PRO, which is controlled by the subject of BEI, as illustrated in (109) (see Hoshi 1991, 1994a, 1994b for a similar analysis of the English *get*-passive and the Japanese *ni*-passive).

(109) *Huang, Li & Li's (2009) short-passive*



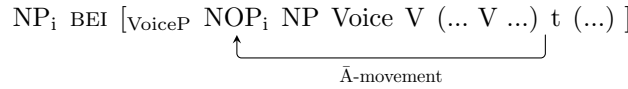
To recapitulate, in section 5.1, I have argued that Huang, Li & Li's (2009) (see also Ting 1995, 1998; Huang 1999; a.o.) analysis of agent-less BEI-constructions fails to account for the well-formedness of certain agent-less BEI-constructions which (apparently) involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI's complement across non-finite clause boundaries (namely, Case 2, with a subject control matrix verb and a PRO subject in the infinitival complement; and Case 3, with an ECM matrix verb and an arbitrary PRO subject in the infinitival complement). In section 5.1.4, I further argued that (voice) restructuring is not required to license long-distance dependencies in agent-less BEI-construction; hence, their assumption that agent-less BEI-constructions (must) involve A-movement dependencies, hence needing a different analysis from overt-agent BEI-constructions which involve $\bar{A}$ -dependencies, is untenable.

In addition, in section 5.2, I have argued that the alternative approach to the BEI-construction involving base-generation of the subject of BEI and NOP movement in BEI's complement generally fails to account for the contrast when the BEI-construction involves a cross-finite-clause dependency between the subject of BEI and a subject vs. object gap. Huang, Li & Li's (2009) analysis is no exception to this general problem.

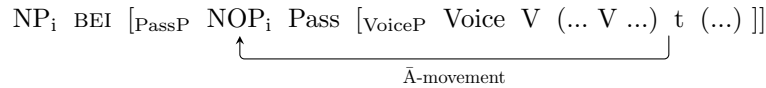
6.2 Bruening & Tran (2015)

Bruening & Tran (2015) also analyze BEI as a two-place predicate, both introducing the subject of BEI as its argument and selecting a secondary predicate of the subject of BEI, and provide different analyses for overt-agent and agent-less BEI-constructions. But Bruening & Tran (2015) propose that BEI selects an active VoiceP in overt-agent BEI-constructions and a passive VoiceP in agent-less BEI-constructions; both overt-agent and agent-less BEI-constructions involve an $\bar{A}$ -moved NOP, which is co-indexed with the subject of BEI, as illustrated in (110-a) and (110-b), respectively.

(110) a. *Bruening & Tran's (2015) active-voice-selecting BEI-construction*

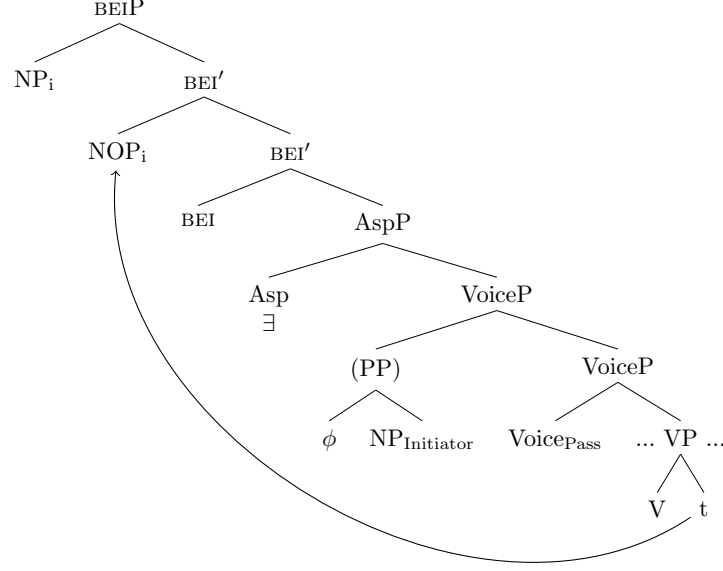


b. *Bruening & Tran's (2015) passive-voice-selecting BEI-construction*



More recently, Ngui (2024) proposes a variant of Bruening and Tran’s (2015) analysis, which is different in two ways: first, the complement of BEI always contains a passive VoiceP in the sense of Legate (2012, 2014) – in particular, in overt-agent BEI-constructions, the agent/external argument of the matrix verb (which is dubbed an *initiator*, following Legate 2012, 2014) is introduced within a PP adjunct of the passive VoiceP; second, it is assumed that NOP movement targets Spec, BEIP, instead of BEI’s complement. This analysis is illustrated in (111).⁴⁵

(111) *Ngui’s (2024) analysis of BEI-construction*



Again, to recapitulate, in section 5.1, I have argued that the analyses proposed by Bruening & Tran (2015) and Ngui (2024) encounter the opposite problem: they fail to account for the ill-formedness of certain agent-less BEI-constructions that involve a long-distance dependency between the subject of BEI and a deeply embedded object gap in BEI’s complement across non-finite clause boundaries (namely, Case 1, with an object control matrix verb; and Case 2, with a subject control matrix verb and an overt controllee in the infinitival complement; and Case 3, with an ECM matrix verb and

⁴⁵Ngui (2024) rejects the evidence from reflexive binding that in overt-agent the BEI-constructions, the agent/external argument of the matrix verb is introduced in Spec, VoiceP, assuming, instead, that in the relevant cases a bare reflexive *ziji* ‘self’ can (always) be bound logophorically (but see section 2.1). He also assumes that the general ban on extracting the agent/external argument of the matrix verb in the BEI-construction could only stem from the ban on preposition stranding (but see 30). In addition, he also refutes any restriction on long-distance dependencies in agent-less BEI-constructions, which was extensively discussed in section 5.1 (see especially the discussion in section 5.1.1), and explicitly sets aside the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI, which was discussed in section 5.2, for his analysis. With these assumptions, a natural concern arises regarding the empirical adequacy of his analysis.

an overt subject in the infinitival complement). Importantly, while both Bruening & Tran (2015) and Ngui (2024) indeed assume that in agent-less BEI-constructions BEI’s complement contains a passive VoiceP, they would not predict the base-generated subject of BEI to be in any way sensitive to any case problem that might arise in BEI’s complement.⁴⁶

In addition, the analyses proposed by Bruening & Tran (2015) and Ngui (2024) are also no exception to the general difficulty of accounting for the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI encountered by the alternative approach to the BEI-construction involving base-generation of the subject of BEI and NOP movement in BEI’s complement, as discussed in section 5.2.

6.3 Liu & Huang (2016)

Lastly, Liu & Huang (2016), as a representative proposal considering both a base-generation analysis and a raising analysis of the subject in the BEI-construction, argue, following Huang (2013), that the subject of BEI may indeed be either base-generated or derived. This is an attempt to reconcile two conflicting arguments regarding the base-generated vs. derived status of the subject of BEI presented in the literature, which I will discuss in section 7.

Specifically, they propose to decompose BEI into a two-place predicate meaning ‘experience’ (Exp) and a raising predicate meaning ‘become’ (Bec), and suggest that overt-agent and agent-less BEI-constructions involving just a simple transitive verbal projection (which they dub *local long-passives* and *short-passives*, respectively) can be analyzed as involving either base-generation of the subject of BEI (as an argument of the Exp head) and A-movement of a PRO (to Spec, BecP), controlled by the subject of BEI, as illustrated in (112-a), or a derived subject of BEI via A-movement (to Spec, BecP, in which case the Exp head is absent), as illustrated in (112-b).

- (112) a. *Liu & Huang’s (2016) control analysis of local long- and short-passives*
- $$[IP \ NP_i \ Infl \ [_{ExpP} \ t \ Exp \ [_{BecP} \ PRO_i \ Bec \ [_{VoiceP} \ t \ (NP \ Voice \ V \ t \ (...)) \]]]]$$
- ↑
↑
↑
- A-movement
A-movement
A-movement
- b. *Liu & Huang’s (2016) raising analysis of local long- and short-passives*
- $$[IP \ NP \ Infl \ [_{BecP} \ t \ Bec \ [_{VoiceP} \ t \ (NP \ Voice \ V \ t \ (...)) \]]]]$$
- ↑
↑
↑
- A-movement
A-movement
A-movement

Note that Liu & Huang (2016) assume with Huang, Li & Li (2009) that agent-less BEI-constructions (must) involve A-movement dependencies – hence, they also fail to account for the possibility of (apparent) long-distance dependencies in agent-less BEI-constructions.

⁴⁶Based on the Vietnamese *bi*-construction, Bruening & Tran (2015: 166, footnote 25) specifically assume that in Vietnamese case assignment does not differ from the active to the passive.

Furthermore, for BEI-constructions involving multiple verbal projections (which they dub *non-local long-passives*), Liu & Huang (2016) maintain Huang, Li & Li's (2009) analysis involving base-generation of the subject of BEI and NOP movement in BEI's complement. Hence, they also fail to account for the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI.

7 More on the base-generated vs. derived status of the subject of bei

In this section, I will reconcile two conflicting arguments regarding the base-generated vs. derived status of the subject of BEI presented in the literature. Specifically, in section 7.1, I will show that the distribution and interpretation of 'deliberately'-type adverbs in the BEI-construction can be accounted for under the proposed analysis of the BEI-construction in which the subject of BEI is derived rather than base-generated, with 'deliberately'-type adverbs having two possible attachment sites in the BEI-construction; in section 7.2, I will provide a critical review of the argument for a raising analysis of the subject in the BEI-construction based on the possibility for the subject of BEI and a (deeply embedded) verb in BEI's complement to form an idiom and the availability of its idiomatic meaning.

7.1 'Deliberately'-type adverbs

In the literature, an assumption has been made that 'deliberately'-type adverbs can only modify a base-generated subject (see e.g., Lakoff 1971; Lasnik & Fiengo 1974; Huang 1999, 2013; Huang, Li & Li 2009; Bruening & Tran 2015; Liu & Huang 2016; a.o.). Under such an assumption, the contrast between the English *be*-passive and the English *get*-passive with respect to whether 'deliberately'-type adverbs can modify the grammatical subject, as seen in (113), is taken to indicate that the subject of a *be*-passive is derived, while the subject of a *get*-passive is base-generated as an argument of *get*.⁴⁷

- (113) a. *English be-passive: 'deliberately' cannot modify grammatical subject*
 *Gillian was hit by that truck deliberately! (where Gillian is deliberate)
 (Bruening & Tran 2015: ex. 33a)
- b. *English get-passive: 'deliberately' can modify grammatical subject*
 Gillian got hit by that truck deliberately! (Bruening & Tran 2015: ex. 33b)

⁴⁷A few speakers, such as Jackendoff (1972: 83), David Pesetsky (p.c.), have reported that 'deliberately'-type adverbs can modify the grammatical subject of a *be*-passive in examples like (i).

- (i) *English be-passive: 'deliberately' can modify grammatical subject*
- a. John was carefully examined by the doctor.
 - b. Fred was carelessly arrested by the police.
 - c. Mary was intentionally seduced by Joe.

In the same vein, the major argument for a base-generation analysis of the subject of BEI has come from the possibility for ‘deliberately’-type adverbs to modify the subject of BEI, as seen in (114) (see e.g., Huang 1999, 2013; Huang, Li & Li 2009; Bruening & Tran 2015; Liu & Huang 2016; a.o.).

- (114) BEI-construction: ‘deliberately’ can modify subject of BEI
 Zhangsan_i guyi bei (Lisi) da-le ___i.
 Zhangsan deliberately BEI Lisi hit-PRF
 ‘Zhangsan deliberately got hit (by Lisi).’ (Huang, Li & Li 2009: 115, ex. 6-7)

Note that *guyi* ‘deliberately’ can also modify the agent/external argument of the matrix verb, whether it is overtly expressed or is non-overt, as seen in (115).

- (115) BEI-construction: ‘deliberately’ can modify subject of BEI
 Zhangsan_i bei (Lisi) guyi da-le ___i.
 Zhangsan BEI Lisi deliberately hit-PRF
 ‘Zhangsan was hit (by Lisi) deliberately.’

However, the assumption that ‘deliberately’-type adverbs can only modify a base-generated subject is falsified by the possibility for ‘deliberately’-type adverbs to modify the derived subject/underlying object of an unaccusative construction, both in English, as seen in (116) (see e.g., Bruening & Tran 2015), and in Mandarin, as seen in (117).

- (116) *English unaccusative construction: ‘deliberately’ can modify grammatical subject*
 a. The Iceman froze solid deliberately. (Bruening & Tran 2015: ex. 44c)
 b. The robot broke open deliberately. (Bruening & Tran 2015: ex. 44d)
- (117) *Mandarin unaccusative construction: ‘deliberately’ can modify grammatical subject*
 Lisi_i guyi bu lai/zou ___i.
 Lisi deliberately not come/leave
 ‘Lisi did not come/leave deliberately.’

Although, as seen in (118), transitivity the unaccusative constructions in (116) removes the possibility for ‘deliberately’-type adverbs to modify the underlying object of the unaccusative matrix verb (see e.g., Bruening & Tran 2015).

- (118) a. Mr. Freeze froze the Iceman solid deliberately. (only Mr. Freeze can be deliberate) (Bruening & Tran 2015: ex. 45b)
 b. Johnson broke the robot open deliberately. (only Johnson can be deliberate) (Bruening & Tran 2015: ex. 45c)

Based on examples like (116) and (118), Bruening & Tran (2015: 144) suggest that “when a ‘deliberately’-type adverb attaches to [a projection of] a predicate, it associates with the structurally highest argument of that predicate”: in (116) and

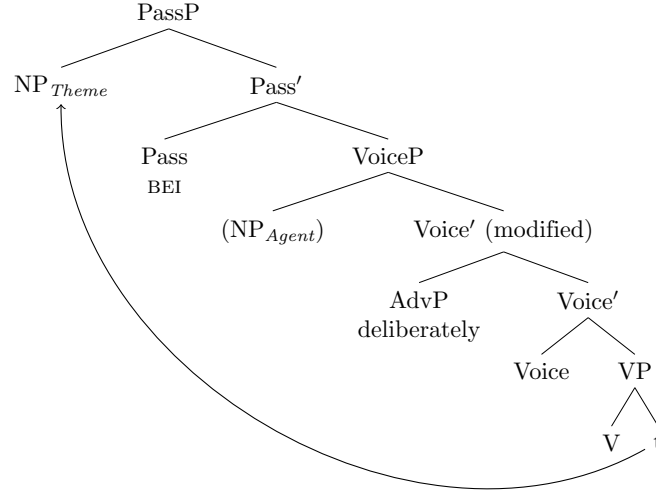
(117), the highest argument of the unaccusative (matrix) verb is the underlying object; in (118), or indeed any transitive construction, it is the agent/external argument. Building on Bruening & Tran (2015), I propose allowing ‘deliberately’-type adverbs to attach to projections of two particular predicates: the Voice head and the passive head/BEI. In the former case, the adverb associates with the agent/external argument of the matrix verb, while in the latter case, it associates with the subject of BEI.

Concretely, I assume the following denotation of *deliberately* in (119).

- (119) *Definition of ‘deliberately’*
 $\text{deliberately}: \lambda x. \lambda e. \text{deliberately}(e, x)$

In the BEI-construction, *guyi* ‘deliberately’ has two attachment sites. When it attaches to a projection of the Voice head, it associates with the agent/external argument of the matrix verb in Spec, VoiceP, as illustrated in (120).

- (120) *‘Deliberately’ modifying agent/external argument of matrix verb*

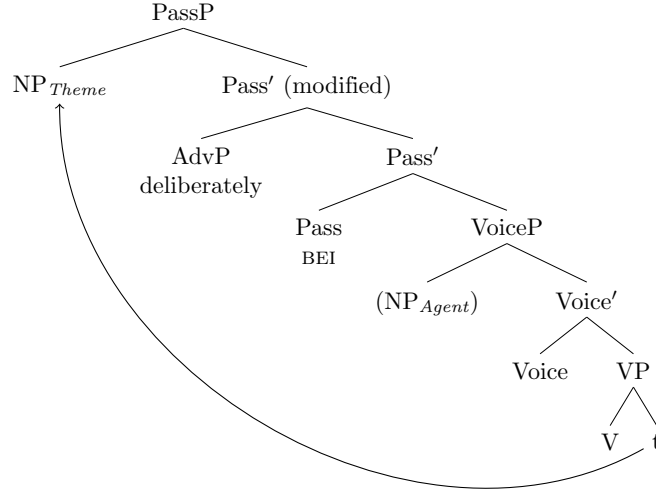


- a. $\text{Voice}': \lambda x. \lambda e. V(e, \text{NP}_{\text{Theme}}) \ \& \ \text{Agent}(e, x)$
- b. $\text{Voice}' \text{ (modified)}: \lambda x. \lambda e. V(e, \text{NP}_{\text{Theme}}) \ \& \ \text{Agent}(e, x) \ \& \ \text{deliberately}(e, x)$
- c. $\text{VoiceP}: \lambda e. V(e, \text{NP}_{\text{Theme}}) \ \& \ \text{Agent}(e, \text{NP}_{\text{Agent}}) \ \& \ \text{deliberately}(e, \text{NP}_{\text{Agent}})$

When *guyi* ‘deliberately’ attaches to a projection of the passive head/BEI, it associates with the subject of BEI in Spec, PassP, as illustrated in (121).⁴⁸

⁴⁸The speaker variation with respect to whether or not the grammatical subject of the English *be*-passive can be modified by ‘deliberately’-type adverbs might suggest that for some/most speakers, ‘deliberately’-type adverbs can only attach to a projection of the Voice head (hence ‘deliberately’-type adverbs can only associate with the (overtly expressed or non-overt) agent/external argument of the passivized verb), but for a few speakers, ‘deliberately’-type adverbs can also attach to a projection of the passive head (hence ‘deliberately’-type adverbs can also associate with the grammatical subject of the English *be*-passive).

(121) ‘*Deliberately*’ modifying subject of BEI



- a. $\text{Pass}' : \lambda e. V(e, \text{NP}_{\text{Theme}}) \ \& \ \text{Agent}(e, \text{NP}_{\text{Agent}})$
- b. $\text{Pass}' \text{ (modified)} : \lambda x. \lambda e. V(e, \text{NP}_{\text{Theme}}) \ \& \ \text{Agent}(e, \text{NP}_{\text{Agent}}) \ \& \ \text{deliberately}(e, x)$
- c. $\text{PassP} : \lambda e. V(e, \text{NP}_{\text{Theme}}) \ \& \ \text{Agent}(e, \text{NP}_{\text{Agent}}) \ \& \ \text{deliberately}(e, \text{NP}_{\text{Theme}})$

For present purposes, it suffices to show that the distribution and interpretation of ‘deliberately’-type adverbs in the BEI-construction can be accounted for under the proposed analysis of the BEI-construction in which the subject of BEI is derived rather than base-generated, with ‘deliberately’-type adverbs having two possible attachment sites in the BEI-construction. I leave it to future research to provide a full account of the distribution and interpretation of ‘deliberately’-type adverbs in broader contexts, which would require both expanding and restricting the possible attachment sites of ‘deliberately’-type adverbs, in order to account for both the possibility for ‘deliberately’-type adverbs to modify the derived subject/underlying object of an unaccusative construction, as seen in (117), and the impossibility of for ‘deliberately’-type adverbs to modify the derived subject/underlying experiencer indirect object of an apparently transitive construction with a hyper-raising predicate, as seen previously in (105) in section 5.1.

7.2 Idioms

The major argument for a raising analysis of the subject in the BEI-construction has come from the possibility for the subject of BEI and a (deeply embedded) verb in BEI’s complement to form an idiom and the availability of its idiomatic meaning. Concretely, in the BEI-constructions in (122), the idiom chunks *niu* ‘cow’ and *pianyi* ‘advantage’ are part of the idioms *chui niu* ‘bluff’ and *zhan pianyi* ‘take advantage’, respectively, and the idiomatic meanings of the idioms are preserved. Hence, Huang

(2013), Liu & Huang (2016), among others, have argued that the subject of BEI must be base-generated in the gap position in BEI’s complement, in order for the idiomatic meanings of the idioms to be available.

- (122) a. Niu_i dou bei (ta yi-ge-ren) chui-guang-le ___i.
 cow DIST BEI 3SG one-CL-person blow-be.empty-PRF
 ‘All the bluffing was done (by him alone).’ (Huang 2013: ex. 17)
- b. Pianyi_i dou bei (ta yi-ge-ren) zhan-guang-le ___i.
 advantage DIST BEI 3SG one-CL-person take-be.empty-PRF
 ‘All the advantage was taken (by him alone).’ (Liu & Huang 2016: ex. 11)

For all speakers I have consulted, including myself, the idiomatic meaning of *zhan pianyi* ‘take advantage’ is also available in (123-a), where the subject of BEI and a deeply embedded verb in BEI’s complement form an idiom (contra Huang 2013; Liu & Huang 2016), as well as in (123-b), which involves long-distance topicalization (see also Huang, Li & Li 2009: 206, ex. 37).⁴⁹

- (123) a. Pianyi_i bei ta jiao ziji-de jiaren zhan-guang-le ___i.
 advantage BEI 3SG order self’s family take-be.empty-PRF
 Lit. ‘Advantage was ordered his family to take by him.’
- b. Pianyi_i, ta jiao ziji-de jiaren zhan-guang-le ___i.
 advantage 3SG order self’s family take-be.empty-PRF
 Lit. ‘Advantage, his ordered his family to take.’

While these idiom facts are consistent with the proposed analysis of the BEI-construction in which the subject of BEI is derived (via A-movement after (successive-cyclic) composite A/ $\bar{A}$ -movement), the compositional nature of the idioms *chui niu* ‘bluff’ and *zhan pianyi* ‘take advantage’ raises a concern: as is pointed out by Nunberg, Sag & Wasow (1994), compositional idioms whose meanings are distributed among their parts – for example, in ‘take advantage’, “‘take’ is assigned a meaning roughly paraphrasable as ‘derive’, and ‘advantage’ means something like ‘benefit’” – are allowed to be separated syntactically, so long as their interpretations are composed in the permitted manner; hence, as is pointed out by Bruening & Tran (2015), basing an argument in favor of a raising analysis of the subject in the BEI-construction on the above idiom facts is particularly weak (if not invalid). Concretely, in Mandarin, it can be argued that the idiom chunks *niu* ‘cow’ and *pianyi* ‘advantage’ in the idioms *chui niu* ‘bluff’ and *zhan pianyi* ‘take advantage’ have idiomatic meanings on their own. Consider (124), where the idiom chunks *zhe-zhong niu* ‘this kind of cow’ and *zhe-zhong pianyi* ‘this kind of advantage’ are base-generated topics linked to a null object inside islands for extraction (via an $\bar{A}$ -moved NOP which is co-indexed with the base-generated topic; see Huang 1984: 570). In these cases, the idioms *chui zhe-zhong niu* ‘this kind of bluffing’ and *zhan zhe-zhong pianyi* ‘take this kind of advantage’ are not constituents (underlyingly), and yet the idiomatic meanings of the idioms are available.

⁴⁹Huang (2013) makes an assumption that the idiomatic meaning of an idiom is preserved only under A-movement, which is simply incorrect.

- (124) a. Zhe-zhong niu_i, wo mei jian-guo [_{NP} yi-ge [_{NOP_i} gan chui ___i]
 this-kind cow 1SG not see-EXP one-CL dare blow
 de ren].
 REL person
 ‘This kind of bluffing, I haven’t seen one single person who dares to
 make (it).’
 b. Zhe-zhong pianyi_i, wo zhidao [_{NP} henduo [_{NOP_i} ai zhan ___i]
 this-kind advantage 1SG know many love take
 de ren].
 REL person
 ‘This kind of advantage, I know many people who love to take (it).’

In contrast to compositional idioms, truly non-compositional idioms, such as *kick the bucket* ‘die’ in English and *ling hefan* ‘die (Lit. take boxed meal)’ in Mandarin, lose their idiomatic interpretations both in a *be*-passive or a BEI-construction and under topicalization, as seen in (125) and (126).

- (125) a. The bucket_i was kicked ___i by Pat. (Nunberg, Sag & Wasow 1994: 508)
 b. The bucket_i, Pat kicked ___i.
 (126) a. He-fan_i (zhongyu) bei (Lisi) ling-le ___i.
 boxed-meal at last BEI Lisi take-PRF
 ‘The boxed meal was taken (by Lisi) (at last).’ (Not ‘Lisi died (at last).’)
 b. He-fan_i, Lisi (zhongyu) ling-le ___i.
 boxed-meal Lisi at last take-PRF
 ‘The boxed meal, Lisi took (it) (at last).’ (Not ‘Lisi died (at last).’)
 c. Lisi (zhongyu) ling-le he-fan.
 Lisi at last take-PRF boxed-meal
 ‘Lisi died (at last).’

Because the availability or unavailability of the idiomatic meaning of an idiom in the BEI-construction ultimately depends on the compositional or non-compositional nature of the idiom, I contend that the idiom facts are not evidence for or against a raising analysis of the subject in the BEI-construction.

8 Indirect object as the subject of bei

Finally, in this section, I will extend the proposed analysis of the BEI-construction to BEI-constructions where the subject of BEI is identified with an indirect object in BEI’s complement (i.e., the so-called indirect passives; see e.g., Huang, Li & Li 2009), without an analysis of which the paper would feel incomplete. Specifically, I will provide an analysis of BEI-constructions where the subject of BEI is identified with the recipient indirect object of a canonical double-object construction or the affectee indirect object of an affective double-object construction. In (127-b), the

subject of BEI is identified with the recipient indirect object of a canonical double-object construction, where the verb is intrinsically ditransitive, sub-categorizing for both a recipient indirect object and a theme direct object.

- (127) a. *Canonical double-object construction*
 Wo gaozhi-le Zhangsan zhe-jian shi.
 1SG inform-PRF Zhangsan this-CL matter
 ‘I informed Zhangsan about this matter.’
 b. *Indirect object (recipient) as the subject of BEI*
 Zhangsan_i bei (wo) gaozhi-le ___i zhe-jian shi.
 Zhangsan BEI 1SG inform-PRF this-CL matter
 ‘Zhangsan was informed about this matter (by me).’

In (128-b) and (129-b), the subject of BEI is identified with the affectee indirect object of an affective double-object construction, where the verb is transitive, sub-categorizing for a theme direct object but not an affectee indirect object. Note that in (128-b) but not (129-b), the affectee indirect object and the theme direct object are in a possessor-possessum relation.⁵⁰

- (128) a. *Affective double-object construction*
 Wo he-guang-le Lisi yi-ping (ta xihuan de) jiu.
 1SG drink-be.empty-PRF Lisi one-bottle 3SG like REL wine
 ‘I drank (from) Lisi a bottle of wine (that he likes).’
 b. *Indirect object (affectee) as the subject of BEI*
 Lisi_i bei (wo) he-guang-le ___i yi-ping (ta xihuan de)
 Lisi BEI 1SG drink-be.empty-PRF one-bottle 3SG like REL
 jiu.
 wine
 ‘Lisi was drunk a bottle of wine (that he likes) (from) (by me).’
 (129) a. *Affective double-object construction*

⁵⁰This kind of affective double-object construction is productive in Mandarin:

- (i) a. Wo jiang/sheng-le ta liang-ci zhi.
 1SG lower/raise-PRF 3SG two-time position
 ‘I lowered/raised his position twice.’
 b. Ta_i bei (wo) jiang/sheng-le ___i liang-ci zhi.
 3SG BEI 1SG lower/raise-PRF two-time position
 ‘He was lowered/raised position twice (by me).’
 (ii) a. Wo bo-le zhe-ge yangcong liang-ceng pi.
 1SG peel-PRF this-CL onion two-layer skin
 ‘I peeled (from) this onion two layer’s of (its) skin.’
 b. Zhe-ge yangcong_i bei (wo) bo-le ___i liang-ceng pi.
 this-CL onion BEI 1SG peel-PRF two-layer skin
 ‘This onion was peeled two layers of (its) skin (from) (by me).’

Wo tou-jin-le Lisi yi-ge san-fen-qiu.
 1SG throw-be.in-PRF Lisi one-CL three-point-goal
 ‘I threw in a three-pointer on Lisi.’

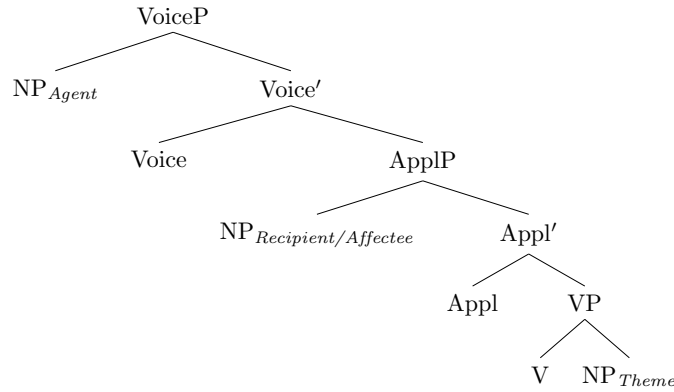
- b. *Indirect object (affectee) as the subject of BEI*

Lisi_i bei (wo) tou-jin-le ___i yi-ge san-fen-qiu.
 Lisi BEI 1SG throw-be.in-PRF one-CL three-point-goal
 Lit. ‘Lisi was thrown a three-pointer on (by me).’ (Based on Huang, Li
 & Li 2009: 140, ex. 66)

In addition, there is a contrast between canonical and affective double-object constructions with respect to whether the theme direct object can be the subject of BEI: in canonical double-object constructions, it can, while in affective double-object constructions, it cannot. There is also a contrast between canonical and affective double-object constructions with respect to whether either the indirect or direct object can be $\bar{A}$ -extracted: for canonical double-object construction, $\bar{A}$ -extraction of either the recipient indirect object or the theme direct object is possible, while for affective double-object construction, $\bar{A}$ -extraction of neither the affectee indirect object nor the theme direct object is possible. These facts are not illustrated with examples here but will be taken into consideration in the analysis.

I assume that both canonical and affective double-object constructions in Mandarin have the structure in (130), where the theme direct object is introduced by the verb, and the recipient or affectee indirect object is introduced by an applicative head which projects above the VP, following Marantz (1993), Bruening (2010), Holmberg, Sheehan & Van der Wal (2019) and others.⁵¹ In addition, I assume that the applicative head can either assign case to the recipient or affectee indirect object (in which case the Voice head assigns case to the theme direct object) or assign case to the theme direct object (in which case the Voice head assigns case to the recipient or affectee direct object), following Holmberg, Sheehan & Van der Wal (2019).

- (130) *Argument structure of double-object construction*



⁵¹In other approaches to the double-object constructions, the indirect and direct objects are contained in a small-clause-like constituent, which is the complement of the verb (see e.g., Harley 1995, 2002; Pesetsky 1995; Pytkänen 2002, 2008).

To account for the contrasts between canonical and affective double-object constructions with respect to whether the theme direct object can be the subject of BEI and whether either object can be $\bar{A}$ -extracted, I further assume that ApplP is not a phase in a canonical double-object construction, while it is a phase without a phase-EPP feature in an affective double-object construction (cf. McGinnis 2001). A direct consequence of such an assumption is that in an affective double-object construction, extraction of the theme direct object is never possible (neither in a BEI-construction nor in cases of $\bar{A}$ -extraction) (cf. Tsai 2018).

In a canonical double-object construction, either the indirect or direct object can be the subject of BEI and can be $\bar{A}$ -extracted. Under the proposed analysis of the BEI-construction, either object of a canonical double-object construction can be the subject of BEI, because either object can be targeted by the composite probe $[\phi + \bar{A}]$ on the passive head/BEI. In particular, either object of a canonical double-object construction can be the subject of BEI in an agent-less BEI-construction (where the Voice head does not assign a theta-role, nor does it assign case), because the applicative head can assign case to either the recipient indirect object (when the theme direct object becomes the subject of BEI) or the theme direct object (when the recipient direct object becomes the subject of BEI). Similarly, either object of a canonical double-object construction can be $\bar{A}$ -extracted, because either object can be targeted by an $\bar{A}$ -probe.⁵²

By contrast, in an affective double-object construction, the affectee indirect object can be the subject of BEI, but cannot be $\bar{A}$ -extracted. Under the assumption that in an affective double-object construction both VoiceP and ApplP are phases, the affectee indirect object must undergo movement from Spec, ApplP to the adjacent Spec, VoiceP – from a phase edge to an adjacent phase edge – to be $\bar{A}$ -extracted. I suggest that such a movement is banned for independent reasons.⁵³ Furthermore, under the proposed analysis of the BEI-construction, in the BEI-construction, it is the passive head/BEI, instead of the Voice head, that heads a phase; in this case, the affectee indirect object would move from Spec, ApplP to Spec, PassP, crossing the VoiceP, which, I suggest, is licit.

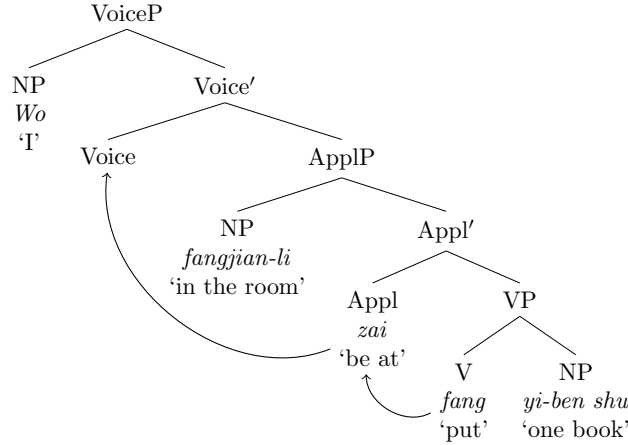
It is worth noting that while the subject of BEI can be identified with an indirect object introduced by an applicative head, it cannot be identified with an NP introduced in a PP adjunct. Concretely, the well-formedness of (131-a) and the ill-formedness of (132-a) suggest that in (131-b), the locative NP *fangjian-li* ‘in the room’ is an indirect object introduced by an applicative head in Spec, ApplP, hence it can be targeted by the composite probe $[\phi + \bar{A}]$ on the passive head/BEI, whereas in (132), the locative NP *fangjian-li* ‘in the room’ is introduced by the preposition *zai*

⁵²Note that in a canonical double-object construction, the recipient indirect object would undergo movement from Spec, ApplP to Spec, VoiceP, crossing ApplP but no other maximal projections, to be $\bar{A}$ -extracted. Such a movement would violate Spec-to-Spec anti-locality, in the sense of Erlewine (2016, 2020) (see also Deal 2019). Here, I will not attempt to resolve this issue, but simply note that cross-linguistically, languages do exhibit variation in whether the recipient indirect object can be $\bar{A}$ -extracted: in standard English, the recipient indirect object cannot be $\bar{A}$ -extracted, whereas in Mandarin, North-West English, Norwegian, Zulu, Lubukusu, and other languages, the recipient indirect object can be $\bar{A}$ -extracted (see e.g., Holmberg, Sheehan, & Van der Wal 2019).

⁵³In particular, Bošković (2015) shows that extraction from a double-phase (or phase-over-phase) configuration is banned cross-linguistically, and proposes an account of the relevant facts under particular assumptions about the domain and timing of spell-out.

‘at’ in a PP adjunct, hence it cannot be targeted by the composite probe $[\phi + \bar{A}]$ on the passive head/BEI.⁵⁴

- (131) a. (*Zai) fangjian-li_i bei (wo) fang-le ___i yi-ben shu.
 at room-in BEI 1SG put-PRF one-CL book
 Lit. ‘In the room was put a book (by me).’
 b. Wo fang-zai-le fangjian-li yi-ben shu.
 1SG put-be.at-PRF room-in one-CL book
 ‘I put a book in the room.’
 c. *Argument structure of (131-b)*

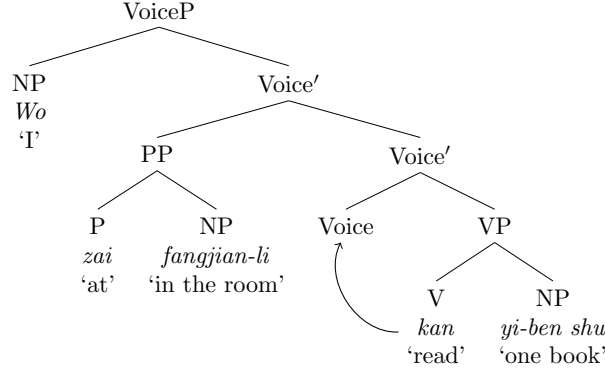


- (132) a. *Zai fangjian(-li)_i/*Fangjian-li_i bei (wo) ___i kan-le yi-ben shu.
 at room-in/room-in BEI 1SG read-PRF one-CL book
 Int. ‘(In) the room was read a book (by me).’
 b. Wo zai fangjian-li kan-le yi-ben shu.
 1SG at room-in read-PRF one-CL book
 ‘I read a book in the room.’

⁵⁴Note that the postposition *-li* ‘in’ is conventionally treated as a marker for locative NPs and crucially does not make the subject *fangjian-li* ‘in the room’ a PP in (131-a) (see Huang, Li & Li 2009: 13-14). A PP such as *zai fangjian-li* ‘in the room’ cannot surface as the subject of BEI or a grammatical subject more generally. Another test that can tease apart (131-b) and (132-b) is *locative inversion*: while a locative indirect object is subject to locative inversion, as seen in (i-a), a locative NP introduced in a PP adjunct cannot undergo locative inversion, as seen in (i-b). Again, note that the subject in (i-a) must be a locative NP and not a PP.

- (i) a. (*Zai) fangjian-li_i fang-le ___i yi-ben shu.
 at room-in put-PRF one-CL book
 Lit. ‘In the room put a book.’
 b. *Zai fangjian-li_i/*Fangjian-li_i ___i kan-le yi-ben shu.
 at room-in/room-in read-PRF one-CL book
 Int. ‘In the room read a book.’

c. *Argument structure of (132-b)*



9 Conclusion

The BEI-construction in Mandarin is a well studied construction known for exhibiting both passive-like properties and *tough*-movement-like properties (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.). I argued for a novel analysis of the BEI-construction in Mandarin as a passive construction where the passive head/BEI hosts a composite probe $[\phi + \bar{A}]$, which triggers composite A/ $\bar{A}$ -movement, in the sense of Van Urk (2015). The subject in the BEI-construction is derived via (successive-cyclic) composite A/ $\bar{A}$ -movement, followed by a terminating step of A-movement, similar to Longenbaugh’s (2017) analysis of English *tough*-movement. Under the proposed analysis, the mixed A/ $\bar{A}$ -properties associated with the BEI-construction are direct consequences of composite A/ $\bar{A}$ -movement (following Van Urk 2015; Longenbaugh 2017).

The proposed analysis of the BEI-construction accounted for two restrictions on long-distance dependencies in the BEI-construction – namely, the requirement that no overt, case-less NPs should intervene between the subject of BEI and the gap in agent-less BEI-constructions, and the subject/object contrast with respect to the possibility of crossing a finite clause boundary to become the subject of BEI. I argued that the ban on any overt, case-less NPs intervening between the subject of BEI and the gap in agent-less BEI-constructions follows from the proposed analysis of the BEI-construction as a passive construction and Burzio’s generalization (Burzio 1986), which states that all and only the verbs that can assign a theta-role to the (logical) subject can assign accusative case to an object. Specifically, in agent-less BEI-constructions, when there is an overt NP that cannot be assigned case by the matrix Voice head, that NP must become the subject of BEI, where it can receive case from Infl; in such cases, it is predicted that long-distance dependencies between the subject of BEI and a deeply embedded gap in BEI’s complement is impossible. I argued that the contrast when the BEI-construction involves a cross-finite-clause dependency between the subject of BEI and a subject vs. object gap follows from the possibility of raising to subject via A-movement to Spec, CP, or hyper-raising to subject (see e.g., Fong 2019; Wurmbrand 2019; Lohninger, Kovač & Wurmbrand 2022; a.o.), and

the ban on improper $\bar{A}$ -movement to Spec, CP followed by composite A/ $\bar{A}$ -movement (see Longenbaugh 2017).

Acknowledgments

This paper develops work presented in my earlier publication (Chen 2022) and in Chapter 2 of my PhD dissertation (Chen 2023). I am deeply grateful to those acknowledged in my thesis, especially my committee members – David Pesetsky, Danny Fox, and Lisa Cheng – for their guidance and support. I also thank the two anonymous reviewers and Julie Legate for their constructive feedback, which has led to a significant improvement of the paper. I am further indebted to Julie Legate and Daniel Harbour for their careful and supportive handling of the manuscript. Special thanks to Ka Fai Yip for insightful discussions and continued support throughout the development of this work. For judgements, I thank Keyi Zhao, Lisa Cheng, Zhouyi Sun, my linguist friends in the Many-Time-Zone Reading Group: Danfeng Wu, Haoming Li, Ka Fai Yip, Minqi Liu, Tommy Tsz-Ming Lee, Xuotong Yuan, Yitong Luo, and my friends in the Boston-Beijing Quad (BBQ): Jiatong Yang, Qi Hong, Wentao Xu, Xueqiu Deng, Yangyang Chen, Yonglin Wang. All remaining errors are mine.

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